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全同态加密方案的研究

作者姓名： 李晨

一级学科： 军队指挥学

二级学 密码学

学科门类： 军事学博士

指导教师姓名、职称： 李晖教授

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Research on Fully Homomorphic Encryption Schemes

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By
Li Chen
(Cryptography)
Supervisor: Li Hui
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摘要

全同态加密在密码学和信息安全领域都有广泛的应用，比如在隐私保护的云计算中，利用全同态加密技术可以保证用户的隐私信息在不用暴露给云服务器的前提下，就能够享受云服务器提供的服务；在安全多方计算中，利用全同态加密方案能够保护参与计算的各方的数据不被泄露给彼此，并且能够完成交互式的计算任务。对全同态加密技术的需求早在二十世纪七十年代就已经提出，此后虽然有一些全同态加密方案设计的尝试，也有一些支持单一同态运算的加密方案被提出，但是在很长一段时期里，如何设计安全的全同态加密作为密码学领域的一个公开问题一直没有得到解决。直到2009年，Gentry 提出第一个具有可证明安全性的全同态加密方案。在此之后一些全同态加密方案被陆续提出。但是这些全同态加密方案的效率都无法满足现实应用中的需要。所以对全同态加密方案的优化和新型设计是非常有意义和有迫切需求的研究课题。本文的工作正是基于这些研究需求而展开的。

本文首先介绍了全同态加密的研究历史和研究现状。我们回顾了现有的几种全同态加密方案，介绍了这些方案的背景知识，对各自的算法进行了形式化描述，对各自的正确性和安全性进行了分析，并且用多层电路的模型对这些方案的同态运算能力做了分析。

通过这些研究，我们了解了现有的全同态加密方案的设计思路，即首先设计一个支持部分同态运算的加密方案（以下称为部分同态加密方案），然后利用对解密算法的同态电路运算，来减少在任意同态电路运算中的噪声积累，从而实现任意电路的同态运算。由此可以看出，部分同态加密方案是构造全同态加密方案的基础。

在对现有的全同态加密方案中的部分同态加密方案的研究基础上，我们发现了这些部分同态加密方案的共同特点，提出了一种用来表示部分同态加密方案的统一模型，即双噪声结构。在这个模型下本文提出了完全全同态和广义全同态加密方案（即目前所说的全同态方案）的区别，并给出了由部分同态加密方案到广义全同态加密方案的构造条件。接下来，我们对一般的基于双噪声结构的广义全同态加密方案提出了一种通用的优化方案（两个版本），该优化方案在电路层面，对解密电路的复杂度进行约减，从而提高了广义全同态加密方案的效率。并且该优化方案还可以与算法层面的优化方案一起使用。

通过对双噪声结构和广义全同态构造的研究，我们指出在双噪声结构基础上是无法构造出完全全同态加密方案的。从而用无噪声的结构来构造完全全同态加密方案成为一种有意义的尝试。考虑到多元多项式环上有良好的代数结构和可以应用的困难问题，我们就基于多元多项式环构造了一个完全全同态加密方案，该方案可以看做是对 Groenber-basis Polly Cracker 方案的改进，与其他改进方案比较，本文所

提出的改进方案并没有引入噪声，而是利用 n 维空间上的仿射/线性变化来隐藏私钥，从而实现完全全同态的加密方案。

最后，本文还给出了一个基于Ring-LWE部分同态加密方案而构造的隐私保护的数据聚合方案，该方案用于保护智能电网通信中的数据隐私。利用该方案用户的个人数据既能得到保护，数据中心还能收集到更多的关于用户数据的统计量，从而为智能电网中的参数分析和优化提供更精细的数据支持。

关键词: 全同态加密，同态电路，双噪声加密。

Abstract

Fully homomorphic encryption (FHE) scheme has widely applications in cryptography and information security. For example, in secure multiparty computation FHE scheme can be used to protect the privacy of each party's data, while the computation involving these data can be interactively done; in privacy-preserving cloud computing, FHE scheme can be used to protect users' data from revealing to the cloud server, while users can also enjoy the service supplied by the cloud. The requirements for FHE scheme had been proposed in 1970s, after that some attempts for designing FHE scheme were presented, and some encryption scheme with single homomorphic property were also proposed. But for a long time, secure FHE scheme which is regarded as an open problem of modern cryptography remained unsolved. Until 2009, Gentry proposed the first FHE scheme with provable security. After that many FHE schemes with provable security have been proposed. But the efficiencies of these FHE schemes do not meet the requirements of practical use. It is strongly required to improve the efficiency of the present FHE scheme and to design new FHE schemes.

In this dissertation, we first review the present somewhat homomorphic encryption schemes. We introduce their background knowledge, describe their algorithms, analyze their correctness and security, and we also investigate the capacity of homomorphic evaluation of each somewhat homomorphic encryption scheme by using a multi-layer circuit model.

Based on the close looking at these somewhat homomorphic encryption schemes, we find their common point in structure, and we propose an uniform model—dual-noise structure—for representing the present somewhat homomorphic encryption schemes. Under this model we show the difference between perfect FHE and generalized FHE. We also give the explicit condition for the existence of generalized FHE. Furthermore, we propose a universal method to optimize generalized FHE scheme. Our optimization scheme is in circuit level. It is applicable for all present FHE schemes and can work together with an algorithm-level optimization.

By the analysis of the present FHE schemes under our dual-noise model, we claim that perfect FHE scheme cannot be achieved by using dual-noise structure. So we try to construct a perfect FHE scheme by using a noise-free structure. We propose a perfect FHE scheme based on multivariate polynomial ring. Our scheme can be viewed as an modified version of Groenber-basis Polly Cracker (GB Polly Cracker) scheme. We use linear/affine transformations instead of adding noise to improve the security of GB

Polly Cracker and remain the perfect fully homomorphic property.

Finally, we propose a privacy-preserving aggregation scheme for smart grid communications. Our aggregation scheme is based on RLWE-FHE scheme. With this aggregation scheme, users' private data can be protect from revealing to the data and control center in smart grid, while the data and control center can obtain some statistical quantities about users' data, so that a fine-grained analysis and optimization can be made for the smart grid.

Keywords: Fully homomorphic encryption, homomorphic circuit, dual-noise encryption.

Notions

$\mathbb{N}$	set of natural number
$\mathbb{Z}$	set of integers
$\mathbb{Z}_{\geq 0}$	set of non-negative integers
$\mathbb{Z}_{>0}$	set of positive integers
$\mathbb{Q}$	set of rational number
x (lowercase letters)	a number
$\boldsymbol{x}$ (bold lowercase letters)	a vector over numbers
$\boldsymbol{A}$ (bold uppercase letters)	a matrix over numbers

Abbreviations

FHE	Fully Homomorphic Encryption
SHE	Somewhat Homomorphic Encryption
LWE	Learning with Error
RLWE	Ring Learning with Error
PPT	Probabilistic Polynomial Algorithm
CPA	Chosen-Plaintext Attack

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Chapter 1 Introduction

1.1 Fully Homomorphic Encryption Scheme – History and Status

Since the present of RSA[1], the first public key encryption (PKE) scheme, the homomorphic property of public key encryption scheme had been concerned by researchers. For some purposes the homomorphic property should be avoided, but in some scenarios it needs to be remained for some special use. The homomorphic property means: given two ciphertexts $\text{Encrypt}(pk, m_1)$ and $\text{Encrypt}(pk, m_2)$, anyone can compute $\text{Encrypt}(pk, m_1 + m_2)$, or $\text{Encrypt}(pk, m_1 \cdot m_2)$, or both. Dual homomorphic public key encryption scheme was mentioned by Rivest et al.[2], but for a long period how to construct dual PKE scheme was an open problem in modern cryptography. Furthermore, given a set of ciphertext $\text{Encrypt}(pk, m_i), i = 1 \dots l$ for some $l \in \mathbb{N}$, if one can compute $\text{Encrypt}(pk, F(m_1, \dots, m_l))$ for an arbitrary function F , we say this encryption scheme fully homomorphic. Let the plaintexts take value in $GF(2)$, then any function can be represented as a circuit over $GF(2)$ which only includes addition and multiplication in $GF(2)$, this implies that dual homomorphic encryption is the basic of fully homomorphic encryption (FHE).

In 1993 the first fully homomorphic encryption scheme was proposed by Kobitz and Fellows[3], which is referred to as Polly Cracker. It is designed based on the problem of solving systems of multivariate polynomial equations, and in a short time a Groebner basis version Polly Cracker was proposed[4]. Although Polly Cracker support homomorphic evaluation of an arbitrary function, its security problem (see [5],[6],[7]) and inefficiency make it inflexible in practice. There are also some other works related to fully homomorphic encryption, such as [8],[9]. These lead us more and more close to fully homomorphic encryption.

Note that before fully homomorphic encryption scheme came into being, there is another technique to achieve circuit privacy or fully homomorphic that is using Yao's "garbled circuits" [10],[11] with a two-flow oblivious transfer protocol. But this technique typically has poor efficiency. Since the garbled circuit of an evaluating circuit will result in a high communication overhead for the secure transfer protocol, the garbled circuit technique is seldom used in practice.

Finally in 2009, Gentry proposed the first fully homomorphic encryption scheme with provable security [12]. Gentry first designs a somewhat homomorphic scheme based on ideal lattice, which is a structure can be viewed both as an ideal and lattice.

The property of the ideal leads to the dual homomorphic property of the scheme, and the hard problem of the lattice guarantees its provable security. Then Gentry uses a technique called bootstrapping to transform the somewhat homomorphic scheme into a fully homomorphic scheme. Gentry's FHE is a landmark in theoretic, but it is too inefficient to be used in practice. After Gentry's FHE, researchers proposed many other FHE schemes intending to design different and efficient FHE schemes. These schemes include:

- **DGHV-FHE scheme**[13]: This FHE scheme is designed over integers, and its security is based on the hard problem of approximating Greatest Common Divisor (A-GCD) problem. It is constructed using a standard technique that transforms a symmetrical encryption scheme into an asymmetrical encryption scheme, which is also used in LWE-based PKE scheme. This scheme has a simple structure and easy to understand. But due to the fast algorithm attacking A-GCD problem, the parameters in DGHV-FHE must be set very large, which results in a big public key size and a big ciphertext expansion. Therefore it is hard to be used in practical use.
- **SV-FHE scheme**[14]: This scheme is designed over algebraic integers, and its security is based on the hard problem of finding small generator of a principle ideal in algebraic integer ring. SV-FHE can be viewed as a special case of Gentry's FHE, its public key size is much smaller than that of Gentry's scheme. But it is still not enough efficient for practice.
- **LWE-FHE scheme**[15]: This scheme can be viewed as a homomorphic version of Regev's LWE encryption scheme. Its security is guaranteed by the hardness of LWE problem. It is a relatively secure and efficient FHE scheme, and many optimization techniques have been proposed for it. Some practical use[16] has shown that LWE-FHE may be the best choice in reality.
- **NTRU-FHE scheme**[17]: This scheme is designed over polynomial ring, its security relies on the hard problem on NTRU-lattice. It is more efficient than LWE-FHE scheme, but it is not as secure as LWE-FHE scheme.
- **PCN-FHE scheme**[18],[19]: This scheme is designed over multivariate polynomial ring, its security is based on hard problem of Groebner basis with noise. PCN is a brief referring of Polly Cracker with Noise, it has been proved as secure

as LWE-FHE, but since its big cost in encrypting and key storing, researchers prefer LWE-FHE than PCN-FHE scheme.

Other research works related to FHE include:

- **On IND-CCA1 secure Somewhat Homomorphic Encryption Scheme:** Since an encryption scheme supporting for any form of homomorphic operation cannot be secure against adaptive chosen ciphertext attacks, so it arises as an important problem that if it can achieve IND-CCA1 security, i.e. resist against lunchtime attacks. This problem is first mentioned in [12] and is still an open problem. Some variants of Gentry's FHE scheme, such as the scheme which is usually be referred to Gentry - Halevi variant [20] has been proved not IND-CCA1 secure[21]. Also in [21], A modified SV-somewhat homomorphic scheme is shown to be IND-CCA1 secure.
- **Single-Instruction Multiple-Data:**For the requirement of evaluating circuit on plaintext, the plaintext space of a FHE scheme is always set to be $\{0,1\}$, and the ciphertext expansion is always great. In a specific FHE scheme, to encrypt a plaintext $b \in \{0,1\}$, it first maps b to an element $\tilde{b}$ in a ring R , which the FHE scheme is based on, and there exists a homomorphism from R to $\{0,1\}$. The decryption is also can be viewed as: first obtains $\tilde{b}$, and then maps $\tilde{b}$ back to b . An interesting question arises as to whether we can directly set ring R as the plaintext space. In this case we can compact multi-data into a single plaintext instruction, and we refer to this operation as Single-Instruction Multiple-Data (SIMD), which is first mentioned in [14]. The original Gentry's somewhat homomorphic encryption does support SIMD operation, while the SV somewhat homomorphic encryption scheme does support SIMD operation. The LWE somewhat homomorphic encryption scheme also support SIMD operation [22]. The applications of SIMD have been realized in some scenarios[23].
- **Noise Reduction Technique:** The present somewhat homomorphic encryption schemes can be modeled as noise encryption, i.e. the ciphertext can be viewed as a plaintext with a carrying noise, correspondingly decryption can be viewed as a noise-eliminating process. But condition of correctly-decrypting is that the noise must remain in a certain small range. With homomorphic evaluating, the noise grows big, and when it exceeds the certain range, the decryption will not output the correct plaintext. This is the reason why FHE is hard to constructed.

Gentry introduced a noise reduction technique referred to “bootstrapping”, and constructed the first FHE with provable security. The bootstrapping technique can also be referred to “key-switching” technique, since it transforms a ciphertext with respect to a given secret key and with big noise into another ciphertext that with small noise but with respect to a new secret key. The advantage of bootstrapping or say key-switching is that one can successive uses it (in a range) to reduce the noise, but its disadvantage is the growing size of middle keys. Another noise reduction technique, known as “modulus-switching”, was introduced by Brakerski[24]. The concept of “modulus-switching” or say “modular reduction” was first mentioned in [15], and it is first be used to reduce the size of ciphertext. By using a “ladder of gradually decreasing moduli”, the noise increasing can be reduced by modulus-switching. But this noise reduction technique can only work for a limited times, therefore it only results in a leveled FHE scheme.

The further researches about fully homomorphic encryption are main from two aspects. On one aspect, researchers try to optimize the present FHE scheme, to make them adoptable in solving real problem. On the other aspect, since all the present FHE scheme are not so efficient enough, researcher also try to find new primitives and models to design new FHE schemes.

1.2 Contributions

In this dissertation we propose some results from the two aspect described above, and in specific, the major contributions are as follows

- **Uniform Representation of FHE:** We propose a dual-noise model to represent all present FHE schemes. Under this model we show the difference between the perfect fully homomorphic encryption and the generalized fully homomorphic encryption. We conclude that perfect FHE cannot be achieved under dual-noise model. We also deduct a theorem for the existence of generalized FHE.
- **Universal Optimization method for FHE:** We give a universal method to optimize a generalized FHE scheme. Our method is in circuit level. It can be adopted together with an algorithm level optimization. Its correctness is also proved.
- **Noise Free FHE scheme:** As shown via our uniform model of FHE, the perfect FHE cannot be achieved under dual-noise model, so we give a noise free FHE

scheme. Our scheme is based on multivariate polynomial ring, it is an improved version of Groebner-basis Polly Cracker, its security relies on the hardness of ideal membership (IM) decision problem.

- **Privacy-preserving Aggregation Scheme:** Based on LWE somewhat homomorphic encryption scheme and the SIMD technique, we proposed a privacy-preserving aggregation scheme for smart grid communications. With our aggregation scheme, users only need to report their personal data once, then the server center can obtain two kinds of data, i.e. the mean and the variance of the uses' data, and while the individual user's data remains secret.

1.3 Organization

The organization of this dissertation is as follows.

In Chapter 2, we introduce several present FHE schemes, including DGHV-FHE scheme, Gentry's FHE scheme, SV-FHE scheme, LWE-FHE scheme. In specific, we respectively introduce their preliminary knowledge, scheme descriptions and parameter settings.

In Chapter 3, we work on the theory and practice of construction of fully homomorphic encryption using bootstrapping technique. Section 3.1 gives a uniform description of the noised FHE scheme. Section 3.2 describes circuit of plaintext and ciphertext, the concept of homomorphic evaluation and the definition of perfect fully homomorphic encryption. Section 3.3 describes the bootstrapping method, gives the definition of generalized fully homomorphic encryption, and gives the theorem for the existence of generalized fully homomorphic encryption. Section 3.4 gives a universal method for optimizing a generalized fully homomorphic encryption, and gives a proof for the correctness of our optimization scheme. Section 3.7 describes the KDM security and a solution for FHE without KDM security.

In Chapter 4, we work on the theory and practice of construction of fully homomorphic encryption without using bootstrapping. Section 4.1 describes the original Polly Cracker scheme, with its correctness and homomorphic property. Section 4.2 gives some basics about multivariate polynomial ring. Section 4.3 describes the Groebner basis Polly Cracker (GB Polly Cracker) scheme, with its correctness and homomorphic property. Section 4.4 gives a new transformation-carrying version of GB Polly Cracker scheme, with its correctness and homomorphic property. Section 4.5 discusses other result and further work in Polly Cracker family schemes.

In Chapter 5, we proposed a privacy-preserving aggregations scheme for smart grid communications. Section 5.1 introduces the basics and requirements on smart grid. Section 5.2 formalizes the problem to solve. Section 5.3 addresses some preliminaries we need in constructing the aggregation scheme. Section 5.4 gives the formal description of our aggregations scheme and its optimized version. Section 5.5 and section 5.6 respectively gives the security and performance analysis of our aggregation scheme. Section 5.7 addresses some related work in privacy-preserving aggregations scheme for smart grid.

The outlines and the relations between each chapter are shown in the following figure:

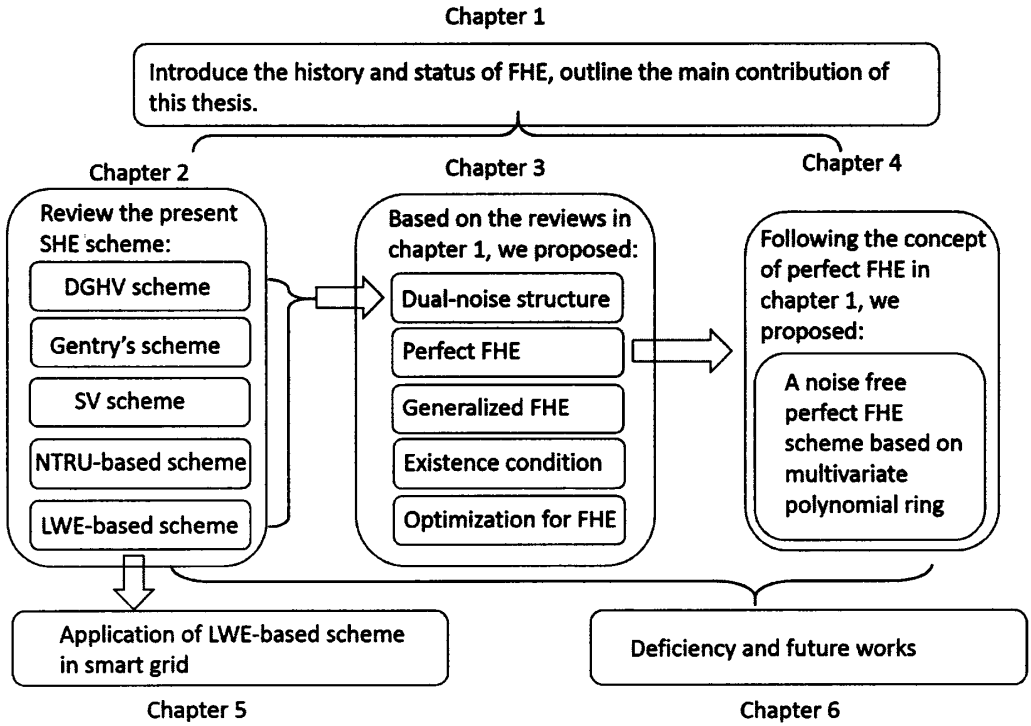


Figure 1.1 Outlines and relations

Chapter 2 Somewhat Homomorphic Encryption Scheme

2.1 Somewhat Homomorphic Encryption Scheme over Integers

The somewhat homomorphic encryption scheme over integers, which is usually referred to as DGHV scheme, was first proposed in [13]. By using the bootstrapping technique introduced by Gentry in [12], DGHV can support homomorphic evaluation for any circuit, i.e., DGHV is finally constructed to be a fully homomorphic encryption scheme.

The security of DGHV somewhat homomorphic encryption scheme can be reduced to the approximate GCD problem, which roughly says: given some integers $x_i = q_i p + r_i$, where q_i, r_i are randomly chosen from a prescribed interval, then to find the p . The approximate GCD problem for the case of given two x_i s was analyzed by Howgrave-Graham [25]. The parameter in DGHV scheme is set to avoid some attacks such as the generalized version attack in [25], and the attack by solving the associated simultaneous Diophantine approximation problem [26], but further attacking to DGHV scheme can be found in [27].

It is commented that DGHV scheme is similar to Regev's first encryption scheme [28], but DGHV scheme does not enjoy the worst-to-average-case security of Regev's first encryption scheme. This is because of the aggressive parameters setting which is chosen in order to support homomorphic property.

We introduce the DGHV somewhat homomorphic scheme as follows (our introduction mainly consults [13]).

2.1.1 Notions

For a real number z , we use the notions $\lfloor z \rfloor$, $\lceil z \rceil$, $\llbracket z \rrbracket$ respectively denotes the rounding down, up or to the nearest integers of z , i.e., respectively denotes the unique integer $z_1 \in [z, z + 1)$, $z_2 \in (z - 1, z]$, $z_3 \in (z - \frac{1}{2}, z + \frac{1}{2}]$ such that $z - z_1, z - z_2, z - z_3 \in [0, 1)$.

For a real number z and a positive integer p , we denote by $Q_p(z)$ and $R_p(z)$ the quotient and remainder of z divided by p , and here the quotient and remainder are respectively defined as $Q_p(z) = \lfloor z/p \rfloor$ and $R_p(z) = z - Q_p(z) \cdot p$, and we also use $z \bmod p$ and $\llbracket z \rrbracket_p$ to refer to the remainder of z divided by p .

In the following γ , η , ρ , τ respectively denotes security parameters who takes value in $\mathbb{N}$.

Given an η -bit odd positive integer p , we denote by $\mathcal{D}_{\gamma,\eta}(p)$ the following distribution over γ -bit integers:

$$\mathcal{D}_{\gamma,\eta}(p) = \{x = pq + r : q \xleftarrow{\$} \mathbb{Z} \cap [0, 2^\gamma/p), r \xleftarrow{\$} \mathbb{Z} \cap (-2^\rho, 2^\rho)\}$$

2.1.2 Scheme Description

DGHV homomorphic encryption scheme is formally defined as 5 PPT algorithms: DGHV.KeyGen, DGHV.Enc, DGHV.Dec, DGHV.Add, DGHV.Mult.

- **DGHV.KeyGen:** Takes as input the security parameter 1^λ , randomly choose an η -bit odd positive integer p , with $p \xleftarrow{\$} 2\mathbb{Z} + 1 \cap [2^{\eta-1}, 2^\eta)$. Sample $x_i \xleftarrow{\$} \mathcal{D}_{\gamma,\eta}(p)$, $i = 0, 1, \dots, \tau$, and relabel these x_i such that x_0 is the largest. Re-sample until x_0 is odd and $r_p(x_0)$ is even. Output $x_0, x_1, \dots, x_\tau$ as the public key, and keep p as the secret key.
- **DGHV.Enc:** Takes as input a plaintext $m \in \{0, 1\}$, randomly choose a subset S of $\{1, 2, \dots, \tau\}$, and randomly choose an integer $r \in (-2^{\rho'}, 2^{\rho'})$. Then the corresponding ciphertext is computed by

$$c = [m + 2r + 2 \sum_{i \in S} x_i]_{x_0}.$$

where ρ' is a secondary noise parameter greater than ρ .

- **DGHV.Dec:** Takes as input a ciphertext c , output the plaintext as

$$m = (c \bmod p) \bmod 2.$$

- **DGHV.Add:** Takes as input two ciphertext c_1, c_2 , output $c_3 = c_1 + c_2$.
- **DGHV.Mult:** Takes as input two ciphertext c_1, c_2 , output $c_3 = c_1 \cdot c_2$.

2.1.3 Correctness and Parameter Setting

Let c be the encryption of $m \in \{0, 1\}$, i.e.,

$$\begin{aligned} c &= [m + 2r + 2 \sum_{i \in S} x_i]_{x_0} \\ &= m + 2r + 2 \sum_{i \in S} x_i + k \cdot x_0 \end{aligned}$$

where $k = Q_{x_0}(m + 2r + 2 \sum_{i \in S} x_i)$. modulo p on both side, we have

$$c \bmod p = (m + 2r + 2 \sum_{i \in S} r_i + k \cdot r_0) \bmod p$$

where r_i is the noise in public key x_i , i.e., $x_i = pq_i + r_i$. We view $\tilde{r} = 2r + 2 \sum_{i \in S} r_i + k \cdot r_0$ as the noise of the ciphertext c , it can be written as $c = m + \tilde{r} \pmod{p}$. Considering $r_0 \bmod 2 = 0$, then $\tilde{r} \bmod 2 = 0$. Since x_0 is the largest one in $x_i, i = 0, 1, \dots, \tau$, it implies $k < \tau$.

We set the parameters η, ρ, ρ', τ satisfying:

- (i) $\rho' \leq 2\rho$,
- (ii) $\log \tau < \rho$,
- (iii) $3 + 2\rho < \eta$.

Then it can guarantee that $(m + \tilde{r}) \in (-\frac{p}{2}, \frac{p}{2})$, and it followed by $c \bmod p = m + \tilde{r}$. Considering $r_0 \bmod 2 = 0$, finally we have

$$(c \bmod p) \bmod 2 = m,$$

which shows that the decryption is correct.

As a homomorphic encryption scheme, it is not only required to correctly decrypt a fresh ciphertext, which means a ciphertext obtain from encrypt a plaintext, but also required to correctly decrypt a homomorphically evaluated ciphertext, i.e., a ciphertext output by a homomorphic circuit whose inputs are fresh ciphertext.

We define the one-layer circuit of ciphertext as shown in figure 2.1, which consist of one U -fan-in addition gate after U 2-fan-in multiplication gates.

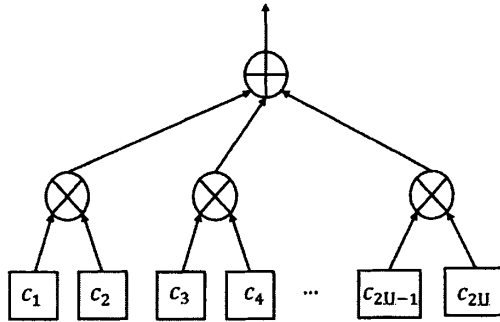


Figure 2.1 One-layer circuit

For any given circuit, we can view it as a multi-layer circuit, with each layer consisting of a bundle of one-layer circuits, and as shown in the follows.

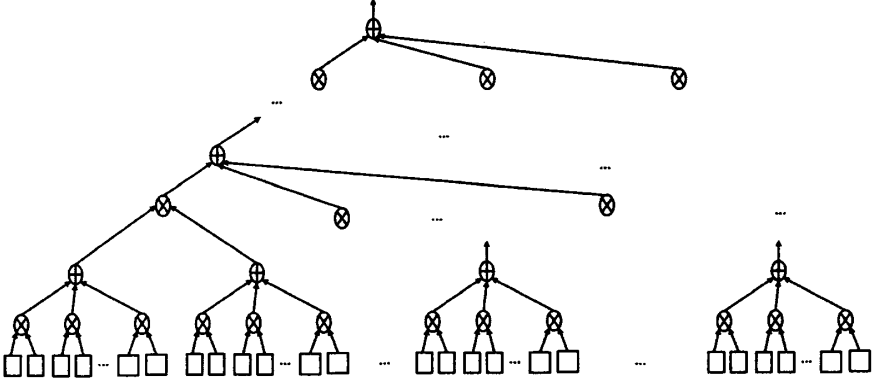


Figure 2.2 Multi-layer circuit

Furthermore, we can also view the one-layer circuit as a symbolic multivariate quadratic function (over $GF(2)$) in form

$$F(x_1, \dots, x_{2U}) = x_1x_2 + \dots + x_{2U-1}x_{2U}$$

Similarly, we can view any circuit as a symbolic multivariate polynomial function combination of the function (over $GF(2)$) in the form

$$\mathcal{F}(\dots) = F(F(F(\dots)), \dots, F(F(\dots)))$$

In fact, to evaluate the one-layer circuit on inputting a set of ciphertexts $c_i, \dots, c_{2U}$ is equivalent to take value of $F(x_1, \dots, x_{2U})$ at $c_i, \dots, c_{2U}$. Similarly, to evaluate any circuit on $c_i, \dots, c_{2U'}$ is equivalent to take value of the corresponding symbolic multivariate polynomial function at $c_i, \dots, c_{2U'}$. (More discussion about circuits and symbolic multivariate polynomial function can be found in Chapter 3.)

Let $c_i \leftarrow \text{DGHV.Enc}(pk, m_i), i = 1, \dots, 2U$, be the inputs of the one-layer circuit. We can write as $c_i = m_i + \tilde{r}_i \pmod{p}$, where $\tilde{r}_i = 2r_i + 2 \sum_{j \in S_i} r_{i,j} + k \cdot r_{i,0}$. Let c' denote the output of the one-layer circuit, by the above analysis we have

$$c' = \sum_{i=1}^U (m_{2i-1}m_{2i} + m_{2i-1}\tilde{r}_{2i} + m_{2i}\tilde{r}_{2i-1} + \tilde{r}_{2i-1}\tilde{r}_{2i}) \pmod{p},$$

Let $d = \max\{\|m_i + \tilde{r}_i\| : i = 1, \dots, N\}$, then

$$\left\| \sum_{i=1}^U (m_{2i-1}m_{2i} + m_{2i-1}\tilde{r}_{2i} + m_{2i}\tilde{r}_{2i-1} + \tilde{r}_{2i-1}\tilde{r}_{2i}) \right\| < Ud^2$$

when $Ud^2 < \frac{p}{2}$, we have

$$c' \bmod p = \sum_{i=1}^U (m_{2i-1}m_{2i} + m_{2i-1}\tilde{r}_{2i} + m_{2i}\tilde{r}_{2i-1} + \tilde{r}_{2i-1}\tilde{r}_{2i}),$$

and considering $\tilde{r}_i = 0 \bmod 2$, it follows that

$$(c' \bmod p) \bmod 2 = \left(\sum_{i=1}^U m_{2i-1}m_{2i} \right) \bmod 2,$$

i.e., the decryption of c' correctly obtain the corresponding circuit evaluation on $m_i, i = 1, \dots, 2U$.

In general, Let a given circuit can be viewed as a l -layer circuit, with some $l \in \mathbb{N}, l \geq 1$. Let $c_i^{(0)}, (i = 1, \dots, (2U)^l)$ be the fresh ciphertexts that input into the first layer circuit, let $c_i^{(1)}$ be the output of the first layer circuit and the input of the second layer circuit, and so on, let $c_i^{(l)}$ be the output of the l -the layer circuit.

Let $c_i^{(0)} = m_i^{(0)} + \tilde{r}_i^{(0)}$, where $\tilde{r}_i^{(0)} = 2r_i^{(0)} + 2 \sum_{j \in S_i} r_{i,j}^{(0)} + k \cdot r_{i,0}^{(0)}$, obviously $\tilde{r}_i^{(0)} \bmod 2 = 0$. Relabel these $c_i^{(0)}$ to $c_{j,i}^{(0)}$ such that $c_{j,i}^{(0)}$ are the inputs of the j -th one-layer circuit of this layer. For clearly, we write $c_{j,i}^{(0)} = m_{j,i}^{(0)} + \tilde{r}_{j,i}^{(0)}$, where $\tilde{r}_{j,i}^{(0)} = 2r_{j,i}^{(0)} + 2 \sum_{t \in S_{j,i}} r_{j,i,t}^{(0)} + k \cdot r_{j,i,0}^{(0)}$.

By the analysis above,

$$c_j^{(1)} = \sum_{i=1}^U m_{j,2i-1}m_{j,2i} + m_{j,2i-1}\tilde{r}_{j,2i} + m_{j,2i}\tilde{r}_{j,2i-1} + \tilde{r}_{j,2i-1}\tilde{r}_{j,2i} \pmod{p}$$

$$\text{let } m_j^{(1)} = \sum_{i=1}^U m_{j,2i-1}m_{j,2i}, \quad \tilde{r}_j^{(1)} = \sum_{i=1}^U m_{j,2i-1}\tilde{r}_{j,2i} + m_{j,2i}\tilde{r}_{j,2i-1} + \tilde{r}_{j,2i-1}\tilde{r}_{j,2i},$$

then

$$c_j^{(1)} = m_j^{(1)} + \tilde{r}_j^{(1)} \pmod{p},$$

with $\tilde{r}_j^{(1)} \bmod 2 = 0$, and $\|c_j^{(1)}\| < N \cdot d^2$.

By recursion, we can finally deduct that

$$c^{(l)} = m^{(l)} + \tilde{r}^{(l)} \pmod{p}$$

with $m^{(l)} = \mathcal{F}(m_1^{(0)}, \dots, m_{(2U)^l}^{(0)})$ and $\tilde{r}^{(l)} \bmod 2 = 0$, and if we define

$$d_j^{(i)} = \|c_j^{(i)}\| = \|m_j^{(i)} + \tilde{r}_j^{(i)}\|$$

and

$$d^{(i)} = \max\{d_j^{(i)}\},$$

where exists the j -th one-layer circuit in the i -th layer. Then we have

$$d^{(l)} < U(d^{(l-1)})^2 < \dots < U^{\frac{2^l-1}{2}} d^{2^l}$$

When $U^{\frac{2^l-1}{2}} d^{2^l} < \frac{p}{2}$,

$$(c^{(l)} \bmod p) \bmod 2 = (m^{(l)} + \tilde{r}^{(l)}) \bmod 2 = m^{(l)} \bmod 2,$$

which means the decryption of the ciphertext $c^{(l)}$ output by the l -layer circuit is correct.

2.1.4 Security and Parameter Setting

For security requirement, the parameters must be set as follows:

- $\rho = \omega(\log \lambda)$, $\rho' = \rho + \omega(\log \lambda)$: To protect against brute-force attacks on the noise ρ .
- $\gamma = \omega(\eta^2 \log \lambda)$: To protect against various lattice-based attacks on the underlying approximate-gcd problem.
- $\tau \geq \gamma + \omega(\log \lambda)$: To use the leftover hash lemma [29] in the reduction of security to approximate-gcd problem.
- $\eta \geq \rho \cdot \Theta(\lambda \log^2 \lambda)$: To support homomorphism for deep enough circuits.

A suggestion of parameter setting in [13] is to set $\rho = \lambda$, $\rho' = 2\lambda$, $\eta = \tilde{O}(\lambda^2)$, $\gamma = \tilde{O}(\lambda^5)$, and $\tau = \gamma + \lambda$, where λ is the security parameter. This setting results in a scheme with public key size $\tilde{O}(\lambda^{10})$, and ciphertext expansion $\tilde{O}(\lambda^5)$. E.g., to achieve a 80-bit security the computation and storage costs of are unbearable. Although the public key can be reduced to $\tilde{O}(\lambda^7)$ [30], it is still hard to use in practice.

2.2 Somwhat Homomorphic Encryption Scheme Based on Ideal Lattice

Homomorphic Encryption scheme Based on Ideal Lattice was first proposed by Gentry[12], and combining with the bootstrapping technique, he gave the first fully homomorphic encryption scheme with IND-CPA security. We estimate Gentry's FHE scheme as follows. (the follow content are based on [12]).

2.2.1 Preliminaries

2.2.1.1 Lattice

Let $\mathbb{R}^m$ be the m -dimensional Euclidean space, and $b_1, b_2, \dots, b_n \in \mathbb{R}^m$ be n linearly independent vectors over $\mathbb{Z}$. The Lattice Λ determined by $b_1, b_2, \dots, b_n$ is

defined as follows:

$$\Lambda = \mathcal{L}(b_1, b_2, \dots, b_n) = \left\{ \sum_{i=1}^n x_i b_i : x_i \in \mathbb{Z} \right\}$$

The integers n and m are called the *rank* and *dimension* of the lattice, respectively. The sequence of vectors $b_1, b_2, \dots, b_n$ is called a *basis* of the lattice. If we view $b_1, b_2, \dots, b_n$ as the column vectors of the matrix B , then the lattice Λ can also be represented by:

$$\Lambda = \mathcal{L}(B) = \{ Bx : x \in \mathbb{Z}^n \}$$

and we also call B is a basis of Λ . In general, the bases of a lattice are not unique, e.g., let $b_1 = (1, 2)^T, b_2 = (1, -1)^T$ and $b'_1 = (2, 1)^T, b'_2 = (5, 1)^T$ then

$$\Lambda = \mathcal{L}(b_1, b_2) = \mathcal{L}(b'_1, b'_2)$$

i.e., b_1, b_2 and b'_1, b'_2 are two bases of the lattice Λ .

Given a lattice $\Lambda = \mathcal{L}(B), B \in \mathbb{R}^{m \times n}$, The *fundamental parallelepiped* $\mathcal{P}(B)$ ($\in \text{span}(B)$) of Λ is defined as the half open parallelepiped

$$\mathcal{P}(B) = \{ Bx : x \in \mathbb{R}^n, 1/2 \leq x_i < 1/2 \}$$

An obvious but important fact is shown as the following property:

Property 2.1 Given a lattice $\Lambda = \mathcal{L}(B)$, no lattice point but zero exists in $\mathcal{P}(B)$, i.e.,

$$\mathcal{L}(B) \cap \mathcal{P}(B) = \{0\}$$

Given a lattice basis $B \in \mathbb{R}^{m \times n}$ and a point $v \in \mathbb{R}^m$, we define $v \bmod B$ as the unique point $v' \in \mathcal{P}(B)$, such that $v - v' \in \mathcal{L}(B)$. By this definition, the following property holds obviously.

Lemma 2.2 Given a lattice basis $B \in \mathbb{R}^{m \times n}$ and a point $v \in \mathbb{R}^m$, then

$$(1) \ v \in \mathcal{P}(B) \iff v \bmod B = v.$$

$$(2) \ v \in \mathcal{L}(B) \iff v \bmod B = 0.$$

The operation of $v \bmod B$ is efficiently computable. When $B \in \mathbb{Z}^{n \times n}$, and $v \in \mathbb{Z}^n$,

we can compute $v \bmod B$ as follows:

$$v - B \lfloor B^{-1}v \rfloor$$

where $B \lfloor B^{-1}v \rfloor$ is known as *Babai's rounding procedure*, $\lfloor (\cdot) \rfloor$ rounds the coefficients of a vector (half up) to the nearest integer.

2.2.1.2 Ideal Lattice

We consider the ring $R = \mathbb{Z}[x]/\langle F(x) \rangle$, where $F(x)$ is a monic n degree polynomial over $\mathbb{Z}$. We can identify an element $v \in R$ with a vector, the i -th component of which is exactly the coefficient of the $(i-1)$ -degree term of v . Let I be an ideal of R , and $I = \langle a_1, a_2, \dots, a_l \rangle$, then the ideal I is identified with the lattice

$$\Lambda = \mathcal{L}(a_i x^j \bmod F(x), 1 \leq i \leq l, 0 \leq j \leq n-1)$$

We call such Λ the ideal lattice. Given two points $v_1, v_2 \in \Lambda$, we can define the addition and multiplication about v_1, v_2 respectively as follows:

$$\text{add}(v_1, v_2) = v_1 + v_2, \quad \text{mult}(v_1, v_2) = v_1 \cdot v_2 \bmod F(x)$$

where “+” is the normal addition of vectors, “ $\cdot$ ” is the normal multiplication of polynomials.

2.2.2 Scheme Description

Gentry's somewhat homomorphic encryption scheme includes the following algorithm: Gentry.KeyGen, Gentry.Enc, Gentry.Dec, Gentry.Add, Gentry.Mult.

- **Gentry.KeyGen:** Take as input a ring R and a basis B_I of an ideal lattice I . The plaintext space P is $R \bmod B_I$. It runs $(B_J^{sk}, B_J^{pk}) \xleftarrow{R} \text{IdealGen}(R, B_I)$. The public key pk includes R, B_I, B_J^{pk} , and secret key sk includes B_J^{sk} .
- **Gentry.Enc:** Take as input a plaintext $\pi \in P$ and public key pk . It samples ψ' from the coset $\pi + I$, i.e., sets $\psi' \leftarrow \text{Sample}(\pi, R, B_I, B_J^{pk})$. output the ciphertext $\psi \leftarrow \psi' \bmod B_J^{pk}$.
- **Gentry.Dec:** Takes as input a ciphertext φ and secret key sk . Outputs the plaintext as

$$\pi \leftarrow (\psi \bmod B_J^{sk}) \bmod B_I$$

- **Gentry.Add:** Takes as input two ciphertexts ψ_1, ψ_2 and public key pk , outputs

$$(\psi_1 + \psi_2) \bmod B_J^{pk}$$

- **Gentry.Mult:** Takes as input two ciphertexts ψ_1, ψ_2 and public key pk , outputs

$$(\psi_1 \times \psi_2) \bmod B_J^{pk}$$

2.2.3 Correctness and Parameter Setting

Given a “fresh” ciphertext ψ , which is obtained by encrypting a plaintext π , then we have $\psi = \pi + L_I + L_J$, where L_I, L_J are respectively the lattice point on $\mathcal{L}(B_I)$ and $\mathcal{L}(B_J^{pk})$. Then we also have

$$\psi \bmod B_J^{sk} = \psi = \pi + L_I + L'_J$$

where L'_J is also a lattice point on $\mathcal{L}(B_J^{pk})$. Therefore $\pi = \pi + L_I + L'_J \bmod B_I$. Since I and J are relatively prime, it implies that $L'_J = 0$, i.e., $\psi' \bmod B_J^{sk} = \psi'$. By lemma (2.2), we conclude that to guarantee the correctness of decryption, it must sample ψ' such that $\psi' \in \mathcal{P}(B_J^{sk})$.

Considering that the shape of $\mathcal{P}(B_J^{sk})$ should not be revealed in encryption algorithm, therefore we define r_{Dec} to be the radius of the sphere inscribed in $\mathcal{P}(B_J^{sk})$. It only need to sample ψ' such that $\|\psi'\|_{l^2} < r_{\text{Dec}}$, then $\psi' \in \mathcal{P}(B_J^{sk})$ holds, and the shape of $\mathcal{P}(B_J^{sk})$ is not be revealed.

Let r_{Samp} denotes the l^2 norm of the vector output by Sample algorithm in encryption. Setting $r_{\text{Samp}} < r_{\text{Dec}}$ can guarantee the correctness of decryption on a “fresh” ciphertext.

While we also need that a not so “fresh” ciphertext, i.e., the ciphertext outputted by some circuit on “fresh” ciphertext, can be correctly decrypted. Before we give the parameter setting to guarantee the correctness of decryption on a not “fresh” ciphertext, we need the following lemma.

Given $v_1, v_2 \in R = \mathbb{Z}[x]/\langle F(x) \rangle$, we consider the l^2 -norm, i.e. the Euclidean length, of $\|v_1 + v_2\|_{l^2}$ and $\|v_1 \times v_2\|_{l^2}$. Using triangle inequality, we have $\|v_1 + v_2\|_{l^2} < \|v_1\|_{l^2} + \|v_2\|_{l^2}$. To bound $\|v_1 \times v_2\|_{l^2}$, we define the following notion:

$$\delta_{l^2}(F(x)) = \sup \left\{ \frac{\|v_1(x) \cdot v_2(x) \bmod F(x)\|_{l^2}}{\|v_1(x)\|_{l^2} \cdot \|v_2(x)\|_{l^2}} : \deg(v_1(x)), \deg(v_2(x)) < \deg(F(x)) \right\}$$

Lemma 2.3 Let $F(x) \in \mathbb{Z}[x]$ and $F(x) = x^n - h(x)$, where $h(x)$ has degree at most $n - (n - 1)/k$ for $k \geq 2$. Then $\delta_{l^2}(F(x)) \leq \sqrt{2n} \cdot (1 + 2n(\sqrt{(k - 1)n}\|F\|_{l^2}^k))$.

We usually set $F(x) = x^n + 1$, and achieve $\delta_{l^2}(F(x)) = \sqrt{n}$.

We consider the ciphertext output by a one-layer circuit. Let $\psi_1, \dots, \psi_{2U}$ be the fresh ciphertexts that input to the one-layer circuit, Let $\psi_i = \text{G.Enc}(pk, \pi_i)$, by above analysis we can write $\psi_i = \pi_i + L_{Ii} + L_{Ji}$, where L_{Ii}, L_{Ji} are respectively the lattice point on $\mathcal{L}(B_I)$ and $\mathcal{L}(B_J^{pk})$, and with $\pi_i + L_{Ii} \in \mathcal{B}(r_{\text{Sample}})$. Here $\mathcal{B}(r_{\text{Sample}})$ denotes a n -dimensional open sphere centered at zero and with radius r_{Sample} . Denoted by $\psi'_i = \pi_i + L_{Ii}$, it is equivalent to $\|\psi'_i\|_{l^2} < r_{\text{Sample}}$.

Let ψ denote the output of the one-layer circuit, by simple calculation we have

$$\begin{aligned} \psi &= \sum_{i=1}^U \psi_{2i-1} \cdot \psi_{2i} \\ &= \sum_{i=1}^U \psi'_{2i-1} \cdot \psi'_{2i} + L'_J \\ &= \sum_{i=1}^U \pi_{2i-1} \cdot \pi_{2i} + L'_I + L'_J. \end{aligned}$$

Denoted by $\psi' = \sum_{i=1}^U \pi_{2i-1} \cdot \pi_{2i} + L'_I$, it satisfies $\|\psi'\|_{l^2} < U \cdot \delta_{l^2} \cdot r_{\text{Sample}}^2$. Then setting $U \cdot \delta_{l^2} \cdot r_{\text{Sample}}^2 < r_{\text{Dec}}$, will result in

$$\begin{aligned} (\psi \bmod B_J^{sk}) \bmod B_I &= ((\sum_{i=1}^U \pi_{2i-1} \cdot \pi_{2i} + L'_I + L'_J) \bmod B_J^{sk}) \bmod B_I \\ &= (\sum_{i=1}^U \pi_{2i-1} \cdot \pi_{2i} + L'_I) \bmod B_I \\ &= \sum_{i=1}^U \pi_{2i-1} \cdots \pi_{2i} \bmod B_I, \end{aligned}$$

i.e. it guarantees the correctness of decryption of ψ .

Then we consider the ciphertext output by a l -layer circuit, with each layer consisting of a bundle of one-layer circuits. Similar to the analysis in Chapter 1, by recursion, we can finally deduct the condition of correctly decrypting a ciphertext output by a l -layer circuit is

$$(U \cdot \delta_{l^2})^{l(l-1)/2} \cdot r_{\text{Sample}}^{2^l} \leq r_{\text{Dec}}$$

2.2.4 Security and Parameter Setting

Gentry's somewhat homomorphic encryption scheme is claimed with IND-CPA security, which is based on the hardness assumption of Ideal Coset Problem (ICP).

Definition 2.4 (Ideal Coset Problem) Let $R = \mathbb{Z}[x]/\langle F(x) \rangle$, I be an ideal of R , and B_I is the ideal lattice corresponding to I . Given algorithms **IdealGen** and **Sample**. The challenger set $(B_J^{sk}, B_J^{pk}) \xleftarrow{R} \text{IdealGen}(R, B_I)$, and send B_J^{pk} to adversary. The adversary choose a plaintext $\pi \neq 0$, and send it to challenger. The challenger

randomly choose $b \xleftarrow{R} \{0, 1\}$, if $b = 0$, it samples ψ' from the closet $\pi + I$ and set $\psi \leftarrow \psi' \bmod B_J^{pk}$, if $b = 1$, it samples ψ uniformly from $R \bmod B_J^{pk}$, and then return ψ to challenger. The problem is given (ψ, B_J^{pk}) guess b .

Base on this hardness assumption, a formal security proof is given in [12], but this assumption does not hold naturally, some parameters must be set specifically to guarantee that the ICP assumption is hard.

To protect against brute attack, r_{Samp} must be set large enough so that Sample can generate $O(2^n)$ samples. A convenient parameter set is $r_{\text{Samp}} = \sqrt{n} \cdot \|B_I\|_{l^\infty}$, where $\|B_I\|_{l^\infty}$ is the maximum l^∞ -norm of the vectors in B_I .

To protect against lattice attack, $r_{\text{Dec}}/r_{\text{Samp}}$ cannot be set too large, e.g., 2^n . Because certainly $\lambda_1(J) < r_{\text{Dec}}$, and using some lattice basis reduction techniques, such as LLL algorithm, one can easily solve the closest vector problem within a factor $2(2/\sqrt{3})^n$, i.e., given a target vector t , one can find a lattice vector v , such that $\|v - t\|_2 \leq 2(2/\sqrt{3})^n \|v' - t\|$, where v' is the vector closest to t . However, it is still unknown that if the closest vector problem can be solved within a sub-exponential factor, so setting $r_{\text{Dec}}/r_{\text{Samp}} = 2^{nc}$ with $c < 1$, is so far believed secure.

Considering Gentry's somewhat homomorphic encryption scheme is additively and multiplicatively homomorphic, so the IND-CCA2 security dose not hold, but if the IND-CCA1 hold is still an open problem so far.

2.3 Somewhat Homomorphic Encryption Scheme based on Algebraic Integers

After the proposition of Gentry's FHE scheme, a new somewhat homomorphic encryption scheme over algebraic integers was proposed in [14], and by using the bootstrapping technique, this somewhat homomorphic encryption scheme can be easily transformed into a FHE scheme. We describe this somewhat homomorphic encryption scheme as follows. (Our description a refers to [14].)

2.3.1 Preliminaries and Notions

2.3.1.1 Ideal in Algebraic Integer Ring

Let $K = \mathbb{Q}(\theta)$ be number field, with θ the root of a monic irreducible polynomial $F(x) \in \mathbb{Z}[x]$ of N degree. Let $\mathcal{O}_K$ be the algebraic integer ring of K , and obviously $\mathbb{Z}[\theta]$ is a subring of $\mathcal{O}_K$. Given an ideal $\mathfrak{a}$ of $\mathbb{Z}[\theta]$, we can view $\mathfrak{a}$ as a N -dimensional $\mathbb{Z}$ -module generated by a $\mathbb{Z}$ -basis $\gamma_1, \gamma_2, \dots, \gamma_N \in \mathbb{Z}[\theta]$. Therefore, to write down a ideal $\mathfrak{a}$, we just need to write down a $\mathbb{Z}$ -basis $\gamma_1, \gamma_2, \dots, \gamma_N$, or if $\gamma_i = \sum_{j=0}^{N-1} \gamma_{i,j} \theta^j$,

then we just need to write down the basis matrix $(\gamma_{i,j})_{N \times N}$. Note that for an ideal $\mathfrak{a}$, its basis matrix is not unique, but these basis matrixes have the same Hermite Normal Form (HNF).

For an ideal $\mathfrak{a}$ of the ring $\mathbb{Z}[\theta]$, there is another way to represent it. There exists $\delta_1, \delta_2 \in \mathbb{Z}[\theta]$, such that $\mathfrak{a}$ can be represented as a two-dimensional $\mathbb{Z}[\theta]$ -module generated by δ_1, δ_2 , and furthermore one of the one element in the basis, say δ_1 , can be chosen as an arbitrary element in $\mathbb{Z}[\theta]$, usually we choose $\delta_1 \in \mathbb{Z}$.

For the ideal $\langle p \rangle = p\mathbb{Z}[\theta]$ of ring $\mathbb{Z}[\theta]$, with p a rational prime, we consider the prime ideal dividing $p\mathbb{Z}[\theta]$. We factor $F(x)$ into irreducible polynomial modulo p as

$$\mathfrak{p}_i = \langle p, F_i(\theta) \rangle$$

then. for p not dividing $[\mathcal{O}_K : \mathbb{Z}[\theta]]$, the prime ideal dividing $p\mathbb{Z}[\theta]$ is given by

$$\mathfrak{p}_i = \langle p, F_i(\theta) \rangle,$$

with p as the norm of $\mathfrak{p}_i$. We define the residue degree of the ideal $\mathfrak{p}_i$ to be the degree d_i of $F_i(x)$. Then reduction modulo $\mathfrak{p}_i$ gives a homomorphism:

$$\iota_{\mathfrak{p}_i} : \mathbb{Z}[\theta] \rightarrow F_{p^{d_i}},$$

i.e., $\forall \gamma \in \mathbb{Z}[\theta]$,

$$\iota_{\mathfrak{p}_i}(\gamma) = \gamma \bmod (p, F_i(\theta)).$$

An ideal $\mathfrak{p}$ with residue degree 1 has a very simple representation $\langle p, \theta - \alpha \rangle$, where p is the norm of $\mathfrak{p}$, and α is a root of $F(x)$ modulo p . For any $\chi(\theta) \in \mathbb{Z}[\theta]$, let $\chi(\theta) = \sum_{i=0}^{N-1} \chi_i \theta^i$, then

$$\iota_{\mathfrak{p}}(\chi(\theta)) = \chi(\alpha) \bmod p.$$

2.3.1.2 Generator of Principle ideal in $\mathbb{Z}[\theta]$

Let $\gamma = G(\theta) \in \mathbb{Z}[\theta]$, such that $\text{resultant}(G(x), F(x)) = p$ with p a rational prime. It is easy to represent the principle ideal $\langle \gamma \rangle = \gamma\mathbb{Z}[\theta]$ as a $\mathbb{Z}[\theta]$ -module generated by two elements in $\mathbb{Z}[\theta]$. Compute the greatest common factor of $G(x), F(x)$ over $\mathbb{F}_p$, denoted by $D(x) = \text{GCD}(G(x), F(x)) \in \mathbb{F}_p[x]$. Since $\text{resultant}(G(x), F(x)) = p$, $D(x)$ at least has a root in $\mathbb{F}_p$, denoted by α , which is also the unique common root of $G(x), F(x)$ in $\mathbb{F}_p$. Let $\mathfrak{p} = \langle p, \theta - \alpha \rangle$, by the following two points, we know that $\langle \gamma \rangle = \mathfrak{p}$.

- (i) $G(\alpha) \equiv 0 \pmod{p}$, i.e., $(\theta - \alpha) | G(\theta) \pmod{p}$, we can write $G(\theta) = F_1(\theta) \cdot p + F_2(\theta) \cdot (\theta - \alpha)$, it implies that $\langle G(\theta) \rangle \subset \langle p, \theta - \alpha \rangle$.
- (ii) Since $\text{resultant}(G(x), F(x)) = p$, there exists $S(x), T(x) \in \mathbb{Z}[x]$ such that $S(x)G(x) + T(x)F(x) = p$, then $S(\theta)G(\theta) = p \pmod{F(\theta)}$, it follows that $G(\theta) | p$. By (i), and notice that $\mathbb{F}_p(\theta)$ is a field, we can deduct $\langle p, \theta - \alpha \rangle \subset \langle G(\theta) \rangle$.

As shown above, given a generator $\lambda \in \mathbb{Z}[\theta]$ of a principle ideal $\mathfrak{p}$, it is easily to construct a 2-dimensional $\mathbb{Z}[\theta]$ -basis $p, \theta - \alpha$, and furthermore we can write down the HNF of the $\mathbb{Z}$ -basis matrix. But given the HNF or the 2-dimensional $\mathbb{Z}[\theta]$ -basis, it is believed hard to find out the generator λ , or to determine whether it is a principle ideal for given N . The best known algorithms to find the generator (which are also used to find the class and unite group of a number field) runs in exponential time in the degree of the field. For fix degree best known algorithm runs in sub-exponential time in the discriminant[31]. Considering the large size of the generator output by known algorithms, even writing it down as a polynomial in θ may take exponential time[32]. Although the quantum algorithms[33] for finding the generator of a principle ideal are relatively efficient, finding and writing down a small generator is still believed a hard problem.

2.3.1.3 Notions

For a polynomial $g(x) = \sum_{i=0}^t g_i x^i \in \mathbb{Q}[x]$, we define its l_2 -norm and l_∞ -norm respectively as:

$$\|g(x)\|_2 = \sqrt{\sum_{i=0}^t g_i^2}, \quad \|g(x)\|_\infty = \max_{i=0, \dots, t} |g_i|.$$

Given an integer r , we define two set as follows:

$$\mathcal{B}_{2,N}(r) = \{A(x) = \sum_{i=0}^{N-1} a_i x^i : \|A(x)\|_2 \leq r\},$$

$$\mathcal{B}_{\infty,N}(r) = \{A(x) = \sum_{i=0}^{N-1} a_i x^i : \|A(x)\|_\infty \leq r\}$$

For a rational number q and a positive integer m , we define $q \bmod m$ as the unique number in $[-m/2, m/2)$ such that $m | (q - q \bmod m)$.

2.3.2 Scheme Description

The SV homomorphic encryption scheme is formally defined as 5 PPT algorithms: **SV.KeyGen**, **SV.Enc**, **SV.Dec**, **SV.Add**, **SV.Mult**. This scheme is parameterized by three values (N, η, μ) which are associated with the security parameter λ , and a typical setting is $(N, 2^{\sqrt{N}}, \sqrt{N})$.

- **SV.KeyGen**: Takes as input the security parameter λ , and runs in the following steps,

- (1) Choose a monic irreducible polynomial $F(x) \in \mathbb{Z}[x]$ of N degree.
- (2) Sample $S(x) \xleftarrow{R} \mathcal{B}_{\infty, N}(\eta/2)$, set $G(x) = 1 + 2S(x)$, and let $p \leftarrow \text{resultant}(G(x), F(x))$. Re-sample unless p is a rational prime.
- (3) Compute $D(x) \leftarrow \text{GCD}(G(x), F(x))$ over $F_p[x]$.
- (4) Find the unique root of $D(x)$ in F_p , denote it by $\alpha \in F_p$.
- (5) Run XGCD algorithm (Extended Euclidean Algorithm) over $\mathbb{Q}[x]$ to find $Z(x) = \sum_{i=0}^{N-1} z_i x^i \in \mathbb{Z}[x]$, such that
$$Z(x)G(x) = p \pmod{F(x)}.$$

- (6) Set $B = z_0 \bmod 2p$.

Output (p, α) as public key, and B as the secret key.

- **SV.Enc**: Take as input the public key (p, B) and ciphertext $M \in \{0, 1\}$, execute the following steps,

- (1) Choose randomly $R(X) \xleftarrow{R} \mathcal{B}_{\infty, N}(\mu/2)$.
- (2) Set $C(x) \leftarrow M + 2R(x)$.
- (3) Evaluate $C(x)$ at α , and set $c = C(\alpha) \bmod p$.

Output c as the ciphertext.

- **SV.Dec**: Take as input secret key B and ciphertext c , Outputs the plaintext as

$$M = (c - \lfloor c \cdot B/p \rfloor) \bmod 2.$$

- **SV.Add**: Take as input public key (p, α) and two ciphertexts c_1, c_2 , Output $c_3 = (c_1 + c_2) \bmod p$.

- **SV.Mult:** Take as input public key (p, α) and two ciphertexts c_1, c_2 , Output $c_4 = (c_1 c_2) \bmod p$.

2.3.3 Correctness and Parameter Setting

Let c be fresh ciphertext by encrypting M , i.e.,

$$c = C(\alpha) \bmod p = C(\theta) \bmod \langle p, \theta - \alpha \rangle$$

Let $\mathfrak{p} = \langle p, \theta - \alpha \rangle$, we have $C(\theta) - c \in \mathfrak{p}$. Since $\mathfrak{p}$ is generated by $\gamma = G(\theta)$, there exists $Q(\theta) \in \mathbb{Z}[\theta]$ such that

$$C(\theta) - c = Q(\theta) \cdot \gamma \quad (2-1)$$

Notice that γ^{-1} is given by $Z[\theta]/p$, where $Z[\theta]$ is computed in **SV.KeyGen** via XGCD algorithm. Multiply $Z[\theta]/p$ on both sides of equation (2-1) and rearrange, we have

$$-c \cdot Z(\theta)/p = Q(\theta) + C(\theta) \cdot Z(\theta)/p$$

If we can guarantee $\|C(\theta) \cdot Z(\theta)/p\|_\infty < 1/2$, then rounding $-c \cdot Z(\theta)/p$ to the nearest integer will obtain $\lfloor -c \cdot Z(\theta)/p \rfloor = Q(\theta)$, this results in

$$C(\theta) = c - \lfloor c \cdot Z(\theta)/p \rfloor \cdot \gamma.$$

Since $M \equiv C(\theta) \pmod{2}$ and $\gamma \equiv 1 \pmod{2}$,

$$M = (c - \lfloor c \cdot Z(\theta)/p \rfloor) \bmod 2$$

Notice that the coefficients of the terms of $C(\theta)$, with degree higher than 1 are all even, so

$$M = (c - \lfloor c \cdot z_0/p \rfloor) \bmod 2 = (c - \lfloor c \cdot B/p \rfloor) \bmod 2$$

By the above analysis, the condition of correctly decryption is $\|C(\theta) \cdot Z(\theta)/p\|_\infty < 1/2$. In the following we will show what a parameter setting can guarantee this condition hold.

To evaluate $\|Z(\theta)\|_\infty$, we need a lemma as follows:

Lemma 2.5 Let $F(x), G(x) \in \mathbb{Z}[x]$, with $F(x)$ a monic polynomial of N degree, and $G(x)$ of degree M , $M < N$. If the resultant of $F(x)$ and $G(x)$, denoted by $\text{resultant}(F(x), G(x))$, is equal to p , then there exists $Z(x) \in \mathbb{Z}[x]$ such that

$Z(x)G(x) = p \pmod{F(x)}$, and

$$\|Z(x)\|_\infty \leq \|G(x)\|_2^{N-1} \cdot \|F(x)\|_2^M$$

Proof: Since $\text{resultant}(F(x), G(x)) = p \neq 0$, $F(x)$ and $G(x)$ have no common factor in $\mathbb{Q}[x]$ with degree higher than 1. It implies that $\text{GCD}(F(x), G(x)) = 1$, and by XGCD algorithm there exists $S(x), T(x) \in \mathbb{Q}[x]$, with $\deg(S(x)) = N - 1$ and $\deg(T(x)) = M - 1$, such that $S(x)G(x) + T(x)F(x) = 1$. Furthermore let $S(x) = \sum_0^{N-1} s_i x^i$ and $T(x) = \sum_0^{M-1} t_i x^i$, these s_i, t_i satisfy

$$\text{Syl}(G(x), F(x))^T \begin{pmatrix} s_{N-1} \\ \vdots \\ s_0 \\ t_{M-1} \\ \vdots \\ t_0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ \vdots \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

where $\text{Syl}(G(x), F(x))$ is the Sylvester matrix of $G(x), F(x)$. By Gramer's rule, and notice $\text{resultant}(F(x), G(x)) = \det(\text{Syl}(G(x), F(x)))$, we have the explicit expression for the coefficient of $S(x)$ as

$$s_{N-i} = \det(\text{Syl}^{(i)}(G(x), F(x))) / \text{resultant}(F(x), G(x))$$

where $\text{Syl}^{(i)}(G(x), F(x))^T$ is the matrix by replacing the i -th column of $\text{Syl}(G(x), F(x))^T$ by $(0, \dots, 1)^T$.

Let $Z(x) = pS(x)$, then $Z(x) \in \mathbb{Z}[x]$ and satisfies $Z(x)G(x) = p \pmod{F(x)}$. Bounding the determinant of $\text{Syl}^{(i)}(G(x), F(x))^T$ via Hadamard's inequality, we finally conclude that.

$$\|Z(x)\|_\infty \leq \|G(x)\|_2^{N-1} \cdot \|F(x)\|_2^M.$$

□

For simplicity, we can set $M = N - 1$, and this setting can be achieved in SV scheme by a possibility very close to 1. Before giving bound on $\|C(\theta) \cdot Z(\theta)/p\|_\infty$, we need a new definition as follows:

$$\delta_\infty = \sup \left\{ \frac{\|G(x) \cdot H(x) \bmod F(x)\|_\infty}{\|G(x)\|_\infty \cdot \|H\|_\infty} : \deg G, \deg H < N \right\}$$

The value of δ_∞ is associated with $F(x)$ closely, which is shown in the following lemma:

Lemma 2.6 Let $F_1(x) = x^N - a$, $F_2(x) = x^N - ax^{N-1}$, with $|a| \geq 1$, then

$$\delta_\infty(F_1(x)) \leq |a|N, \quad \delta_\infty(F_2(x)) \leq |a|^{N-1}N$$

Proof: Let $G(x) = \sum_{i=0}^{N-1} g_i x^i$, $H(x) = \sum_{i=0}^{N-1} h_i x^i$, for $i \in [0, N-1] \cap \mathbb{Z}$ set $c_i = \sum_{j=0}^i g_j h_{i-j}$, $c_{N-1+i} = \sum_{j=0}^{N-1} g_j h_{N-1+i-j}$, then we have

$$G(x) \cdot H(x) = \sum_{i=0}^{2N-2} c_i x^i,$$

and

$$G(x) \cdot H(x) \bmod F_1(x) = \sum_{i=0}^{N-1} (c_i + a \cdot c_{N+i}) x^i$$

it follows that $\delta_\infty(F_1(x)) \leq |a|N$.

similarly,

$$G(x) \cdot H(x) \bmod F_2(x) = \sum_{i=0}^{N-2} c_i x^i + \left(\sum_{i=1}^{N-1} a^i \cdot c_{N-1+i} \right) x^{N-1}$$

it follows that $\delta_\infty(F_2(x)) \leq |a|^{N-1}N$. $\square$

Via the definition of δ_∞ and lemma (2.5), the bound of $\|C(\theta) \cdot Z(\theta)/p\|_\infty$ is given by

$$\|C(\theta) \cdot Z(\theta)/p\|_\infty \leq 1/p \cdot \delta_\infty(F(x)) \cdot \|C(x)\|_\infty \cdot \|G(x)\|_2^{N-1} \cdot \|F(x)\|_2^{N-1}$$

Combined with the correctly-decryption condition $\|C(\theta) \cdot Z(\theta)/p\|_\infty < 1/2$, we conclude the correctly-decryption radius as

$$r_{\text{Dec}} = \frac{p}{2 \cdot \delta_\infty(F(x)) \cdot \|G(x)\|_2^{N-1} \cdot \|F(x)\|_2^{N-1}}$$

i.e., when $\|C(x)\|_\infty < r_{\text{Dec}}$, the ciphertext $C(\alpha)$ can be correctly decrypted.

Furthermore, since $p = \text{resultant}(G(x), F(x)) \approx \|G(x)\|_2^N \cdot \|F(x)\|_2^{N-1}$, we have $r_{\text{Dec}} \approx \|G(x)\|_2 / (2\delta_\infty(F(x)))$. By choosing of $G(x)$ in **SV.KeyGen**, we have $\|G(x)\|_2 \approx \sqrt{N} \cdot \eta$. When we choose $F(x) = x^N + 1$, we have $r_{\text{Dec}} \approx \eta / (2\sqrt{N})$. By the choosing of $C(x)$ in **Encrypt**, it implies $\|C(x)\|_\infty \leq \mu + 1$. Therefore it is only need to set the parameters such that $\mu + 1 \leq \eta / (2\sqrt{N})$, the ciphertext, obtained by encryption algorithm, will be correctly decrypted.

The parameters setting should also guarantee the correctness of decryption on a ciphertext output by a evaluation circuit. We give the details as follows

Let c_1, c_2 be ciphertext (not limited to fresh ciphertexts) of $M_1, M_2 \in \{0, 1\}$, i.e., there exists $C_1(x), C_2(x) \in \mathbb{Z}[x]$, with $\|C(x)\|_{\infty} r_{\text{Dec}}$, such that $c_i = C_i(\alpha) \bmod p, i = 1, 2$ and $M_i = C_i(\theta) \bmod 2$. Let $c_3 = (c_1 + c_2) \bmod p$, then there exists $C_3 = C_1 + C_2$ such that $c_3 = C_3(\alpha) \bmod p$ and $M_3 = (M_1 + M_2) \bmod 2 = C_3(\theta) \bmod 2$. Let $c_4 = (c_1 \cdot c_2) \bmod p$, then there exists $C_4 = (C_1 \cdot C_2) \bmod F(x)$, such that $c_4 = C_4(\alpha) \bmod p$ and $M_4 = (M_1 \cdot M_2) \bmod 2 = C_4(\theta) \bmod 2$. suppose $C_i(x) \in \mathcal{B}_{\infty, N}(r_i), i = 1, 2$, then $C_3(x) \in \mathcal{B}_{\infty, N}(r_3 = r_1 + r_2)$ and $C_4(x) \in \mathcal{B}_{\infty, N}(r_4 = N(r_1 \cdot r_2))$. By the condition of correctly decryption, when $r_1 + r_2 < r_{\text{Dec}}$, the decryption of c_3 will correctly obtain M_3 , and similarly, when $N(r_1 \cdot r_2) < r_{\text{Dec}}$, the decryption of c_4 will correctly obtain M_4 .

We consider the ciphertext output by a one-layer circuit, with input $c_1, c_2, \dots, c_{2U}$, and output c' . Let $c_1, c_2, \dots, c_{2U}$ be the encryption of $M_1, M_2, \dots, M_{2U}$, then there exists $C_1(x), C_2(x), \dots, C_{2U} \in \mathcal{B}_{\infty, N}(r)$, such that $c_i = C_i(\alpha) \bmod p$ and $M_i = C_i(\theta) \bmod 2$. By the definition of one-layer circuit, we have

$$c' = \sum_{i=1}^U c_{2i-1} c_{2i},$$

and we set

$$C'(x) = \sum_{i=1}^n C_{2i-1}(x) C_{2i}(x),$$

it satisfies $c' = C'(\alpha) \bmod p$ and $M' = C'(\theta) \bmod 2$, where $M' = \sum_{i=1}^U M_{2i-1} M_{2i}$. By a simple calculation, we have $C'(x) \in \mathcal{B}_{\infty, N}(UNr^2)$. Therefore by setting $UNr^2 < \eta/2\sqrt{N}$, we can guarantee c' will be correctly decrypted to M' .

Let c'' be the ciphertext output by a l -layer circuit, with input a bundle of fresh ciphertext. By recursion, we can finally deduct the condition of correctly decrypting of c'' is

$$(UN)^{(1+l)l/2}(\mu + 1)^{2^l} \leq \eta/2\sqrt{N}.$$

2.4 Somewhat Homomorphic Encryption Scheme Based on LWE

Learning with error (LWE) problem which can be viewed as an extension of the learning parity with noise (LPN), was first introduced by Regev [34], and Regev gave a quantum reduction from the worst-case of GapSVP and SIVP to LWE, i.e., an efficient solution to the learning problem implies a quantum algorithm for GAPSVP and SIVP.

Later a classical (non-quantum) reduction is given by peikert [35]. These works makes sense of constructing encryption scheme based on LWE problem.

The first encryption scheme based on LWE is given in [34], and a more efficient scheme is proposed in [36], which is based on Ring-LWE problem. Based on LWE and ring-LWE, Brakerski et al proposed two fully homomorphic encryption scheme (FHE) in [15] and [37], and the later is recognised as the most efficient FHE so far. Following in this chapter, we will introduce the basics of homomorphic encryption scheme based on LWE (Ring-LWE). The most results in this chapter due to Regev and Brakersk.

2.4.1 Preliminaries

Definition 2.7 (LWE Problem) Let integers $n, q \geq 2$, given a vector $\mathbf{s} \in \mathbb{Z}_q^n$ and an ‘error’ distribution χ on $\mathbb{Z}_q$, we construct a LWE distribution (with respect to $\mathbf{s}$) as

$$A_{n,p,\chi}(\mathbf{s}) = \{(\mathbf{a}, \langle \mathbf{a}, \mathbf{s} \rangle + e)\}$$

where $\mathbf{a}$ is randomly chosen from the uniform distribution over $\mathbb{Z}_q^n$, e is chosen from χ , and $\langle \mathbf{a}, \mathbf{s} \rangle$ is the inner product of $\mathbf{a}, \mathbf{s}$. The learning with error problem with parameters n, q, χ denoted by $\text{LWE}_{n,q,\chi,m}$ is defined as: given m independent samples from $A_{n,q,\chi}(\mathbf{s})$, output $\mathbf{s}$. The decision variant of $\text{LWE}_{n,q,\chi,m}$ is defined as: given m independent samples from $\mathbb{Z}_q^n \times \mathbb{Z}_q$, to decide whether these samples are from $A_{n,q,\chi}(\mathbf{s})$ or from the uniform distribution over $\mathbb{Z}_q^n \times \mathbb{Z}_q$.

There exist quantum reduction [34] and classic reduction[35] from the worst-case approximating shortest vector problem on lattice, also known as GapSVP, to Decision- $\text{LWE}_{n,q,\chi,m}$. These reductions take χ to be discrete Gaussian distribution, which is B -bounded for an appropriate B . So far, the best known algorithms for solving Decision- $\text{LWE}_{n,q,\chi,m}$ run in nearly exponential time in the dimension n [38],[39], and for the noise magnitude being $1/2^k$ fraction of the modulus q , the best algorithm runs in time $2^{\tilde{O}(n/k)}$.

Ring version LWE problem, or RLWE in short, was first proposed in[36], later in[37] a fully homomorphic encryption scheme was proposed based on a special case of RLWE problem, and we describe this special case of RLWE as follows.

Let $R = \mathbb{Z}[x]/\langle F(x) \rangle$, where $F(x) \in \mathbb{Z}[x]$ is a monic polynomial with degree n . Let $q \in \mathbb{Z}$ be a prime, and $R_q = R/qR$. Noted that R_q is isomorphic to $\mathbb{Z}_q[x]/\langle F(x) \rangle$, each element in R_q can be written as a $n - 1$ degree polynomial in $\mathbb{Z}_q[x]$, the addition is done component-wise in coefficients modulo q , and the multiplication is the polynomials

multiplication modulo $F(x)$ and modulo q in coefficients. We can also view an element in R_q as an n -dimension vector in $\mathbb{Z}_q$, in which the i -th component is exactly the coefficient of $(i - 1)$ -degree.

Definition 2.8 (RLWE Problem) Let integers $n, q \geq 2$, given $F(x) \in \mathbb{Z}[x]$, and an ‘error’ distribution χ over R . Sample $s \xleftarrow{\$} R_q$, we construct a LWE distribution (with respect to s) as

$$A_{n,p,F(x),\chi}(s) = \{(a, a \cdot s + e)\}$$

where a is randomly chosen from the uniform distribution over R_q , e is chosen from χ , and $a \cdot s$ is computed in R_q . The learning with error problem with parameters $n, q, F(x), \chi$ over ring, denoted by $\text{LWE}_{n,q,F(x),\chi,m}$, is defined as: given m independent samples from $A_{n,q,F(x),\chi}(s)$, output s . The decision variant of $\text{LWE}_{n,q,F(x),\chi,m}$ is defined as: given m independent samples from $R_q \times R_q$, to decide whether these samples are from $A_{n,q,F(x),\chi}(s)$ or from the uniform distribution over $R_q \times R_q$.

2.4.2 Scheme Description

The first encryption scheme based on LWE is given in [34], by properly setting parameters, this scheme is somewhat homomorphic, We describe the original Regev’s scheme and a variant as follows.

The original Regev’s LWE-based encryption scheme is formally defined by 3 algorithms: Regev.KeyGen , Regev.Enc , Regev.Dec .

- $\text{Regev.KeyGen}(1^n)$: takes as input the security parameter n . Samples $s \xleftarrow{\$} \mathbb{Z}_q^n$. For $i = 1, \dots, m$, Samples $\mathbf{a}_i \xleftarrow{\$} \mathbb{Z}_q^n$, $e_i \xleftarrow{\$} \chi$, then computes $b_i = \langle \mathbf{a}_i, s \rangle + e_i \bmod q$. Let $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_m]^T$, $\mathbf{b} = [b_1, \dots, b_m]^T$, and $\mathbf{e} = [e_1, \dots, e_m]^T$, then it also can be written as $\mathbf{b} = [\mathbf{A} \cdot \mathbf{s} + \mathbf{e}]_q$. Output $sk = s$, $pk = \mathbf{A}, \mathbf{b}$.
- $\text{Regev.Enc}(pk, m)$: takes as input the public key pk and plaintext $m \in \{0, 1\}$. Samples $\mathbf{r} \xleftarrow{\$} \{0, 1\}^n$. Computes

$$\mathbf{v} = \mathbf{A}^T \cdot \mathbf{r} \quad \text{and} \quad w = \mathbf{b}^T \cdot \mathbf{r} + \left\lfloor \frac{q}{2} \right\rfloor \cdot m$$

Output $\mathbf{c} = (\mathbf{v}, w)$ as the ciphertext.

- $\text{Regev.Dec}(sk, \mathbf{c})$: takes as input the secret key sk and ciphertext $\mathbf{c} = (\mathbf{v}, w)$. Computes

$$m = \left\lfloor \left\lfloor \frac{2 \cdot [w - \langle \mathbf{v}, s \rangle]_q}{q} \right\rfloor \right\rfloor_2$$

The variant Regev's LWE-based encryption scheme is formally defined by 3 algorithms: $\text{Regev}^*. \text{KeyGen}$, $\text{Regev}^*. \text{Enc}$, $\text{Regev}^*. \text{Dec}$.

- $\text{Regev}^*. \text{KeyGen}(1^n)$: takes as input the security parameter n . Samples $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_q^n$. For $i = 1, \dots, m$, Samples $\mathbf{a}_i \xleftarrow{\$} \mathbb{Z}_q^n$, $e_i \xleftarrow{\$} \chi$, then computes $b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + 2e_i \bmod q$. Let $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_m]^T$, $\mathbf{b} = [b_1, \dots, b_m]^T$, and $\mathbf{e} = [e_1, \dots, e_m]^T$, then it also can be written as $\mathbf{b} = [\mathbf{A} \cdot \mathbf{s} + 2\mathbf{e}]_q$. Output $sk = \mathbf{s}$, $pk = \mathbf{A}, \mathbf{b}$.
- $\text{Regev}^*. \text{Enc}(pk, m)$: takes as input the public key pk and plaintext $m \in \{0, 1\}$. Computes

$$\mathbf{v} = \mathbf{A}^T \cdot \mathbf{r} \quad \text{and} \quad w = \mathbf{b}^T \cdot \mathbf{r} + m$$

Output $\mathbf{c} = (\mathbf{v}, w)$ as the ciphertext.

- $\text{Regev}^*. \text{Dec}(sk, \mathbf{c})$: takes as input the secret key sk and ciphertext $\mathbf{c} = (\mathbf{v}, w)$. Computes

$$\mathbf{m} = \llbracket [w - \langle \mathbf{v}, \mathbf{s} \rangle]_q \rrbracket_2$$

Both the original and variant Regev's scheme can be used to construct FHE, as did in [15] and [40], here we mainly use the variant Regev's scheme to show the correctness can be guarantee, and how the homomorphic properties can be achieved.

Compared with the variant Regev's scheme, the homomorphic encryption scheme based on variant Regev's scheme has two more algorithms: $\text{Regev}^*. \text{Add}$, $\text{Regev}^*. \text{Mult}$. The homomorphic addition algorithm is simple given as follows:

- $\text{Regev}^*. \text{Add}(pk, \mathbf{c}, \mathbf{c}')$: takes as input the secret key pk and ciphertext $\mathbf{c} = (\mathbf{v}, w)$, $\mathbf{c}' = (\mathbf{v}', w')$. Computes

$$\mathbf{c}_{\text{add}} = \llbracket (\mathbf{v} + \mathbf{v}', w + w') \rrbracket_q$$

The homomorphic multiplication algorithm is a little complicated, it need include extra key. Except the secret key $\mathbf{s}$ in above, it samples $\mathbf{s}_1, \dots, \mathbf{s}_L \xleftarrow{\$} \mathbb{Z}_q^n$, and let $\mathbf{s}_0 = \mathbf{s}$. For all $l \in [L]$, $0 \leq i, j \leq n$, and $\tau \in \{0, \dots, \lceil \log q \rceil\}$, it samples

$$\mathbf{a}_{l,i,j,\tau} \xleftarrow{\$} \mathbb{Z}_q^n \quad \text{and} \quad e_{l,i,j,\tau} \xleftarrow{\$} \chi$$

and computes

$$b_{l,i,j,\tau} = \langle \mathbf{a}_{l,i,j,\tau}, \mathbf{s}_l \rangle + 2e_{l,i,j,\tau} + 2^\tau \cdot \mathbf{s}_{l-1}[i] \cdot \mathbf{s}_{l-1}[j]$$

where $\mathbf{s}_{l-1}[i]$ is the i -th component of $\mathbf{s}_{l-1}$, and we define $\mathbf{s}_{l-1}[0] = 1$.

Given two ciphertexts $\mathbf{c} = (\mathbf{v}, w)$, $\mathbf{c}' = (\mathbf{v}', w')$, let $\mathbf{x} = (\mathbf{x}[1], \dots, \mathbf{x}[n])$ be an unknown vector, we define an n -variant symbol polynomial over $\mathbf{x}$ as:

$$\phi(\mathbf{x}) = (w - \langle \mathbf{v}, \mathbf{x} \rangle)(w' - \langle \mathbf{v}', \mathbf{x} \rangle)$$

open the parenthesis, and define $x[0] = 1$, it obtains that

$$\phi(\mathbf{x}) = \sum_{0 \leq i \leq j \leq n} h_{i,j} \cdot \mathbf{x}[i] \cdot \mathbf{x}[j]$$

Further more, let $h_{i,j,\tau}$ be the τ -th bit in its binary representation of $h_{i,j}$, i.e., $h_{i,j,\tau} \in \{0, 1\}$ and satisfy

$$h_{i,j} = \sum_{\tau \in \{0, \dots, \lceil \log q \rceil\}} h_{i,j,\tau} \cdot 2^\tau$$

Then homomorphic multiplication algorithm is given as follows:

- **Regev*.Mult($pk, \mathbf{c}, \mathbf{c}'$):** takes as input the secret key pk and ciphertext $\mathbf{c} = (\mathbf{v}, w)$, $\mathbf{c}' = (\mathbf{v}', w')$. Suppose that $\mathbf{c}, \mathbf{c}'$ can be decrypted by secret key $\mathbf{s}_{l-1}$, then Computes

$$\begin{aligned} \mathbf{v}_{\text{mult}} &= \sum_{\substack{0 \leq i \leq j \leq n \\ \tau \in \{0, \dots, \lceil \log q \rceil\}}} h_{i,j,\tau} \cdot \mathbf{a}_{l+1,i,j,\tau} \\ \mathbf{w}_{\text{mult}} &= \sum_{\substack{0 \leq i \leq j \leq n \\ \tau \in \{0, \dots, \lceil \log q \rceil\}}} h_{i,j,\tau} \cdot b_{l+1,i,j,\tau} \end{aligned}$$

It is easy to check that $w_{\text{mult}} - \langle \mathbf{v}_{\text{mult}}, \mathbf{s}_l \rangle = (w - \langle \mathbf{v}, \mathbf{s}_{l-1} \rangle) \cdot (w' - \langle \mathbf{v}', \mathbf{s}_{l-1} \rangle)$, i.e., the decryption on the product of $(\mathbf{v}, w)$ and $(\mathbf{v}', w')$ equals to the product of the decryptions respectively on $(\mathbf{v}, w)$ and $(\mathbf{v}', w')$. It is also noticed that the secret key to decrypt $(\mathbf{v}, w)$ and $(\mathbf{v}', w')$ is $\mathbf{s}_{l-1}$, while the secret key to decrypt $(\mathbf{v}_{\text{mult}}, \mathbf{w}_{\text{mult}})$ is $\mathbf{s}_l$. Therefore, it is need to indicate the secret key by which a ciphertext can be decrypted. This can be done by attaching a label to a ciphertext, e.g., $c = (w, \mathbf{v}, i)$, which means $(w, \mathbf{v})$ can be decrypted using $\mathbf{s}_i$.

From above analysis, a homomorphic encryption scheme based on LWE can be formally represented as five algorithms as follows:

- **SH.KeyGen(1^n):** takes as input the security parameter n . Samples $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_q^n$. For $i = 1, \dots, m$, Samples $\mathbf{a}_i \xleftarrow{\$} \mathbb{Z}_q^n$, $e_i \xleftarrow{\$} \chi$, then computes $b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + 2e_i \bmod q$.

Let $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_m]^T$, $\mathbf{b} = [b_1, \dots, b_m]^T$, and $\mathbf{e} = [e_1, \dots, e_m]^T$, then it also can be written as $\mathbf{b} = [\mathbf{A} \cdot \mathbf{s} + 2\mathbf{e}]_q$.

Samples $\mathbf{s}_1, \dots, \mathbf{s}_L \xleftarrow{\$} \mathbb{Z}_q^n$. For all $l \in [L]$, $0 \leq i, j \leq n$, and $\tau \in \{0, \dots, \lfloor \log q \rfloor\}$, it samples

$$\mathbf{a}_{l,i,j,\tau} \xleftarrow{\$} \mathbb{Z}_q^n \text{ and } e_{l,i,j,\tau} \xleftarrow{\$} \chi$$

and computes

$$b_{l,i,j,\tau} = \langle \mathbf{a}_{l,i,j,\tau}, \mathbf{s}_l \rangle + 2e_{l,i,j,\tau} + 2^\tau \cdot \mathbf{s}_{l-1}[i] \cdot \mathbf{s}_{l-1}[j] \bmod q$$

where $\mathbf{s}_{l-1}[i]$ is the i -th component of $\mathbf{s}_{l-1}$, and we define $\mathbf{s}_{l-1}[0] = 1$. Let $\psi_l = \{\mathbf{a}_{l,i,j,\tau}, b_{l,i,j,\tau}\}_{i,j,\tau}$, and $\Psi = \{\psi_l\}_l$.

Outputs $sk = \{\mathbf{s}_0 = \mathbf{s}, \mathbf{s}_1, \dots, \mathbf{s}_L\}$ as the secret key, $pk = (\mathbf{A}, \mathbf{b})$ as the public key, and $evk = \Psi$ as the evaluate key.

- **SH.Enc**(pk, w): takes as input the public key pk and plaintext $w \in \{0, 1\}$. Samples $\mathbf{r} \xleftarrow{\$} \{0, 1\}^n$. Computes

$$\mathbf{v} = \mathbf{A}^T \cdot \mathbf{r} \text{ and } u = \mathbf{b}^T \cdot \mathbf{r} + w$$

Outputs $\mathbf{c} = (u, \mathbf{v}, 0)$ as the ciphertext.

- **SH.Dec**($sk, \mathbf{c}$): takes as input the secret key sk and ciphertext $\mathbf{c} = (u, \mathbf{v}, l)$. Outputs

$$w = \lceil [u - \langle \mathbf{v}, \mathbf{s} \rangle]_q \rceil_2$$

- **SH.Add**($pk, \mathbf{c}, \mathbf{c}'$): takes as input the secret key pk and ciphertext $\mathbf{c} = (u, \mathbf{v}, l)$, $\mathbf{c}' = (u', \mathbf{v}', l)$. Outputs

$$\mathbf{c}_{\text{add}} = ([u + u']_q, [\mathbf{v} + \mathbf{v}']_q, l)$$

- **SH.Mult**($pk, evk, \mathbf{c}, \mathbf{c}'$): takes as input the secret key pk and ciphertext $\mathbf{c} = (u, \mathbf{v}, l)$, $\mathbf{c}' = (u', \mathbf{v}', l)$. Computes

$$\begin{aligned} \mathbf{v}_{\text{mult}} &= \sum_{\tau \in \{0, \dots, \lfloor \log q \rfloor\}} \sum_{0 \leq i \leq j \leq n} h_{i,j,\tau} \cdot \mathbf{a}_{l+1,i,j,\tau} \bmod q \\ u_{\text{mult}} &= \sum_{\tau \in \{0, \dots, \lfloor \log q \rfloor\}} \sum_{0 \leq i \leq j \leq n} h_{i,j,\tau} \cdot b_{l+1,i,j,\tau} \bmod q \end{aligned}$$

Outputs $(u_{\text{mult}}, \mathbf{v}_{\text{mult}}, l + 1)$.

2.4.3 Correctness and Parameter Setting

Let $c = (u, \mathbf{v}, 0)$ be a “fresh” ciphertext of the plaintext $w \in \{0, 1\}$, i.e., $\mathbf{v} = \mathbf{A}^T \cdot \mathbf{r}$, $u = \mathbf{b}^T \cdot \mathbf{r} + w$, where $\mathbf{r} \xleftarrow{\$} \{0, 1\}^m$. Then

$$\begin{aligned} u - \langle \mathbf{v}, \mathbf{s} \rangle \mod q &= \mathbf{b}^T \cdot \mathbf{r} + m - \mathbf{A}^T \cdot \mathbf{r} \mod q \\ &= (\mathbf{A} \cdot \mathbf{S} + 2\mathbf{e})^T \cdot \mathbf{r} + m - \mathbf{A}^T \cdot \mathbf{r} \mod q \\ &= 2\langle \mathbf{e}, \mathbf{r} \rangle + m \mod q \\ &= 2e + m \mod q \end{aligned}$$

We define the decryption noise norm of $c = (u, \mathbf{v}, 0)$ to be $2Bm + 1$, and denote it as $N_{\text{dec}}(c)$. When $N_{\text{dec}} < \frac{q}{2}$, we have $u - \langle \mathbf{v}, \mathbf{s} \rangle \mod q = 2\langle \mathbf{e}, \mathbf{r} \rangle + m$, and $[[u - \langle \mathbf{v}, \mathbf{s} \rangle]_q]_2 = m$.

Let $c_i = (u_i, \mathbf{v}_i, l)$ be the ciphertext of the plaintext $w_i \in \{0, 1\}$, $i = 1, 2$. Let $c_{\text{add}} = (u_{\text{add}}, \mathbf{v}_{\text{add}}, l) = \text{SH.Add}(c_1, c_2)$. Then

$$\begin{aligned} u_{\text{add}} - \langle \mathbf{v}_{\text{add}}, \mathbf{s}_l \rangle \mod q &= 2e_1 + m_1 + 2e_2 + m_2 \mod q \\ &= 2e_{\text{add}} + m_1 + m_2 \mod q \end{aligned}$$

We define the decryption noise norm $N_{\text{dec}}(c_{\text{add}}) = N_{\text{dec}}(c_1) + N_{\text{dec}}(c_2)$. When $N_{\text{dec}}(c_{\text{add}}) < \frac{q}{2}$, we have $\text{SH.Dec}(c_{\text{add}}) = m_1 + m_2$.

Let $c_{\text{mult}} = (u_{\text{mult}}, \mathbf{v}_{\text{mult}}, l + 1) = \text{SH.Mult}(c_1, c_2)$. Then

$$\begin{aligned} u_{\text{mult}} - \langle \mathbf{v}_{\text{mult}}, \mathbf{s}_{l+1} \rangle \mod q &= \left(\sum_{\tau \in \{0, \dots, \lfloor \log q \rfloor\}} \sum_{0 \leq i \leq j \leq n} h_{i,j,\tau} \cdot b_{l+1,i,j,\tau} \right) \\ &\quad - \left(\left\langle \sum_{\tau \in \{0, \dots, \lfloor \log q \rfloor\}} \sum_{0 \leq i \leq j \leq n} h_{i,j,\tau} \cdot \mathbf{a}_{l+1,i,j,\tau}, \mathbf{s}_{l+1} \right\rangle \right) \mod q \\ &= 2 \cdot \sum_{\tau \in \{0, \dots, \lfloor \log q \rfloor\}} \sum_{0 \leq i \leq j \leq n} h_{i,j,\tau} \cdot e_{i,j,\tau} \\ &\quad + (w_1 - \langle \mathbf{v}_1, \mathbf{s}_l \rangle)(w_2 - \langle \mathbf{v}_2, \mathbf{s}_l \rangle) \mod q \end{aligned}$$

We define the decryption noise norm $N_{\text{dec}}(c_{\text{mult}}) = N_{\text{dec}}(c_1) \cdot N_{\text{dec}}(c_2) + 2n^2 \lfloor \log q + 1 \rfloor \cdot B$.

When $N_{\text{dec}}(c_{\text{mult}}) < \frac{q}{2}$, we have $\text{SH.Dec}(c_{\text{mult}}) = m_1 \cdot m_2$.

2.4.4 Security and Parameter Setting

The security of Regev.Scheme is based on the following lemmas.

Lemma 2.9 (Matrix-Vector Leftover Hash Lemma) Let $\kappa, n, m, q \in \mathbb{N}$, and $m \geq n \log q + 2\kappa$. Let $\mathbf{A} \xleftarrow{\$} \mathbb{Z}^{m \times n}$ be an uniformly random matrix, and let $\mathbf{r} \xleftarrow{\$} \{0, 1\}^m$, $\mathbf{y} \xleftarrow{\$} \mathbb{Z}_q^n$, then

$$\Delta((\mathbf{A}, \mathbf{A}^T \cdot \mathbf{r}), (\mathbf{A}, \mathbf{y})) \leq 2^{-\kappa}$$

where $\Delta(\mathbf{A}, \mathbf{B})$ denotes the statistical distribution distance between $\mathbf{A}$ and $\mathbf{B}$.

Lemma 2.10 (LWE Hardness Assumption) Let $q = q(n) = \prod_i q_i \in \mathbb{N}$ be a product of co-prime integers such that for all i , $q_i = \text{poly}(n)$, and Let $B \geq n$. There exist an efficiently sampleable B -bound distribution χ such that if there exist an efficient algorithm that solves $\text{DLWE}_{n,q,\chi}$ problem, then

- There exist a quantum algorithm that solves $\text{SIVP}_{\tilde{O}(n\sqrt{n} \cdot q/B)}$ and $\text{gapSVP}_{\tilde{O}(n\sqrt{n} \cdot q/B)}$ on any n dimension lattice, and run in $\text{poly}(n)$ time.[34]
- There exist a classical algorithm that solves ζ -to- γ decisional $\text{gapSVP}_{\zeta,\gamma}$, where $\zeta = \tilde{O}(q\sqrt{n})$ and $\gamma = \tilde{O}(n\sqrt{n} \cdot q/B)$ on any n dimension lattice, and run in $\text{poly}(n)$ time.[35]

The parameters must be set such that the leftover hash lemma and LWE hardness assumption hold. Take the correctness into consideration, we give a suggested parameter setting as follows:

- $n \geq \kappa$: the dimension of the LWE problem should not less the security parameter κ .
- $m = O(n \log q)$: to satisfies the leftover hash lemma, which is need in the IND-CPA security reduction.
- $q = O(2^{n^\epsilon})$, $\epsilon < 1$: on one side q should be large enough to guarantee a ciphertext though a complex evaluation circuit can be correctly decrypted, on the other side we should not let q be so large, e.g. be exponential in n , since at that case gapSVP can be solved via some lattice basis reduction techniques, such LLL algorithm, but it is still safe to set q subexponential in n .
- $B = n$: considering the best algorithm for $\text{LWE}_{n,q,\chi}$ with χ being a B -bound distribution and B being $1/2^k$ fraction of the modulus q , runs in time $2^{\tilde{O}(n/k)}$.

2.5 Somewhat Homomorphic Encryption Scheme Based on NTRU

The NTRU encryption scheme[41], which was first introduced by Hoffstein, Pipher and Silverman[42] at the rump session of Crypto'96, is the fastest known lattice-based encryption scheme. Although there exists various attack to NTRU [43],[44],[45],[46], the recent work[47] shows us that by setting the parameter properly the NTRU encryption

scheme can be as secure as worst-case problems over ideal lattices. Thus building FHE scheme based on NTRU is a reasonable choice. As follows, we describe a somewhat homomorphic encryption scheme based on NTRU, note that the multi-key version of NTRU FHE has been proposed [17], our description partly consults [17].

2.5.1 Preliminaries and Notions

The NTRU encryption scheme works over the ring $R = \mathbb{Z}/\langle F(x) \rangle$ and $R_q = R/qR$, where $F(x) \in \mathbb{Z}[x]$ is monic polynomial of n degree, and $q \in \mathbb{Z}$ is a prime. Notice that $R_q \equiv \mathbb{Z}_q[x]/\langle F(x) \rangle$, i.e., the ring of n -degree polynomials modulo $F(x)$ with coefficients in $\mathbb{Z}_q$. Addition in R (R_q) is the polynomial addition which is done component-wise in their coefficient (and modulo q). Multiplication is the polynomial multiplication modulo $F(x)$ (and modulo q). Notice that the additive group of R (R_q) is isomorphic to $\mathbb{Z}^n$ ($\mathbb{Z}_q^n$), an element in R (R_q) can be both viewed as an n -degree polynomial over $\mathbb{Z}$ ($\mathbb{Z}_q$) and an n -dimensional vector in $\mathbb{Z}^n$ ($\mathbb{Z}_q^n$).

For a rational number z , we use the notions $\lfloor z \rfloor$ to denote the rounding down of z , i.e., denote the unique integer $z' \in [z - 1, z)$, such that $z - z' \in [0, 1)$.

For an integer $z \in \mathbb{Z}$ and a prime $q \in \mathbb{Z}$, we use $z \bmod q$ to refer to the unique integer $z' \in [-\lfloor \frac{q}{2} \rfloor, \lfloor \frac{q}{2} \rfloor]$ such that $q|z - z'$. Subsequently by $\mathbb{Z}_q$, we refer to $\{-\lfloor \frac{q}{2} \rfloor, \dots, \lfloor \frac{q}{2} \rfloor\}$.

For an element $a(x) \in R$, let $a(x) = a_0 + a_1(x) + \dots + a_{n-1}x^{n-1}$, we use $\|a(x)\|$ to denote the l^2 -norm of $(a_0, a_1, \dots, a_{n-1})$, i.e. $\|a(x)\| = \sqrt{\sum_{i=0}^{n-1} a_i^2}$. we use $\|a(x)\|_\infty$ to denote the l^∞ -norm of $(a_0, a_1, \dots, a_{n-1})$, i.e. $\|a(x)\|_\infty = \max\{|a_i|\}$.

In this NTRU somewhat homomorphic encryption scheme, we set $F(x) = x^n + 1$ to be the n -th cyclotomic polynomial, with n a power of two. Thus R and R_q are respectively $\mathbb{Z}[x]/\langle x^n + 1 \rangle$ and $\mathbb{Z}_q[x]/\langle x^n + 1 \rangle$.

Definition 2.11 (B -Bounded Polynomial) A polynomial $e \in R$ is called B -bounded, if $\|e\|_\infty \leq B$.

Definition 2.12 (B -Bounded Distribution) A distribution ensemble $\{\chi_n\}_n \in \mathbb{N}$, supported over R , is called B -bounded (for $B = B(n)$) if for all $e \in \chi_n$, e is B -bounded, i.e. $\|e\|_\infty \leq B$.

2.5.2 Scheme Description

The NTRU somewhat homomorphic encryption scheme is formally defined as 5 PPT algorithm: HNTRU.KeyGen, HNTRU.Enc, HNTRU.Dec, HNTRU.Add, HNTRU.Mult. This scheme is parameterized by $(n(\lambda), q(\lambda), B(\lambda))$, which are associated with the security parameter λ .

- **HNTRU.KeyGen:** Takes as input the security parameter 1^λ . Sample polynomials $f', g \xleftarrow{\$} \chi$, set $f := 2f' + 1$, so that $f \equiv 1 \pmod{2}$. Set $h = 2gf^{-1} \in R_q$. Resample f' if it is not invertible in R_q . Output $pk = \{h\}$ as the public key, and $sk = \{f\}$ as the secret key.
- **HNTRU.Enc:** Takes as input the public key $pk\{h\}$ and a plaintext $m \in \{0, 1\}$. Sample polynomials $s, e \in \chi$, set

$$c = hs + 2e + m \in R_q.$$

Output c as the ciphertext.

- **HNTRU.Dec:** Takes as input a ciphertext $c \in R_q$ and secret key $sk\{f\}$. It computes

$$\mu = cf \in R_q.$$

Output $m := \mu \bmod 2$ as the plaintext.

- **HNTRU.Add:** Takes as input two ciphertext $c_1, c_2 \in R_q$, output $c_1 + c_2 \in R_q$.
- **HNTRU.Mult:** Takes as input two ciphertext $c_1, c_2 \in R_q$, output $c_1 \cdot c_2 \in R_q$.

2.5.3 Correctness and Parameter Setting

Let c be the encryption of m , i.e. there exists $s, e \in \chi$ such that $c = hs + 2e + m$. The first step of the decryption results in:

$$\begin{aligned} \mu &= cf \bmod (F(x), q) \\ &= (sh + 2e + m) \bmod (F(x), q) \\ &= (2sgf^{-1}f + 2ef + mf) \bmod (F(x), q) \\ &= (2sg + 2ef + mf) \bmod (F(x), q) \\ &= [(2sg \bmod F(x)) + (2ef \bmod F(x)) + mf] \bmod q. \end{aligned} \tag{2-2}$$

Let μ' denote $(2sg \bmod F(x)) + (2ef \bmod F(x)) + mf$. Considering that $F(x) = x^n + 1$, so $\bmod F(x)$ will not change the parity, and if the $\mu' \bmod q$ operation does not happen, i.e. $\mu = \mu'$, then we will correctly obtain the plaintext by μ modulo two. Therefore the correctly decryption condition is $\|\mu'\|_\infty \leq \lfloor \frac{q}{2} \rfloor$. The crucial part of ensuring the correctness of decryption is to bound the l^∞ -norm of μ' . To give the bounds, we need the following lemma.

Lemma 2.13 ([48],[12]) Let $R = \mathbb{Z}[x]/\langle F(X) \rangle$ with $n \in \mathbb{N}$ and $F(x) = x^n + 1$. For any $s, t \in R$,

$$\begin{aligned}\|s \cdot t \bmod F(x)\| &\leq \sqrt{n} \cdot \|s\| \cdot \|t\| \\ \|s \cdot t \bmod F(x)\|_\infty &\leq n \cdot \|s\|_\infty \cdot \|t\|_\infty.\end{aligned}$$

Since g, s, e are bounded by B , f is bounded by $2B + 1$, and by lemma 2.13, we have

$$\begin{aligned}\|2sg \bmod F(x)\|_\infty &\leq 2n \cdot \|s\|_\infty \cdot \|t\|_\infty = 2nB^2 \\ \|2ef \bmod F(x)\|_\infty &\leq 2n \cdot \|e\|_\infty \cdot \|f\|_\infty = 2nB(2B + 1).\end{aligned}$$

Therefore μ' is bounded by $2nB^2 + 4nB^2 + 2nB + 2B + 1 < 10nB^2$. Finally, by setting $10nB^2 \leq \lfloor \frac{q}{2} \rfloor$, i.e. $q > 20nB^2$, we can guarantee the fresh ciphertext c to be correctly decrypted.

We also need to consider the correctness of decryption of a ciphertext output by an homomorphic evaluation circuit. For simplicity we define the noise of a fresh ciphertext $c = sh + ef + mf \in R_q$ as $\mu' = (2sg \bmod F(x)) + (2ef \bmod F(x)) + mf$ in (2-2), and denoted it by $N(c)$.

Given two fresh ciphertexts $c_1 = s_1h + e_1f + m_1f \in R_q$ and $c_2 = s_2h + e_2f + m_2f \in R_q$, we defined $N(c_1 + c_2) := N(c_1) + N(c_2)$, and $N(c_1 \cdot c_2) := N(c_1) \cdot N(c_2)$. Obviously, when it satisfies $N(c_1 + c_2) \leq \lfloor \frac{q}{2} \rfloor$ and $N(c_1 \cdot c_2) \leq \lfloor \frac{q}{2} \rfloor$ respectively, $c_1 + c_2$ and $c_1 \cdot c_2$ can be correctly decrypted to $m_1 + m_2 \in GF(2)$ and $m_1 \cdot m_2 \in GF(2)$. For any ciphertext output by a homomorphic circuit, we can define its noise by recursively, and the condition of correctly decryption is $N(c) \leq \lfloor \frac{q}{2} \rfloor$.

2.5.4 Security and Parameter Setting

The security of NTRU somewhat homomorphic encryption scheme is based on the following hardness assumptions.

Definition 2.14 (Ring-LWE Assumption [36]) For a given $n \in \mathbb{N}$, let $F(x) \in \mathbb{Z}[x]$ be a polynomial of n degree, and let $q \in \mathbb{Z}$ be a prime. Let $R = \mathbb{Z}[x]/\langle F(x) \rangle$ and $R_q = R/qR$. Let χ is a distribution over the ring R_q . Then for a randomly chosen $s \xleftarrow{\$} R_q$, the decisional Ring LWE assumption $\text{RLWE}_{F,q,\chi}$ states that for any $l = \text{poly}(n)$ it holds that

$$\{(a_i, b_i = a_i \cdot s + e_i)\}_{i \in [l]} \approx \{(a_i, u_i)\}$$

where a_i, u_i are uniformly from R_q , e_i is drawn from the error distribution χ , and b_i is computed in R_q .

Definition 2.15 (Ring-LWE Assumption - Hermite Normal Form [37],[36]) For a given $n \in \mathbb{N}$, let $F(x) \in \mathbb{Z}[x]$ be a polynomial of n degree, and let $q \in \mathbb{Z}$ be a prime. Let $R = \mathbb{Z}[x]/\langle F(x) \rangle$ and $R_q = R/qR$. Let χ is a distribution over the ring R_q . Then for a randomly chosen $s \xleftarrow{\$} \chi$, the decisional Ring LWE assumption with Hermite Normal Form HNF – RLWE $_{F,q,\chi}$ states that for any $l = \text{poly}(n)$ it holds that

$$\{(a_i, b_i = a_i \cdot s + e_i)\}_{i \in [l]} \approx \{(a_i, u_i)\}$$

where a_i, u_i are uniformly from R_q , e_i is drawn from the error distribution χ , and b_i is computed in R_q .

The only difference between RLWE $_{F,q,\chi}$ and HNF – RLWE $_{F,q,\chi}$ is in the choosing of s . For RLWE $_{F,q,\chi}$, s is required to choose uniformly from R_q , while for HNF – RLWE $_{F,q,\chi}$, s is required to choose from a error distribution χ .

Definition 2.16 (Decisional Small Polynomial Ratio (DSPR) problem. [47],[17]) For a given $n \in \mathbb{N}$, let $F(x) \in \mathbb{Z}[x]$ be a polynomial of n degree, and let $q \in \mathbb{Z}$ be a prime. Let $R = \mathbb{Z}[x]/\langle F(x) \rangle$ and $R_q = R/qR$. Let χ is a distribution over the ring R_q . The decisional small polynomial ratio problem DSPR $_{F,q,\chi}$ is to distinguish between the following two distribution ensembles

$$\{h_i = g_i/f_i : g_i, f_i \leftarrow \chi\}_{i \in \mathbb{N}}$$

and

$$\{h_i : h_i \xleftarrow{\$} R_q\}_{i \in \mathbb{N}}$$

The DSPR $_{F,q,\chi}$ problem has been shown [47] as hard as the worst-case hard problem on ideal lattice as long as setting $F(x) = x^n + 1$ to be an n -th cyclotomic polynomial with n a power of two, and setting χ to be discrete Gaussian $D_{\mathbb{Z}^n, r}$, with $r > \sqrt{q} \cdot \text{poly}(n)$.

Based on the HNF – RLWE $_{F,q,\chi}$ hardness assumption and DSPR $_{F,q,\chi}$ hardness assumption, the semantic security of NTRU somewhat homomorphic encryption scheme can be proved in the following two setps[17]:

- (1) Given a ciphertext $c = 2sh + 2e + m$, where $h = g/f$. Then DSPR $_{F,q,\chi}$ assumption allows us to change c to $c' = 2sh' + 2e + m$ drawn uniformly from R_q .
- (2) By HNF – RLWE $_{F,q,\chi}$, $c' = 2sh' + 2e + m$ and $c'' = u + m$, where u follows uniform distribution over R_q , are undistinguished.

Note that the $\text{DSPR}_{F,q,\chi}$ is proved to be hard under the parameter setting $r > \sqrt{q} \cdot \text{poly}(n)$, this will not guarantee the correctness of decryption of NTRU somewhat homomorphic encryption scheme. Thus the proof the security of NTRU somewhat homomorphic encryption scheme is in fact based on a more aggressive $\text{DSPR}_{F,q,\chi}$ assumption which is not proved.

Chapter 3 Fully Homomorphic Encryption Scheme Using Bootstrapping

3.1 Dual Noise Structure over Ring – A Unified Representation of Somewhat Homomorphic Encryption Scheme

3.1.1 From Group to Ring

The former single homomorphic encryption schemes, such as the unpadded RSA scheme, ElGamal scheme, and Paillier scheme[49], are all group-based scheme. These schemes usually involve two groups G and G^* . These group-based encryption can be viewed as a mapping $\mathcal{E}$ from G to G^* . Let a plaintext $m \in G$, the corresponding ciphertext will be $\mathcal{E}(m) \in G^*$. If the mapping $\mathcal{E}$ is also homomorphic, i.e. for any $m_1, m_2 \in G$, there holds $\mathcal{E}(m_1 \cdot m_2) = \mathcal{E}(m_1) \circ \mathcal{E}(m_2)$, where $\cdot$ and $\circ$ are respectively the operations in G and G^* , then this encryption scheme is naturally homomorphic. For example the encryption of the Paillier encryption scheme is a mapping $\mathcal{E}$ from $G = (\mathbb{Z}_N, +)$ to $G^* = (\mathbb{Z}_{N^2}^*, \cdot)$, such that for any $m \in G$, $\mathcal{E}(m) = g^m \cdot r^N \bmod N^2$, where $N = pq > 0$ is the product of two large prime integers p, q , and g is a generator of the the group $\mathbb{Z}_{N^2}^*$, $r \in \mathbb{Z}_N$ is a random chosen number. For any $m_1, m_2 \in G$, there holds $g^{(m_1+m_2)} \cdot r^N = g^{m_1} \cdot r_1^N \cdot g^{m_2} \cdot r_2^N \pmod{N^2}$, i.e., $\mathcal{E}(m_1 + m_2) = \mathcal{E}(m_1) \cdot \mathcal{E}(m_2)$.

It is hard to build a somewhat homomorphic encryption scheme based on a group, since there is only one operation in a group, while a somewhat homomorphic scheme needs to support both additive and multiplicative homomorphic operations. Thus a natural idea is to build a somewhat homomorphic encryption scheme based on a ring.

Let R be a ring, and I be an ideal of R . There exists a natural homomorphism ϕ from R to R/I . Since for any $a, b \in R$ it holds that $(a + I) + (b + I) = (a + b) + I$ and $(a + I) \cdot (b + I) = (a \cdot b) + I$, so if we can viewed ϕ as the decryption of a encryption scheme, then this encryption scheme is naturally homomorphic. Indeed, we can construct a encryption scheme such that the decryption is exactly the homomorphic mapping ϕ . we just need to fix the plaintext space to be R/I , the ciphertext space to be R , given a plaintext $M \in R/I$, let the encrypting of M be randomly sampling $c \xleftarrow{\$} M$, or equivalently we fix the plaintext space $\mathcal{P}$ to be a set of representation of R/I , for any plaintext $m \in \mathcal{P}$, the corresponding ciphertext is $c \xleftarrow{\$} m + I$. Then the decrypting of c is to find the coset c belongs to, and we identify Dec with ϕ .

Let $c_1 \in m_1 + I$, $c_2 \in m_2 + I$ be two ciphertexts. The following equations show

the encryption scheme informally constructed above is somewhat homomorphic.

$$\begin{aligned}
\text{Dec}(c_1 + c_2) &= \text{Dec}(m_1 + I + m_2 + I) \\
&= \text{Dec}(m_1 + m_2 + I) \\
&= m_1 + m_2 \\
&= \text{Dec}(m_1 + I) + \text{Dec}(m_2 + I) \\
&= \text{Dec}(c_1) + \text{Dec}(c_2)
\end{aligned}$$

$$\begin{aligned}
\text{Dec}(c_1 \cdot c_2) &= \text{Dec}((m_1 + I) \cdot (m_2 + I)) \\
&= \text{Dec}(m_1 \cdot m_2 + I) \\
&= m_1 \cdot m_2 \\
&= \text{Dec}(m_1 + I) \cdot \text{Dec}(m_2 + I) \\
&= \text{Dec}(c_1) \cdot \text{Dec}(c_2)
\end{aligned}$$

3.1.2 Fix Plaintext in $GF(2)$

Considering that at most time the somewhat homomorphic encryption is used to evaluate homomorphic circuits, and noted that any circuit can be translated to a multivariate polynomial over $GF(2)$, so it is required to set the plaintext space to be $GF(2)$. Let R be a ring, we consider the ring $\{0 + 2R, 1 + 2R\} \subset R/2R$, where $0, 1$ are respectively the zero and identity elements of R . It is easy to prove that $\{0 + 2R, 1 + 2R\}$ is isomorphic to $GF(2)$, or equivalently $\{0, 1\}$ with respect to the addition and multiplication of coset is isomorphic to $GF(2)$. For clearly, we denote the elements in $GF(2)$ by $\bar{0}, \bar{1}$, and denote the isomorphism from $\{0, 1\}$ to $GF(2)$ by ϕ , i.e., $\phi(b) = \bar{b}, b = 0, 1$. This isomorphism also conduce a homomorphism ψ form $(0 + 2R) \cup (1 + 2R)$ to $GF(2)$ such that

$$\phi'(a) = \begin{cases} \bar{0}, & \text{if } a \in 0 + 2R, \\ \bar{1}, & \text{if } a \in 1 + 2R, \end{cases}$$

For simplicity, we usually identify $\bar{0}, \bar{1}$ respectively with $0, 1$.

Given an ideal I of a ring R , we define $\mathcal{L}$ to be the set of the representation elements of R/I , and for any $e \in R$, we define $e \bmod I$ as the unique element in $\mathcal{L}$, such that $e \in (e \bmod I) + I$.

Defining $\mathcal{P} = [(0 + 2R) \cup (1 + 2R)] \cap \mathcal{L}$, then for a plaintext $b \in GF(2)$ we can encrypt it as follows:

(1) choose $m \xleftarrow{\$} (b + 2R) \cap \mathcal{L} \subset \mathcal{P}$,

(2) choose $c \xleftarrow{\$} m + I$.

where I is a ideal of the ring R . Then the decryption of c is correspondingly done in the following two steps:

(1) Find the representation of the coset with respect to I , such that c exists in, denote as

$$m = c \bmod I \in (b + 2R) \cap \mathcal{L}.$$

(2) Find the representation of the coset with respect to $2R$, such that m exists in, denote as

$$b = m \bmod 2R \in \{0, 1\}.$$

Given a ciphertext c , we denote the decryption of c as $\text{Dec}(c)$, i.e. $\text{Dec}(c) = (c \bmod I) \bmod 2R$.

By such encryption scheme, the ciphertext c can be written as

$$c = b + \eta + \rho,$$

where $\eta \in 2R$, $\rho \in I$. When we view η and ρ as the noises, then the encryption of a plaintext $b \in \{0, 1\}$ is to add two noises to b , and the decryption of a ciphertext $c = b + \eta + \rho$ is to remove the two noise.

3.1.3 Somewhat Homomorphic Encryption

Let c_1, c_2 be the encryption of m_1, m_2 , i.e. $c_1 = b_1 + \eta_1 + \rho_1$ and $c_2 = b_2 + \eta_2 + \rho_2$, where $\eta_1, \eta_2 \in 2R$, $\rho_1, \rho_2 \in I$ and $b_1 + \eta_1, b_2 + \eta_2 \in \mathcal{P}$. Then we have

$$c_1 \bmod I = b_1 + \eta_1, c_2 \bmod I = b_2 + \eta_2,$$

and

$$(b_1 + \eta_1) \bmod 2R = b_1, (b_2 + \eta_2) \bmod 2R = b_2.$$

which means the decrypting of c_1 and c_2 can correctly result in b_1 and b_2 . But we can not guarantee that the decrypting of $c_1 + c_2$ and $c_1 \cdot c_2$ can correctly result in $b_1 + b_2$ and $b_1 \cdot b_2$.

Note that ϕ will map an element $e \in R$ to the representation element of the coset that e exists in, so we have

$$\begin{aligned}(c_1 + c_2) \bmod I &= (b_1 + b_2 + \eta_1 + \eta_2 + \rho_1 + \rho_2) \bmod I \\ &= (b_1 + b_2 + \eta_1 + \eta_2) \bmod I\end{aligned}$$

and

$$\begin{aligned}(c_1 \cdot c_2) \bmod I &= [(b_1 + \eta_1)(b_2 + \eta_2) + (b_2 + \eta_2)\rho_1 + (b_1 + \eta_1)\rho_2] \bmod I \\ &= (b_1 + \eta_1)(b_2 + \eta_2) \bmod I \\ &= (b_1b_2 + b_1\eta_2 + b_2\eta_1) \bmod I\end{aligned}$$

If $(b_1 + b_2 + \eta_1 + \eta_2)$ is in $\mathcal{P}$, then $(c_1 + c_2) \bmod I = b_1 + b_2 + \eta_1 + \eta_2$, and $((c_1 + c_2) \bmod I) \bmod 2R = (b_1 + b_2) \bmod 2$, which means

$$\text{Dec}(c_1 + c_2) = \text{Dec}(c_1) + \text{Dec}(c_2).$$

Similarly, if $(b_1b_2 + b_1\eta_2 + b_2\eta_1)$ is in $\mathcal{P}$, then $(c_1 \cdot c_2) \bmod I = b_1b_2 + b_1\eta_2 + b_2\eta_1$, and $((c_1 \cdot c_2) \bmod I) \bmod 2R = (b_1 \cdot b_2) \bmod 2$, which means

$$\text{Dec}(c_1 \cdot c_2) = \text{Dec}(c_1) \cdot \text{Dec}(c_2).$$

By the analysis above, this encryption scheme is not additive and multiplicative homomorphic for any ciphertexts, but it is for those ciphertexts c_1, c_2 with $c_1 = b_1 + \eta_1 + \rho_1$ and $c_2 = b_2 + \eta_2 + \rho_2$, such that $(b_1 + b_2 + \eta_1 + \eta_2) \in \mathcal{P}$ and $(b_1b_2 + b_1\eta_2 + b_2\eta_1) \in \mathcal{P}$. Therefore, in order to support for the homomorphic evaluation, it need to set a encrypt-set $\mathcal{N}_{\text{Enc}} \in \mathcal{P}$, such that for any fresh ciphertext $c = b + \eta + \rho$, $b + \eta \in \mathcal{N}_{\text{Enc}}$. When $\mathcal{N}_{\text{Enc}}$ is set small enough to satisfies

- (1) For any $m_1, m_2 \in \mathcal{N}_{\text{Enc}}$, $m_1 + m_2 \in \mathcal{P}$,
- (2) For any $m_1, m_2 \in \mathcal{N}_{\text{Enc}}$, $m_1 \cdot m_2 \in \mathcal{P}$,

then this encryption scheme is homomorphic for one time addition or one time multiplication. Furthermore, if we need this encryption scheme support for more complex homomorphic evaluation, $\mathcal{N}_{\text{Enc}}$ should be set more smaller.

3.1.4 Formal Description of Dual-Noise-Structure Homomorphic Encryption Scheme

We give a formal description of the dual-noise homomorphic encryption scheme, which we have discussed above. As shown above this dual-noise homomorphic is usually based on a ring, the encryption can be viewed as a noise-adding process, and the decryption can be viewed as a noise-removing process. In specific, the noise-adding process Enc will add two noises, a inner noise N_{in} and a outer noise N_{out} to a plaintext w , and obtain the ciphertext $w + N_{in} + N_{out}$. The noise-removing process Dec usually has two sub processes, Dec_1 and Dec_2 , Dec_1 takes as input $w + N_{in} + N_{out}$, outputs $w + N_{in}$, and Dec_2 takes as input $w + N_{in}$, outputs w .

A dual-noise homomorphic encryption is defined as a 5-tuple $\text{DNH} = (\text{DNH.KeyGen}, \text{DNH.Enc}, \text{DNH.Dec}, \text{DNH.Add}, \text{DNH.Mult})$, and it is associated with public parameters $R, I, \mathcal{N}_{\text{enc}}, \mathcal{N}_{\text{dec}}$, where R is a ring, I is an ideal of R such that I and $2R$ are co-prime, i.e., $I + 2R = R$. Let $\mathcal{L}$ be the set of the representative elements of R/I . $\mathcal{N}_{\text{enc}}$ and $\mathcal{N}_{\text{dec}}$ are two set related to some measure of the ciphertexts. Noticed that there always holds $\mathcal{N}_{\text{enc}} \subset \mathcal{N}_{\text{dec}} \subset (\{0 + 2R\} \cup \{1 + 2R\}) \cap \mathcal{L}$.

- **DNH.KeyGen:** Takes as input the system and security parameter, output the secret key sk , the public key pk .
- **DNH.Enc:** Takes as input a plaintext $b \in \{0, 1\}$ and public key pk . Sample $m \xleftarrow{\$} (b + 2R) \cap \mathcal{N}_{\text{enc}}$, i.e. $m = b + \eta \in (b + 2R) \cap \mathcal{N}_{\text{enc}}$. Then sample $c \xleftarrow{\$} m + I$, i.e. $c = b + \eta + \rho \in (b + \eta) + I$. Outputs as the ciphertext $c = b + \eta + \rho \in R$.
- **DNH.Dec:** Takes as input a ciphertext $c = b + \eta + \rho \in R$ and secret key sk , outputs b on the condition of $b + \eta \in \mathcal{N}_{\text{dec}}$. Usually, in formal, DNH.Dec consists of two algorithms, DNH.Dec1 and DNH.Dec2 . DNH.Dec1 takes as input a ciphertext $b + \eta + \rho$, outputs $m = b + \eta$. DNH.Dec2 takes as input m , outputs b .
- **DNH.Add:** Takes as input two ciphertexts $c_1 = b_1 + \eta_1 + \rho_1$ and $c_2 = b_2 + \eta_2 + \rho_2$. Output $c_3 = (c_1 + c_2) \bmod I$.
- **DNH.Mult:** Takes as input two ciphertexts $c_1 = b_1 + \eta_1 + \rho_1$ and $c_2 = b_2 + \eta_2 + \rho_2$. Output $c_4 = (c_1 \cdot c_2) \bmod I$.

It is easily to check that all the somewhat homomorphic encryption schemes we introduced in last chapter are of dual-noise-structure.

3.2 Homomorphic Circuits, Homomorphic Evaluation and Perfect Fully Homomorphic Encryption

3.2.1 Homomorphic Circuits – From Plaintext Circuits to Ciphertext Circuits

Any logic circuit can be represented by a boolean function, the variables and value of the boolean function is respectively the input and output of the circuit. When we view the logic value “0” and “1” as the value 0 and 1 in $GF(2)$, then a n -variate boolean function can be transformed to an n -variate polynomial function over $GF(2)$, vice versa, an n -variate polynomial function over $GF(2)$ can be transformed to an n -variate boolean function. Therefore, we can view a polynomial F over $GF(2)$ as a circuit, or we say F is a circuit over $GF(2)$. This circuit consists of additive gate “ $\oplus$ ” and multiplicative gate “ $\odot$ ”, which are respectively the ordinary addition and multiplication in $GF(2)$.

If there exists a homomorphism φ from a ring R to $GF(2)$, then given a circuit F over $GF(2)$, replacing the add gate and multiply gate of F by the addition “+” and multiplication “ $\cdot$ ” in R , we obtain a polynomial $\tilde{F}$ over R . Let $b_i \in GF(2)$, and $w_i \in \varphi^{-1}(a_i) \in R$, for $i = 1, 2, \dots, l$, then we have

$$\varphi(\tilde{F}(w_1, w_2, \dots, w_l)) = F(a_1, a_2, \dots, a_l).$$

This means that to evaluate $F(b_1, b_2, \dots, b_l)$, we can first evaluate $\tilde{F}(w_1, w_2, \dots, w_l)$ in R , then map it to $GF(2)$. Thus we call polynomial $\tilde{F}$ over R the homomorphic circuit of F .

Fix a dual-noise-structure homomorphic encryption scheme DNH over the ring R , since the plaintext is in $\{0, 1\}$, we call a circuit F over $GF(2)$ a plaintext circuit. Let $\tilde{F}$ be the homomorphic circuit over R , we call $\tilde{F}$ the ciphertext circuit. Let $b_1, b_2, \dots, b_l \in GF(2)$ be the plaintext input into F , $\mu_1, \mu_2, \dots, \mu_l$ be the corresponding ciphertexts, i.e., $\mu_i = b_i + \eta_i + \rho_i$. Input $\mu_1, \mu_2, \dots, \mu_l$ into the circuit $\tilde{F}$, we have

$$\tilde{F}(\mu_1, \mu_2, \dots, \mu_l) = \tilde{F}(b_1, b_2, \dots, b_l) + \eta' + \rho'$$

When $\tilde{F}(b_1, b_2, \dots, b_l) + \eta' \in \mathcal{N}_{\text{dec}}$, we decrypt $\tilde{F}(\mu_1, \mu_2, \dots, \mu_l)$ to obtain $\tilde{F}(b_1, b_2, \dots, b_l)$, then we can obtain $F(a_1, a_2, \dots, a_l)$ by map $\tilde{F}(b_1, b_2, \dots, b_l)$ back to $GF(2)$. We write this whole process as

$$F(b_1, b_2, \dots, b_l) = \text{DNH.Dec}(\tilde{F}(\mu_1, \mu_2, \dots, \mu_l))$$

This means that to evaluate $F(b_1, b_2, \dots, b_l)$, we can first encrypt $b_1, b_2, \dots, b_l$ to the corresponding ciphertexts $\mu_1, \mu_2, \dots, \mu_l$, and evaluate $\tilde{F}(\mu_1, \mu_2, \dots, \mu_l)$ over R , and then decrypt $\tilde{F}(\mu_1, \mu_2, \dots, \mu_l)$, and map it to $GF(2)$.

3.2.2 Homomorphic Evaluatable and Perfect Fully Homomorphic

By the discussion above we conduce the following definitions.

Definition 3.1 Given an encryption scheme $\mathcal{E}$, a circuit F over $GF(2)$ homomorphic is evaluatable with respect to DNH, if for any $b_1, b_2, \dots, b_l \in \{0, 1\}$, let $\mu_1, \mu_2, \dots, \mu_l$ be the corresponding ciphertexts respect to $\mathcal{E}$, it holds that

$$F(a_1, a_2, \dots, a_l) = \mathcal{E}.\text{Dec}(\tilde{F}(\mu_1, \mu_2, \dots, \mu_l)) \quad (3-1)$$

where $\tilde{F}$ is the homomorphic circuit of F , and $\mathcal{E}.\text{Dec}$ denotes the decryption algorithm of the scheme $\mathcal{E}$.

Definition 3.2 An encryption scheme $\mathcal{E}$ is called perfect fully homomorphic, if for arbitrary circuit F over $GF(2)$, F is homomorphic evaluatable with respect to $\mathcal{E}$, i.e., for arbitrary $b_1, b_2, \dots, b_l \in \{0, 1\}$, let $\mu_1, \mu_2, \dots, \mu_l$ be the corresponding ciphertexts respect to $\mathcal{E}$, it holds that

$$F(b_1, b_2, \dots, b_l) = \mathcal{E}.\text{Dec}(\tilde{F}(\mu_1, \mu_2, \dots, \mu_l)) \quad (3-2)$$

where $\tilde{F}$ is the homomorphic circuit of F , and $\mathcal{E}.\text{Dec}$ denotes the decryption algorithm of the scheme $\mathcal{E}$.

Fully homomorphic encryption scheme is very useful in many scenarios, but it is difficult to design. For example, we consider a dual-noise-structure homomorphic scheme DNH. The crucial step is

$$\tilde{F}(w_1, w_2, \dots, w_l) \leftarrow \text{DNH}.\text{Dec}_1(\tilde{F}(\mu_1, \mu_2, \dots, \mu_l)) \quad (3-3)$$

where $w_i = b_i + \eta_i$, and $\mu_i = w_i + \rho_i$. Since $\tilde{F}(\mu_1, \mu_2, \dots, \mu_l) = \tilde{F}(w_1, w_2, \dots, w_l) + \rho'$, the $\text{DNH}.\text{Dec}_1$ step is equivalent to removing the noise ρ' from $\tilde{F}(\mu_1, \mu_2, \dots, \mu_l)$, and this can be correctly done if and only if

$$\tilde{F}(w_1, w_2, \dots, w_l) \in \mathcal{N}_{\text{dec}} \quad (3-4)$$

Unfortunately, for the dual-noise-structure homomorphic encryption scheme, the above condition dose not hold for all circuit F over $GF(2)$. Thus perfect fully homomorphic can not be achieved based on dual-noise-structure. The essential reason is that $\mathcal{N}_{\text{dec}}$ is not closed under addition and multiplication. What it can achieve is that for any $b_1, b_2, \dots, b_l \in \{0, 1\}$, limits $w_1, w_2, \dots, w_l$ into $\mathcal{N}_{\text{dec}} \subset \mathcal{N}_{\text{dec}}$, such that for some circuit

F over $GF(2)$, it satisfies $\tilde{F}(w_1, w_2, \dots, w_l) \in \mathcal{N}_{\text{dec}}$. This is what we call “somewhat homomorphic”.

3.3 Bootstrapping and Generalized Fully Homomorphic Encryption

Perfect fully homomorphic encryption scheme can not be achieved from a dual-noise somewhat homomorphic encryption scheme, because the decryption condition dose not always hold for arbitrary circuit F , therefore to achieve a generalised fully homomorphic encryption scheme a directly idea is to make the decryption condition hold. For simplicity, we first introduce a measure on the ciphertexts.

3.3.1 Measure on the Ciphertexts

For some kinds of ring R , such as $\mathbb{Z}$ and $\mathbb{Z}[x]$, given an element $u \in R$, there always exists an nonnegative integer associated with u , we define it as the norm of u , denoted by $\|u\|$, and satisfying

$$\begin{aligned} \|u_1 + u_2\| &\leq \|u_1\| + \|u_2\| \\ \|u_1 \cdot u_2\| &\leq \lambda \|u_1\| \cdot \|u_2\| \end{aligned} \quad (3-5)$$

where λ is a constant related to the ring R .

Given a dual-noise-homomorphic encryption scheme **DNH** over a ring R , the $\mathcal{N}_{\text{dec}}$ can be determined by an positive integer r_{dec} , such that $\forall u \in R, u \in \mathcal{N}_{\text{dec}}$ if and only if $\|u\| \leq r_{\text{dec}}$. Similarly, the $\mathcal{N}_{\text{enc}}$ can be determined by an positive integer r_{enc} , such that $\forall u \in R, u \in \mathcal{N}_{\text{enc}}$ if and only if $\|u\| \leq r_{\text{enc}}$, and since $\mathcal{N}_{\text{enc}} \subset \mathcal{N}_{\text{dec}}$, we have $r_{\text{enc}} < r_{\text{dec}}$.

Let $\mu_i = b_i + \eta_i + \rho_i, i = 1, 2$. be two “fresh” ciphertext, and let $\mu_3 = \text{DNH.Add}(\mu_1, \mu_2)$, $\mu_4 = \text{DNH.Mult}(\mu_1, \mu_2)$, i.e., $\mu_3 = \mu_1 + \mu_2$, $\mu_4 = \mu_1 \cdot \mu_2$. For $i = 1, 2$. we define the noise norm of μ_i to be $\|b_i + \rho_i\|$, denoted by $N(\mu_i) = \|b_i + \rho_i\|$. For $i = 3, 4$. we define noise norm of μ_i as follow:

$$\begin{aligned} N(\mu_3) &= \|(w_1 + \rho_1) + (w_2 + \rho_2)\| \\ N(\mu_4) &= \|(w_1 + \rho_1) \cdot (w_2 + \rho_2)\| \end{aligned} \quad (3-6)$$

Obviously, we have a following noise norm inequalities:

$$\begin{aligned} N(\mu_1 + \mu_2) &\leq N(\mu_1) + N(\mu_2) \\ N(\mu_1 \cdot \mu_2) &\leq \lambda \cdot N(\mu_1) \cdot N(\mu_2) \end{aligned} \quad (3-7)$$

which means the up bound of the noise norm increases with evaluation by the additive gate and multiplicative gate.

For a ciphertext $\mu = \tilde{F}(\mu_1, \dots, \mu_l)$, its noise norm can be recursively computed by formula (3-6), and it is possible that $N(\mu)$ exceeds the bound r_{dec} , and in this case

$\tilde{F}(\mu_1, \dots, \mu_l)$ can not be correctly decrypted to $\tilde{F}(w_1, \dots, w_l)$ as shown in equation (3-3), therefore the plaintext circuit $F(b_1, \dots, b_l)$ can not be evaluated homomorphically.

3.3.2 Bootstrapping – homomorphic decryption

To successfully homomorphically evaluate $F(b_1, \dots, b_l)$, we need to reduce the noise norm of $\mu = \tilde{F}(\mu_1, \dots, \mu_l)$. A simple idea is to “refresh” μ , i.e., first decrypt it to a plaintext, and then encrypt the plaintext to a new ciphertext. Decryption algorithm involves the secret key, but in some scenario, the secret key is not accessible to a third party, which provide the homomorphic evaluation service. A technique called bootstrapping was proposed in [12] to solve this problem, with this technique the ciphertext can be refreshed by a third part while the secret key remains unknown for the third part. Essentially, the bootstrapping technique is to evaluate the homomorphic circuit of the decryption circuit on the ciphertext that need to be “refreshed”.

we first estimate the decryption algorithm and its corresponding decryption circuit. Obviously the decryption algorithm DNH.Dec can be represented as a boolean circuit, and let its corresponding circuit over $GF(2)$ be F_{dec} .

Given a ciphertext μ , let its binary representation be $\mu[1], \mu[2], \dots, \mu[l]$, and let the secret key be s , with binary representation $s[1], s[2], \dots, s[k]$, we have

$$\text{DNH.Dec}(\mu, s) = F_{\text{dec}}(\mu[1], \dots, \mu[l], s[1], \dots, s[k]) = b \in \{0, 1\}$$

Let the homomorphic circuit of F_{dec} over R be $\tilde{F}_{\text{dec}}$, and let μ_i, s_i be the encryption of $\mu[i], s[i]$ using the secret key s . Suppose the noise norm of $\mu' = \tilde{F}_{\text{dec}}(\mu_1, \dots, \mu_l, s_1, \dots, s_k)$ less than r_{dec} , follow the equation (??), we have then

$$\begin{aligned} \text{DNH.Dec}(\mu, s) &= F_{\text{dec}}(\mu[1], \dots, \mu[l], s[1], \dots, s[k]) \\ &= \text{DNH.Dec}(\tilde{F}_{\text{dec}}(\mu_1, \dots, \mu_l, s_1, \dots, s_k), s) \end{aligned} \quad (3-8)$$

which means μ and μ' are respectively the ciphertexts of a same plaintext. Thus to refresh the ciphertext μ , we need not to decrypt it and re-encrypt, we can just use μ' to replace μ , and this refreshing always occur under the following conditions:

$$\begin{cases} N(\mu) \leq r_{\text{dec}} \\ N(\mu') < N(\mu) \end{cases} \quad (3-9)$$

and for simplicity, if $s_1, \dots, s_k$ are fixed, we also denote this refresh circuit $\tilde{F}_{\text{dec}}(\mu_1, \dots, \mu_l, s_1, \dots, s_k)$ by $\text{Refresh}(\mu)$.

3.3.3 Fully Homomorphic Encryption Achieved

Given an arbitrary circuit F over $GF(2)$ and inputs $b_1, b_2, \dots, b_l$, let $\mu_1, \mu_2, \dots, \mu_l$ be the encryption of $a_1, a_2, \dots, a_l$, and $\tilde{F}$ be the homomorphic circuit of F . We can view the circuit $\tilde{F}$ as a binary tree shown as follows. The leaf nodes are input nodes, taking as inputs $\mu_1, \mu_2, \dots, \mu_l$. The middle nodes and root node are computing node, taking two ciphertexts as inputs, and outputting the result (sum or product, which determined by the node itself). The output from root node is $\tilde{F}(\mu_1, \mu_2, \dots, \mu_l)$.

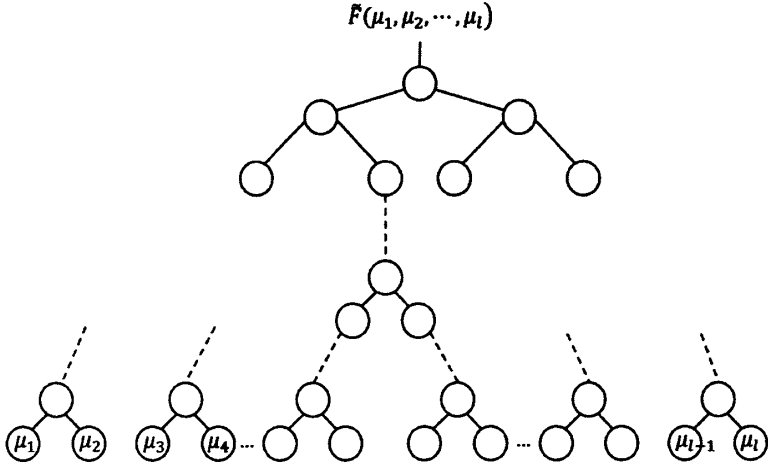


Figure 3.1 Any circuit

For arbitrary middle node u , let μ be the output of u , suppose μ satisfy the condition (3-9), we replace μ by $\mu' = \text{Refresh}(\mu, s)$, which is equivalent to insert a refresh circuit between the node u and its present node, as shown in Figure 3.2.

Finally, we obtain a composed circuit $\tilde{F}'$, such that

$$F(b_1, b_2, \dots, b_l) = \text{DNH.Dec}(\tilde{F}'(\mu_1, \mu_2, \dots, \mu_l), s) \quad (3-10)$$

it means to evaluate the plaintext circuit $F(b_1, b_2, \dots, b_l)$, we can first evaluate the ciphertext circuit $\tilde{F}'(\mu_1, \mu_2, \dots, \mu_l)$, and then decrypt it to recover $F(b_1, b_2, \dots, b_l)$. This is also a kind of homomorphic evaluation, but for distinguishing from the definition of perfect fully homomorphic encryption, we define as follows:

Definition 3.3 An encryption scheme $\mathcal{E}$ is called (Generalized) fully homomorphic, if for arbitrary circuit F over $GF(2)$, F is Generalized homomorphic evaluable with respect to $\mathcal{E}$, i.e., for arbitrary $b_1, b_2, \dots, b_l \in \{0, 1\}$, let $\mu_1, \mu_2, \dots, \mu_l$ be the

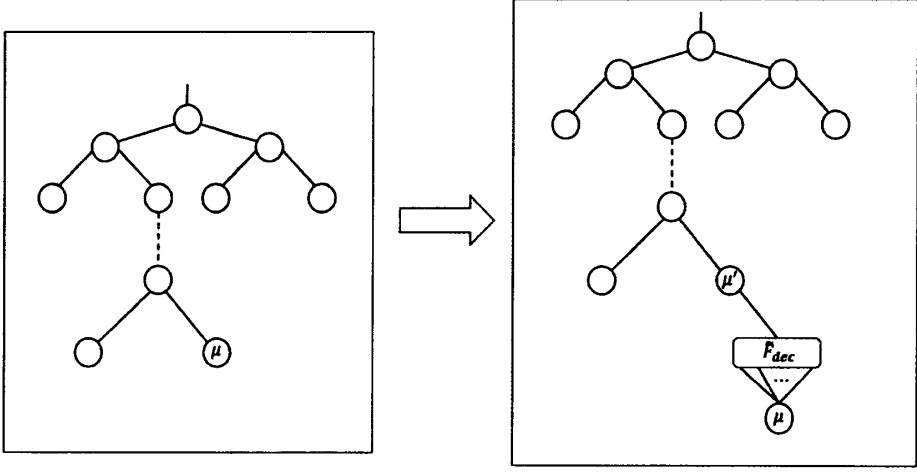


Figure 3.2 Ciphertext refreshing

corresponding ciphertexts respect to $\mathcal{E}$, it holds that

$$F(b_1, b_2, \dots, b_l) = \mathcal{E}.\text{Dec}(\tilde{F}'(\mu_1, \mu_2, \dots, \mu_l)) \quad (3-11)$$

where $\tilde{F}'$ is a circuit determined by F , $\mathcal{E}.\text{Dec}$ is the decryption algorithm of $\mathcal{E}$, and $\tilde{F}'$ can be constructed without know the secret key of $\mathcal{E}$.

As follows, we show that a dual-noise somewhat homomorphic encryption scheme can be fully homomorphic. Given a dual-noise homomorphic encryption scheme DNH, to homomorphically evaluate an arbitrary circuit F , we must guarantee that the refresh process can be continuously executed in the homomorphic circuit $\tilde{F}$, and finally obtain an equivalent circuit $\tilde{F}'$ without using secret key s . It is obvious, as shown in Figure 3.3, that for arbitrary refreshed ciphertexts μ'_1, μ'_2 , i.e., μ'_1, μ'_2 are outputted by the refresh circuit, if $\text{DNH.Add}(\mu'_1, \mu'_2)$ and $\text{DNH.Mult}(\mu'_1, \mu'_2)$ are also can be refreshed, then the refresh process can be continuously executed.

We use the notion r_{refresh} to denote the maximum noise norm of the output by the refresh circuit of DNH, i.e.,

$$r_{\text{refresh}} = \max\{N(\text{Refresh}(\mu)) : \forall \mu \in R\} \quad (3-12)$$

Combined by the refresh condition (3-9) and the noise inequation (3-7), we have the following theorem:

Theorem 3.4 A DNH encryption scheme is generalized fully homomorphic, if it satisfies

$$2 \cdot r_{\text{refresh}} \leq r_{\text{dec}} \quad (3-13)$$

$$\lambda \cdot r_{\text{refresh}}^2 \leq r_{\text{dec}}$$

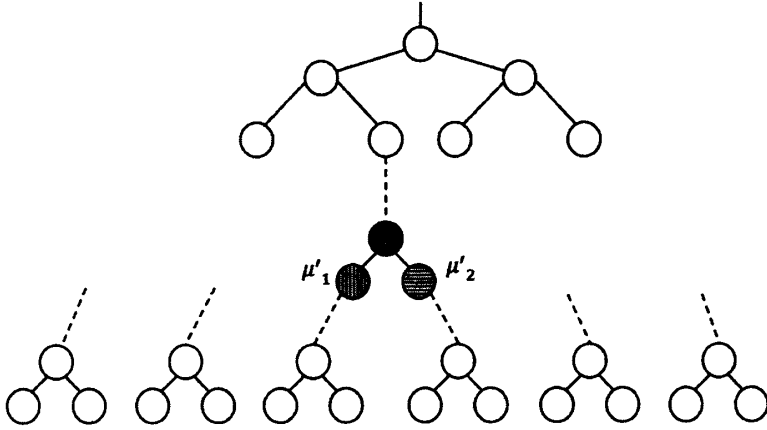


Figure 3.3 Continue refreshing

where λ is a constant related to DNH.

3.4 Optimization for Generalized Fully Homomorphic Encryption

The decryption circuit $\tilde{F}_{\text{dec}}$ of DNH plays an important role in the construction of fully homomorphic encryption scheme based on DNH. If r_{refresh} is too large, condition (3-13) may not hold, and fully homomorphic encryption will not be achieved. Even if r_{refresh} is set such that fully homomorphic encryption can be achieved, the larger r_{refresh} is, the more frequent r_{refresh} need to be executed, and the lower efficient the fully homomorphic encryption will be. Thus it is worthwhile to reduce r_{refresh} .

3.4.1 First Optimization

By the definition (3-12), and the noise norm inequality (3-7), we know that the value of r_{refresh} is determined by the complexity of $\tilde{F}_{\text{dec}}$, where the complexity means the number of gates, especially the multiplicative gates, included in $\tilde{F}_{\text{dec}}$. Therefore, to reduce the size of r_{refresh} , we need to reduce the complexity of $\tilde{F}_{\text{dec}}$.

We have two techniques to reduce the complexity of $\tilde{F}_{\text{dec}}$, one is to decrease the number of the gates, the other is to replace the a multiplicative gate by a few additive gates. The first technique is usually executed in algorithm level, i.e., we optimize the decryption algorithm, then the gate number of he corresponding decryption circuit will be reduced. The second technique is usually done in circuit level, i.e., we can modify the structure of the circuit, including the input, such that we can use a few additive gates to replace a multiplicative gate, finally reduce the complexity of decryption circuit.

Gentry proposed a method to squash the decryption circuit[12], which is a optimization in algorithm level, he use hardness assumption of spares subset problem,

reduce the complexity of the decryption algorithm, and therefore reduce the complexity of decryption circuit. In the following, we propose an optimization in the circuit level, this technique does not rely on hardness of sparse subset problem, it can be used to optimize any fully homomorphic encryption scheme.

Let $\tilde{F}_{\text{dec}}$ be the decryption circuit with inputs $\mu_1, \dots, \mu_l, s_1, \dots, s_k$, where μ_i is the encryption of $\mu[i]$, the i -th bit in the binary representation of the ciphertext μ , s_i is the encryption of $s[i]$, the i -th bit in the binary representation of secret key sk . We can write $\tilde{F}_{\text{dec}}$ as a symbol multivariate polynomial about $\mu_1, \dots, \mu_l, s_1, \dots, s_k$, i.e.,

$$\tilde{F}_{\text{dec}}(\mu_1, \dots, \mu_l, s_1, \dots, s_k) = \sum_i (\mu_1, \dots, \mu_l)^{\mathbf{x}_i} \cdot (s_1, \dots, s_k)^{\mathbf{y}_i} \quad (3-14)$$

where $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{il}) \in \{0, 1\}^l$ and $\mathbf{y}_i = (y_{i1}, y_{i2}, \dots, y_{ik}) \in \{0, 1\}^k$ are respectively the monomial order of $(\mu_1, \dots, \mu_l)^{\mathbf{x}_i}$ and $(s_1, \dots, s_k)^{\mathbf{y}_i}$, and

$$\begin{aligned} (\mu_1, \dots, \mu_l)^{\mathbf{x}_i} &= \mu_1^{x_{i1}} \cdot \mu_2^{x_{i2}} \dots \mu_l^{x_{il}} \\ (s_1, \dots, s_k)^{\mathbf{y}_i} &= s_1^{y_{i1}} \cdot s_2^{y_{i2}} \dots s_l^{y_{ik}} \end{aligned} \quad (3-15)$$

Let $\|\mathbf{x}_i\|_1$, $\|\mathbf{y}_i\|_1$ denote the l^1 -norm, the total degree of $\tilde{F}_{\text{dec}}$ is defined as $\|\mathbf{x}_i\|_1 + \|\mathbf{y}_i\|_1$.

From the definition of r_{refresh} (3-12), and the noise norm inequalities (3-7), we know r_{refresh} is mainly bounded by the total degree of $\tilde{F}_{\text{dec}}$. So our purpose is to reduce the total degree of $\tilde{F}_{\text{dec}}$, and we reduce it in two steps shown as follows:

- In the step of initialization: fix a positive integer D_1 , for all i appears in (3-14), suppose there exist and only exist d_1 $\mathbf{y}_i$, such that $\|\mathbf{y}_i\| > D_1$, w.l.g., let these $\mathbf{y}_i$ be $\mathbf{y}_1, \dots, \mathbf{y}_{d_1}$. Then for $j = 1, 2, \dots, d_1$, we set

$$s_{k+j} = \text{DNH.Enc}((sk[1], sk[2], \dots, sk[k])^{\mathbf{y}_j}, sk)$$

where k is the length of the binary representation of secret key sk , $sk[i]$ is its i -th bit. We view these s_{k+j} , $j = 1, \dots, d_1$ as new variables, then the decryption circuit $\tilde{F}_{\text{dec}}$ is reduce to

$$\tilde{F}'_{\text{dec}}(\mu_1, \dots, \mu_l, s_1, \dots, s_{k+d_1}) = \sum_i (\mu_1, \dots, \mu_l)^{\mathbf{x}_i} \cdot (s_1, \dots, s_{k+d_1})^{\mathbf{y}'_i} \quad (3-16)$$

where $\mathbf{y}'_i = (y_{i1}, y_{i2}, \dots, y_{i_{k+d_1}}) \in \{0, 1\}^{k+d_1}$, and for all i , $\|\mathbf{y}'_i\| < D_1$.

- In the step of refreshing: fix a positive integer D_2 , for all i appears in (3-16), suppose there exists and only exists d_2 $\mathbf{x}_i$, such that $\|\mathbf{x}_i\| > D_2$, w.l.g., let these

$\mathbf{x}_i$ be $\mathbf{x}_1, \dots, \mathbf{x}_{d_2}$. Then for $j = 1, 2, \dots, d_2$, we set

$$\mu_{l+j} = \text{DNH.Enc}((\mu[1], \mu[2], \dots, \mu[l])^{\mathbf{x}_j}, sk)$$

where l is the length of the binary representation of the ciphertext μ , $\mu[i]$ is its i -th bit. We view these $\mu_{l+j}, j = 1, \dots, d_2$ as new variables, then the decryption circuit $\tilde{F}_{\text{dec}}$ is reduce to

$$\tilde{F}_{\text{dec}}''(\mu_1, \dots, \mu_{l+d_2}, s_1, \dots, s_{k+d_1}) = \sum_i (\mu_1, \dots, \mu_{l+d_2})^{\mathbf{x}'_i} \cdot (s_1, \dots, s_{k+d_1})^{\mathbf{y}'_i} \quad (3-17)$$

where $\mathbf{x}'_i = (x_{i_1}, x_{i_2}, \dots, x_{i_{l+d_2}}) \in \{0, 1\}^{l+d_2}$, and for all i , $\|\mathbf{x}'_i\| < D_2$.

This optimization can be viewed as a pre-evaluating for a high degree monomial in the decryption circuit. In first step, since it is executed by the system, who owns the secret key, it can pre-evaluate a monomial $(sk[1], sk[2], \dots, sk[k])^{\mathbf{y}_j}$, with $\|\mathbf{y}_j\| > D_1$ and then encrypt it to sk_{k+j} and output it as a part of the evaluate key. The small the D_1 is set, the lower degree of the monomials about $sk[1], sk[2], \dots, sk[k]$ will have, but the cost is that the number of evaluate key increases. So, the system need to balance on the setting of D_1 . In the second step, which is usually executed by a third party, since the third party owns the ciphertext μ that to be refreshed, so he can pre-evaluate a monomial $(\mu[1], \mu[2], \dots, \mu[l])^{\mathbf{x}_j}$, with $\|\mathbf{x}_j\| > D_2$ and then encrypt it to μ_{l+j} for use in the evaluation. Since these μ_{l+j} are only stored and used temporarily, so the third party can even pre-evaluate all monomial $(\mu[1], \mu[2], \dots, \mu[l])^{\mathbf{x}_j}$, with $\|\mathbf{x}_j\| > 2$, i.e., he can set $D_2 = 2$ without worrying about the increasing in memory cost. After the two steps, the total degree of the decryption circuit is reduced to at most $D_1 + D_2$.

Finally, we give a proof for the correctness of our optimization scheme.

Proof: We only need to proof that for all μ and sk , it satisfies

$$\text{DNH.Dec}(\tilde{F}_{\text{dec}}(\mu_1, \dots, \mu_l, s_1, \dots, s_k), sk) = \text{DNH.Dec}(\tilde{F}_{\text{dec}}''(\mu_1, \dots, \mu_{l+d_2}, s_1, \dots, s_{k+d_1}), sk)$$

On the left,

$$\begin{aligned}
 & \text{DNH.Dec}(\tilde{F}_{\text{dec}}(\mu_1, \dots, \mu_l, s_1, \dots, s_k), sk) \\
 &= \text{DNH.Dec}(\sum_i (\mu_1, \dots, \mu_l)^{\mathbf{x}_i} \cdot (s_1, \dots, s_k)^{\mathbf{y}_i}, sk) \\
 &= \sum_i [(\text{DNH.Dec}(\mu_1, sk), \dots, \text{DNH.Dec}(\mu_l, sk))^{\mathbf{x}_i} \\
 &\quad \cdot (\text{DNH.Dec}(s_1, sk), \dots, \text{DNH.Dec}(s_k, sk))^{\mathbf{y}_i}] \\
 &= \sum_i (\mu[1], \dots, \mu[l])^{\mathbf{x}_i} \cdot (sk[1], \dots, sk[k])^{\mathbf{y}_i} \\
 &= F_{\text{dec}}(\mu[1], \dots, \mu[l], sk[1], \dots, sk[k]) \\
 &= \text{DNH.Dec}(\mu, sk)
 \end{aligned}$$

On the right,

$$\begin{aligned}
 & \text{DNH.Dec}(\tilde{F}_{\text{dec}}''(\mu_1, \dots, \mu_{l+d_2}, s_1, \dots, s_{k+d_1}), sk) \\
 &= \text{DNH.Dec}(\sum_i (\mu_1, \dots, \mu_{l+d_1})^{\mathbf{x}'_i} \cdot (s_1, \dots, s_{k+d_2})^{\mathbf{y}'_i}, sk) \\
 &= \sum_i [(\text{DNH.Dec}(\mu_1, sk), \dots, \text{DNH.Dec}(\mu_{l+d_1}, sk))^{\mathbf{x}'_i} \\
 &\quad \cdot (\text{DNH.Dec}(s_1, sk), \dots, \text{DNH.Dec}(s_{k+d_2}, sk))^{\mathbf{y}'_i}]
 \end{aligned}$$

note that

$$(1) \ s_{k+j} = \text{DNH.Enc}((sk[1], sk[2], \dots, sk[k])^{\mathbf{y}_j}, sk), j = 1, \dots, d_1.$$

$$(2) \text{ For all } i, \mathbf{y}'_i[j] = \mathbf{y}_i[j], j = 1, \dots, k.$$

$$(3) \text{ For } i = 1, \dots, d_1,$$

$$\begin{cases} \mathbf{y}'_i[k+j] = 1, j = i. \\ \mathbf{y}'_i[k+j] = 0, \text{ otherwise.} \end{cases}$$

$$(4) \text{ For } i > d_1, \mathbf{y}'_i[k+j] = 0, j = 1, \dots, d_1.$$

and note that

$$(1) \ \mu_{l+j} = \text{DNH.Enc}((\mu[1], \mu[2], \dots, \mu[l])^{\mathbf{x}_j}, sk), j = 1, \dots, d_2.$$

$$(2) \text{ For all } i, \mathbf{x}'_i[j] = \mathbf{x}_i[j], j = 1, \dots, l.$$

$$(3) \text{ For } i = 1, \dots, d_2,$$

$$\begin{cases} \mathbf{x}'_i[l+j] = 1, j = i. \\ \mathbf{x}'_i[l+j] = 0, \text{ otherwise.} \end{cases}$$

$$(4) \text{ For } i > d_2, \mathbf{x}'_i[k+j] = 0, j = 1, \dots, d_2.$$

then $\text{DNH.Dec}(\tilde{F}_{\text{dec}}''(\mu_1, \dots, \mu_{l+d_2}, s_1, \dots, s_{k+d_1}), sk)$ is finally equal to

$$\begin{aligned} & \sum_i (\mu[1], \dots, \mu[l])^{\mathbf{x}_i} \cdot (sk[1], \dots, sk[k])^{\mathbf{y}_i} \\ &= F_{\text{dec}}(\mu[1], \dots, \mu[l], sk[1], \dots, sk[k]) \\ &= \text{DNH.Dec}(\mu, sk) \end{aligned}$$

□

3.4.2 Second Optimization

The first optimization described above indeed reduces the complexity of the decryption circuit via adding new evaluate keys. The complexity of the decryption circuit is mainly determined by the total degree of the multivariate polynomial about the secret key bits $sk[1], sk[2], \dots, sk[k]$. By setting a monomial $(sk[1], sk[2], \dots, sk[k])^{\mathbf{y}_j}$ with its multi-degree greater than D_1 to be a new variables, we can reduce the total degree of the decryption polynomial to at most degree D_1 . While the cost is that we need to encrypt these new variables and output them as new evaluate keys. Our second optimization aim at using less new variables to reduce the total degree of the decryption polynomial.

Given a polynomial $f(x_1, x_2, x_3) = x_1x_2x_3 + x_1x_3x_4 + x_1x_3x_5 + x_1x_3 + x_2x_4$, whose total degree is 3. By the first optimization, if we set

$$x_6 = x_1x_2x_3$$

$$x_7 = x_1x_3x_4$$

$$x_8 = x_1x_3x_5$$

then the polynomial will be reduced to

$$f'(x_1, x_2, \dots, x_8) = x_1x_3 + x_2x_4 + x_6 + x_7 + x_8,$$

with total degree 2, and the cost is we have 3 new variables. But if we set

$$x_6 = x_1x_3$$

then the polynomial will be reduced to

$$f''(x_1, x_2, \dots, x_8) = x_6 + x_2x_4 + x_2x_6 + x_4x_6 + x_5x_6,$$

which is also with total degree 2, but we only need to add one new variable.

Based on the above discussion, we propose the second optimization scheme as follows:

- In the step of initialization: fix a positive integer D_1 , for all i appears in (3-14), suppose there exists and only exists d_1 $\mathbf{y}_i$, such that $\|\mathbf{y}_i\| > D_1$, w.l.g., let these $\mathbf{y}_i$ be $\mathbf{y}_1, \dots, \mathbf{y}_{d_1}$, and set $\mathcal{D} = \{\mathbf{y}_1, \dots, \mathbf{y}_{d_1}\}$. Then find out a set $\mathcal{D}' = \{\mathbf{y}'_1, \dots, \mathbf{y}'_{d'_1}\}$ with $d'_1 \leq d_1$ such that for all $\mathbf{y}_i \in \mathcal{D}$, there exists some $\mathbf{y}'_j \in \mathcal{D}'$ and $\bar{\mathbf{y}}'_{i,j} \in \{0, 1\}^k$ with $\|\bar{\mathbf{y}}'_{i,j}\| < D_1 - 1$ and $\mathbf{y}_i = \bar{\mathbf{y}}'_{i,j} + \mathbf{y}'_j$.

For $i = 1, 2, \dots, d_1$, let

$$\mathbf{z}_i = \bar{\mathbf{y}}'_{i,j} \| \mathbf{0} + \mathbf{I}_j \in \{0, 1\}^{k+d'_1}$$

where $\bar{\mathbf{y}}'_{i,j} \| \mathbf{0}$ denotes padding d'_1 0 after $\bar{\mathbf{y}}'_{i,j}$, and $\mathbf{I}_j$ is the j -th identity vector in $\mathbb{Z}^{k+d'_1}$.

For other i appears in (3-14), let

$$\mathbf{z}_i = \mathbf{y}_i \| \mathbf{0} \in \{0, 1\}^{k+d'_1}$$

Then for $j = 1, 2, \dots, d'_1$, we set

$$s_{k+j} = \text{DNH.Enc}((sk[1], sk[2], \dots, sk[k])^{\mathbf{z}_j}, sk)$$

where k is the length of the binary representation of secret key sk , $sk[i]$ is its i -th bit. We view these $s_{k+j}, j = 1, \dots, d'_1$ as new variables, then the decryption circuit $\tilde{F}_{\text{dec}}$ is reduce to

$$\tilde{F}'_{\text{dec}}(\mu_1, \dots, \mu_l, s_1, \dots, s_{k+d_1}) = \sum_i (\mu_1, \dots, \mu_l)^{\mathbf{x}_i} \cdot (s_1, \dots, s_{k+d'_1})^{\mathbf{z}_i} \quad (3-18)$$

where $\mathbf{z}_i = (z_{i_1}, z_{i_2}, \dots, z_{i_{k+d'_1}}) \in \{0, 1\}^{k+d'_1}$, and for all i , $\|\mathbf{z}_i\| < D_1$.

- In the step of refreshing: fix a positive integer D_2 , for all i appears in (3-16), suppose there exists and only exists d_2 $\mathbf{x}_i$, such that $\|\mathbf{x}_i\| > D_2$, w.l.g., let these $\mathbf{x}_i$ be $\mathbf{x}_1, \dots, \mathbf{x}_{d_2}$. Then for $j = 1, 2, \dots, d_2$, we set

$$\mu_{l+j} = \text{DNH.Enc}((\mu[1], \mu[2], \dots, \mu[l])^{\mathbf{x}_j}, sk)$$

where l is the length of the binary representation of the ciphertext μ , $\mu[i]$ is its i -th bit. We view these $\mu_{l+j}, j = 1, \dots, d_2$ as new variables, then the decryption circuit $\tilde{F}_{\text{dec}}$ is reduce to

$$\tilde{F}''_{\text{dec}}(\mu_1, \dots, \mu_{l+d_2}, s_1, \dots, s_{k+d_1}) = \sum_i (\mu_1, \dots, \mu_{l+d_2})^{\mathbf{x}'_i} \cdot (s_1, \dots, s_{k+d_1})^{\mathbf{y}'_i} \quad (3-19)$$

where $\mathbf{x}'_i = (x_{i_1}, x_{i_2}, \dots, x_{i_{l+d_2}}) \in \{0, 1\}^{l+d_1}$, and for all i , $\|\mathbf{x}'_i\| < D_2$.

The only difference between the first optimization and second optimization is that for the set $\mathcal{M}$ of all monomials with degree great or equal to D_1 , the first optimization directly sets each of them as a new variables, while the second optimization first finds a another set $\mathcal{M}'$ of monomials such that for any $\mathbf{s}^y \in \mathcal{M}$, it can be write as $\mathbf{s}^y = \mathbf{s}^{y'} \cdot \mathbf{s}^{y''}$, where $\mathbf{s}^{y'} \in \mathcal{M}'$ and the degree of $\mathbf{s}^{y'}$ less than $D_1 - 1$, and then the sets each monomial in $\mathcal{M}'$ as a new variables. In general, we have method to find a $\mathcal{M}'$ such that it has less monomials, but how to find such a set that has least monomial seems not a easy problem. In fact this problem is related (but not equivalent) to the minimum set cover problem[50], which is NP-complete. We give a formal description of problem that our second optimization related to as follows:

Definition 3.5 ((Minimum) Monomial Set Cover (MSC) problem) Let $\alpha = (a_1, \dots, a_n)$ be a n -tuple of variables over a field, for any $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{Z}_{\geq 0}^n$, let $\alpha^{\mathbf{x}} = a_1^{x_1} \cdots a_n^{x_n}$. Given a set of monomials $\mathcal{M} = \{\alpha^{\mathbf{x}^i}, i = 1, \dots, l\}$, find another set $\mathcal{M}' = \{\alpha^{\mathbf{x}'^i}, i = 1, \dots, l'\}$ satisfying:

- i) $d' \leq d$.
- ii) For any $\alpha^{\mathbf{x}^i} \in \mathcal{M}$, there exists $\alpha^{\mathbf{x}'^i} \in \mathcal{M}'$ and a monomial $\alpha^{\mathbf{x}''^i}$ whose multidegree less than or equal to d , such that $\alpha^{\mathbf{x}^i} = \alpha^{\mathbf{x}'^i} \cdot \alpha^{\mathbf{x}''^i}$.

Parameterized by n, l, d , we denote this problem as $\text{MSC}_{n,l,d}$.

Obviously, the set $\mathcal{M}$ itself is a solution for SMLC problem, but this solution is trivial. How to find a solution $\mathcal{M}'$ with the minimum elements is not an easy problem, it is even not known whether this problem is in NP. But we propose a algorithm for computing a feasible solution as follows:

Algorithm 3.6 (Algorithm for MSC problem) Input: $\mathcal{M} = \{\alpha^{\mathbf{x}^i} = (a_1, \dots, a_n)^{\mathbf{x}^i}, i = 1, \dots, l\}$, $\mathcal{M}' = \emptyset$.

- 1: Choose two different monomials $\alpha^{\mathbf{x}^i}, \alpha^{\mathbf{x}^j} \in \mathcal{M}$, compute

$$\alpha^{\mathbf{y}} = \text{GCD}(\alpha^{\mathbf{x}^i}, \alpha^{\mathbf{x}^j}).$$

- 2: If $\deg(\alpha^{\mathbf{x}^i}/\alpha^{\mathbf{y}}) < D$ and $\deg(\alpha^{\mathbf{x}^j}/\alpha^{\mathbf{y}}) < D$, then $\mathcal{M}' \leftarrow \mathcal{M}' \cup \{\alpha^{\mathbf{y}}\}$; $\mathcal{M} \leftarrow \mathcal{M} \setminus \{\alpha^{\mathbf{x}^i}, \alpha^{\mathbf{x}^j}\}$.

3: Repeat step 1, until all $\alpha^{x_i}, \alpha^{x_j}$ have been enumerated.

OUTPUT: $\mathcal{M}' \cup \mathcal{M}$.

3.5 KDM Security or Secret Key Sequence

In the definition of generalized fully homomorphic encryption, it requires that the equivalent homomorphic $\tilde{F}'$ must not involve the secret key s , but the refresh circuit indeed takes as inputs the ciphertext of the secret key bits. Therefore we must suppose that the encryption of sk using the sk itself as the secret key can not leak the information of sk , this is so called key dependant message (KDM) security. The formal definition of KDM security is as follows:

Definition 3.7 A public-key encryption scheme is KDM- $\mathcal{F}$ secure (where $\mathcal{F}$ is a class of functions), if it is secure even against an adversary who is given public keys $pk_1, \dots, pk_n$ and has access to encryptions of key dependent messages $f(sk_1, \dots, sk_n)$ for adaptively selected functions $f \in \mathcal{F}$.

The decryption circuit is executed by a third party, which are assumed to be untrusted. He can directly access the encryption of the secret key itself, so it has potential dangers if the DNH is not proven with KDM security.

unfortunately, all the proposed dual-noise type homomorphic encryption schemes we introduce in chapter 2 are not KDM secure. But we can use a new different secret key to encryption the prior secret key, we call this method the key-sequence bootstrapping, which is first proposed in [12], and is adopted in all homomorphic encryption schemes using bootstrapping.

Key-sequence bootstrapping requires the following modifications to DNH encryption scheme.

- **DNH.KeyGen***: takes as input the system and security parameter, output the public-secret key pairs sequence as:

$$((pk_0, sk_0), (pk_1, sk_1), \dots, (pk_l, sk_l)) \xleftarrow{\$} \text{DNF.KeyGen}$$

and the evaluate key $vk_i = \{v_{i1}, v_{i2}, \dots, v_{il}\}, i = 0, \dots, l-1$, such that

$$v_{ij} \leftarrow \text{DNH.Enc}(sk_i[j], pk_{i+1})$$

where $sk_i[j]$ is the j -th bit of the binary representation of sk_i .

- **DNH.Enc***: takes as input a plaintext $b \in \{0, 1\}$ and public key pk_0 , output the ciphertext as $(w, 0)$, such that $w \leftarrow \text{DNH.Enc}(b, pk_0)$.
- **DNH.Dec***: takes as input a ciphertext $(\mu, k(< l))$ and secret key sk_k , outputs $\varphi(w)$, such that $w \leftarrow \text{DNH.Dec}(\mu, sk_k)$.
- **DNH.Add***: takes as input two ciphertext (μ_1, t) and (μ_2, t) . Output (μ_3, t) , such that $\mu_3 \leftarrow \text{DNH.Add}(\mu_1, \mu_2)$.
- **DNH.Mult***: takes as input two ciphertext (μ_1, t) and (μ_2, t) . Output (μ_4, t) , such that $\mu_4 \leftarrow \text{DNH.Mult}(\mu_1, \mu_2)$.
- **DNH.Refresh***: takes as input a ciphertext (μ, t) and the refresh key vk_t , output $(\mu', t + 1)$, such that

$$\mu' = \text{Refresh}(\mu, vk_t) = \tilde{F}_{\text{dec}}(\mu_1, \dots, \mu_l, v_{t1}, \dots, v_{tk})$$

where μ_i is the encryption of the binary bit $\mu[i]$ of μ under the public key pk_t .

Chapter 4 Fully homomorphic Encryption scheme without Bootstrapping

4.1 polly cracker – the first fully homomorphic encryption scheme

In 1993 Kobitz and Fellows [3] proposed a public key encryption scheme, Polly Cracker, based on the problem of solving systems of multivariate polynomial equations. Due to its inefficiency and some security problems, Polly Cracker can not be used in practice, but with its fully homomorphic property, Polly Cracker shows us some hint in constructing fully homomorphic encryption scheme without using bootstrapping. In this section we give a brief description of Polly Cracker.

The public key encryption scheme, Polly Cracker, is formally defined as a 3-tuple $\text{PollyCracker} = \{\text{PollyCracker.KeyGen}, \text{PollyCracker.Enc}, \text{PollyCracker.Dec}\}$, which is described as follows:

- **PollyCracker.KeyGen:** Randomly choose a point (a vector) $s \in \mathbb{F}^n$, where $\mathbb{F}$ is a finite field. Randomly choose a set $f_i \in \mathbb{F}[x_1, x_2, \dots, x_n], i = 1, \dots, m$, such that $f_i(s) = 0$. Where $x_1, x_2, \dots, x_n$ are n symbols variables and $\mathbb{F}[x_1, x_2, \dots, x_n]$ denotes the ring of multivariate polynomial about $x_1, x_2, \dots, x_n$ over $\mathbb{F}$. Then output s as the the secret key, and $pk = \{f_1, \dots, f_m\}$ as the public key.
- **PollyCracker.Enc:** Given a plaintext $b \in \{0, 1\}$, randomly choose $h \in \langle f_1, \dots, f_m \rangle$, output $c = b + h$ as the ciphertext.
- **PollyCracker.Dec:** Given a ciphertext $c \in \mathbb{F}[x_1, x_2, \dots, x_n]$, evaluate c at s , and output $b = c(s) \bmod 2$ as the plaintext.

Correctness & homomorphic property:

For any c outputted by PollyCracker.Enc , we have

$$c - b \in \langle f_1, \dots, f_m \rangle$$

so $(c - b)(s) = 0$, and it follows that $b = c(s) \bmod 2$.

For any $c_1, \dots, c_l$, with $c_i \leftarrow \text{PollyCracker.Enc}(pk, b_i)$, we have

$$c_i \in b_i + \langle f_1, \dots, f_m \rangle,$$

for an arbitrary symbol polynomial F , by the definition of addition and multiplication of coset, we have

$$F(c_1, \dots, c_l) \in F(b_1, \dots, b_l) + \langle f_1, \dots, f_m \rangle,$$

finally, we have

$$\text{GB.PollyCracker.Dec}(sk, F(c_1, \dots, c_l)) = F(c_1, \dots, c_l)(s) \bmod 2 = F(b_1, \dots, b_l) \bmod 2.$$

by the definition (3.2), it follows that **PollyCracker** is perfect fully homomorphic.

The security of the basic Polly Cracker encryption scheme is partly based on the hardness of solving multivariate polynomial equations over finite field. This problem is proved to be a NP-complete problem even when these polynomials are all quadratic, in which case, it is called MQ (Multivariate Quadratic equations) problem.

But its security is not equivalent to hardness of solving multivariate polynomial equations over finite field, since an attacker may recover the plaintext from the ciphertext without using secret key. For example, given a ciphertext c , if the attacker can correctly decide that c is a member of the ideal $\langle f_1, \dots, f_m \rangle$, then he knows that the plaintext is 0, and in general, if the attacker can correctly decide that c is in the coset $b + \langle f_1, \dots, f_m \rangle$, then he knows that the plaintext is b .

There also exists a simple chosen-ciphertext attack to basic Polly Cracker encryption scheme, the attacker just need to inquire a ciphertext $h + x_i, i = 1, \dots, m$, then the responded plaintext is the value of $s[i]$, i.e. the i -th component of the secret key s .

Since the basic Polly Cracker has many security problems, researchers tried to modified it, before we introduce these modifications, we first introduce the background about multivariate polynomial ring.

4.2 Preliminaries in Multivariate Polynomial Ring

4.2.1 Monomial, Polynomial, and Division

Definition 4.1 (Monomial) Let $x_1, x_2, \dots, x_n$ be n symbols, a monomial in $x_1, x_2, \dots, x_n$ is a product of the form

$$x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdots x_n^{\alpha_n}$$

where all exponents $\alpha_1, \dots, \alpha_n \in \mathbb{Z}_{\geq 0}$ are nonnegative integers. The total degree of this monomial is defined as $\alpha_1 + \dots + \alpha_n$. Let $x = (x_1, \dots, x_n)$ and $\alpha = (\alpha_1, \dots, \alpha_n)$,

we can simply denote this monomial by

$$x^\alpha = x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdots x_n^{\alpha_n}$$

and the total degree of x^α is denoted by $\|\alpha\| = \alpha_1 + \cdots + \alpha_n$.

Definition 4.2 (Polynomial) Let $x_1, x_2, \dots, x_n$ be n symbols, a polynomial f in $x_1, x_2, \dots, x_n$ with coefficients in K is a finite linear combination (with coefficients in K) of monomials, and we can write f in the form

$$f = \sum_{\alpha \in \mathcal{A}} a_\alpha x^\alpha, \quad a_\alpha \in K,$$

where $\mathcal{A}$ is a finite subset of $\mathbb{Z}_{\geq 0}^n$. The set of all polynomial in $x_1, x_2, \dots, x_n$ with coefficients in K is denoted as $K[x_1, x_2, \dots, x_n]$.

Definition 4.3 Let $f = \sum_{\alpha} a_\alpha x^\alpha$ be a polynomial in $K[x_1, x_2, \dots, x_n]$.

- (i) If $a_\alpha \neq 0$, we call $a_\alpha x^\alpha$ a term of f .
- (ii) We call a_α the coefficient of the term $a_\alpha x^\alpha$ (the monomial x^α).
- (iii) The total degree of f , denoted $\deg f$ is defined by the maximum $\|\alpha\|$ such that $a_\alpha \neq 0$.

Definition 4.4 (Monomial Order) A monomial order $>$ on $K[x_1, x_2, \dots, x_n]$ is any relation $>$ on $\mathbb{Z}_{\geq 0}^n$, or equivalently, any relation on the set of monomials $x^\alpha, \alpha \in \mathbb{Z}_{\geq 0}^n$, satisfying:

- (i) $>$ is a total (or linear) order on $\mathbb{Z}_{\geq 0}^n$, i.e., for arbitrary $\alpha, \beta \in \mathbb{Z}_{\geq 0}^n$, $\alpha > \beta$ or $\beta > \alpha$.
- (ii) If $\alpha > \beta$, then for $\gamma \in \mathbb{Z}_{\geq 0}^n$, $\alpha + \gamma > \beta + \gamma$.
- (iii) $>$ is a well-ordering on $\mathbb{Z}_{\geq 0}^n$, i.e., for arbitrary nonempty subset of $\mathbb{Z}_{\geq 0}^n$, there exists a smallest element under $>$.

We introduce some monomial orders as follows.

Definition 4.5 (Lexicographic Order) Let $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n) \in \mathbb{Z}_{\geq 0}^n$. We say $\alpha >_{lex} \beta$ or equivalently say $x^\alpha >_{lex} x^\beta$, if in the vector $\alpha - \beta \in \mathbb{Z}_{\geq 0}^n$, the leftmost nonzero entry is positive.

Definition 4.6 (Reverse Lex Order) Let $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n) \in \mathbb{Z}_{\geq 0}^n$. We say $\alpha >_{\text{revlex}} \beta$ or equivalently say $x^\alpha >_{\text{revlex}} x^\beta$, if in the vector $\alpha - \beta \in \mathbb{Z}_{\geq 0}^n$, the rightmost nonzero entry is negative.

Definition 4.7 (Graded Lex Order) Let $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n) \in \mathbb{Z}_{\geq 0}^n$. We say $\alpha >_{\text{grlex}} \beta$ or equivalently say $x^\alpha >_{\text{grlex}} x^\beta$, if

$$\|\alpha\| > \|\beta\|, \text{ or } \|\alpha\| = \|\beta\| \text{ and } \alpha >_{\text{lex}} \beta$$

Definition 4.8 (Graded Reverse Lex Order) Let $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n) \in \mathbb{Z}_{\geq 0}^n$. We say $\alpha >_{\text{grevlex}} \beta$ or equivalently say $x^\alpha >_{\text{grevlex}} x^\beta$, if

$$\|\alpha\| > \|\beta\|, \text{ or } \|\alpha\| = \|\beta\| \text{ and } \alpha >_{\text{revlex}} \beta$$

Property 4.9 $>_{\text{lex}}, >_{\text{revlex}}, >_{\text{grlex}}, >_{\text{grevlex}}$ are all monomial orders.

Definition 4.10 Let $f = \sum_{\alpha} a_{\alpha} x^{\alpha}$ be a nonzero polynomial in $K[x_1, x_2, \dots, x_n]$ and let $>$ be a monomial order.

(i) The multidegree of f is

$$\text{multideg}(f) = \max\{\alpha \in \mathbb{Z}_{\geq 0}^n, a_{\alpha} \neq 0\}$$

with the maximum taken with respect to $>$.

(ii) The leading coefficient of f is

$$LC(f) = a_{\text{multideg}(f)} \in K$$

(iii) The leading monomial of f is

$$LM(f) = x^{\alpha_{\text{multideg}(f)}}$$

(iv) The leading term of f is

$$LT(f) = LC(f) \cdot LM(f)$$

Having the above definitions, we can define division in $K[x_1, x_2, \dots, x_n]$.

Definition 4.11 Fix monomial order $>$ on $\mathbb{Z}_{\geq 0}^n$, given a polynomial f and a nonzero polynomial g in $K[x_1, x_2, \dots, x_n]$, the quotient and remainder of f divided by

g is respectively defined as the unique $a, r \in K[x_1, x_2, \dots, x_n]$, such that

$$f = a \cdot g + r,$$

and either $r = 0$ or the monomials of r can not be divisible by $LT(g)$, and furthermore, if $a \neq 0$,

$$\text{multideg}(f) \geq \text{multideg}(a \cdot g).$$

Theorem 4.12 Fix a monomial order on $\mathbb{Z}_{\geq 0}^n$, let $F = (f_1, \dots, f_s)$ be an ordered s -tuple of nonzero polynomials in $K[x_1, x_2, \dots, x_n]$. Then every $f \in K[x_1, x_2, \dots, x_n]$ can be written as

$$f = a_1 f_1 + \dots + a_s f_s + r,$$

where $a_i, r \in K[x_1, x_2, \dots, x_n]$, and either $r = 0$ or the monomials of r can not be divisible by any of $LT(f_1), \dots, LT(f_s)$. we call r a remainder of f divided by F . Furthermore, if $a_i \neq 0$,

$$\text{multideg}(f) \geq \text{multideg}(a_i \cdot f_i).$$

Note that the remainder of f divided by F is not unique.

Example 4.13 Let $f = xy^2 - x$, and $f_1 = xy + 1, f_2 = y^2 - 1$, then we check that

$$f = y \cdot f_1 + 0 \cdot f_2 + (-x - y)$$

and

$$f = 0 \cdot f_1 + x \cdot f_2 + 0$$

The second equation shows that $f \in \langle f_1, f_2 \rangle$, but from the first equation, we can not decide whether f is a member of the ideal $\langle f_1, f_2 \rangle$.

4.2.2 Ideal and Groebner Basis

We consider the ideal membership problem, i.e., given an ideal I of $K[x_1, x_2, \dots, x_n]$, how can we decide an element $f \in K[x_1, x_2, \dots, x_n]$ in I or not. We first introduce the Hilbert Basis Theorem.

Theorem 4.14 Every ideal $I \subset K[x_1, x_2, \dots, x_n]$, has a finite generating set, i.e., $I = \langle g_1, \dots, g_t \rangle$ for some $g_1, \dots, g_t \in I$, and we call $g_1, \dots, g_t$ a basis of I .

The basis of an ideal $I \subset K[x_1, x_2, \dots, x_n]$ is usually not unique, and among these bases, a special kind of bases have interesting properties.

Definition 4.15 (Groebner Basis) Fix a monomial order, a finite set $G = \{g_1, \dots, g_t\}$ of an ideal I is said to be a Groebner basis (or standard basis) if

$$\langle LT(g_1), \dots, LT(g_t) \rangle = \langle LT(I) \rangle,$$

where $LT(I)$ denotes the set of leading terms of elements in I , i.e.,

$$LT(I) = \{cx^\alpha : \exists f \in I \text{ with } LT(f) = cx^\alpha\},$$

and $\langle LT(I) \rangle$ denotes the ideal generated by the elements of $LT(I)$.

Theorem 4.16 Fix a monomial order, any nonzero ideal $I \in K[x_1, x_2, \dots, x_n]$ has a Groebner basis, and any Groebner basis for an ideal I is a basis of I .

This theorem guarantees the existence of a Groebner basis, but its constructing is a more complicate problem.

Definition 4.17 Let $f, g \in K[x_1, x_2, \dots, x_n]$ be nonzero polynomials.

- (i) Let $\text{multideg}(f) = \alpha$ and $\text{multideg}(g) = \beta$, let $\gamma = (\gamma_1, \dots, \gamma_n)$, where $\gamma_i = \max(\alpha_i, \beta_i)$ for each i . We call x^γ the least common multiple of $\text{LM}(f)$ and $\text{LM}(g)$, written as

$$x^\gamma = \text{LCM}(\text{LM}(f), \text{LM}(g)).$$

- (ii) The S-polynomial of f and g is

$$S(f, g) = \frac{x^\gamma}{\text{LT}(f)} \cdot f - \frac{x^\gamma}{\text{LT}(g)} \cdot g$$

Note that the leading coefficient has inverse since K is a field.

Theorem 4.18 (Buchberger's Criterion) Let I be a polynomial ideal. Then a basis $G = (g_1, \dots, g_t)$ of I is a Groebner basis of I if and only if for all pairs $(i, j), i \neq j$, the remainder of $S(g_i, g_j)$ divided by G (listed in some order) is zero. Fix a monomial order, any nonzero ideal $I \in K[x_1, x_2, \dots, x_n]$ has a Groebner basis, and any Groebner basis for an ideal I is a basis of I .

As a natural consequent of Buchberger's Criterion, Buchberger in 1965 gave an algorithm to construct a Groebner basis from any basis of an ideal. So far, the Buchberger family algorithms ([51],[52],[53]) are still the most effective algorithm. But the

complexity of Buchberger algorithm is hard to evaluate. It is generally believed that Buchberger algorithm runs in exponential space in n , when the variate number and the number of generators in a given basis are both round n . Although it is hard to drive a Groebner basis from a given basis of an ideal, we can easily write down or say create a Groebner basis via Macaula's trick[54].

Theorem 4.19 Let I be an ideal of $K[x_1, x_2, \dots, x_n]$, and $G = (g_1, \dots, g_t)$ be a Groebner basis of I , then for any $f \in K[x_1, x_2, \dots, x_n]$, $f \in I$ if and only if the remainder of f divided by G (listed in some order) is zero.

This theorem give us a method to solve the ideal membership decision problem.

4.3 Groebner basis Polly Cracker

The Groebner basis version Polly Cracker is a natural extended of the original Polly Cracker, it was present in [4]. It is usually referred to GB Polly Cracker and has be investigated in many times, such as in [18],[19]. The GB Polly Cracker public key encryption scheme is formally defined as a 3-tuple $\text{GB.PollyCracker} = \{\text{GB.PollyCracker.KeyGen}, \text{GB.PollyCracker.Enc}, \text{GB.PollyCracker.Dec}\}$, which is described as follows:

- $\text{GB.PollyCracker.KeyGen}(1^n)$: Takes as input the security parameter 1^n , runs the Groebner base generation algorithm $\text{GBGen}(1^n)$ to generate a Groebner basis $G = (g_1, \dots, g_t)$ with $g_i \in K[x_1, x_2, \dots, x_n]$, where $t = c \cdot n$ for a constant $c \geq 1$. Randomly chooses $f_1, \dots, f_t$, and set $f_i \leftarrow f_i - (f_i \bmod G)$ for $i = 1, \dots, t$. Then outputs $f_1, \dots, f_t$ as the public key, and keep G as the secret key.
- $\text{GB.PollyCracker.Enc}$: Takes as input a plaintext $b \in \{0, 1\}$, randomly choose $h \in \langle f_1, \dots, f_m \rangle$, output $c = b + h$ as the ciphertext.
- PollyCracker.Dec : Takes as input a ciphertext $c \in \mathbb{K}[x_1, x_2, \dots, x_n]$, and output $b = (c \bmod G) \bmod 2$ as the plaintext.

Correctness & homomorphic property:

For any c outputted by $\text{GB.PollyCracker.Enc}$, we have

$$c - b \in \langle f_1, \dots, f_m \rangle \subset \langle g_1, \dots, g_t \rangle,$$

so $(c - b) \bmod G = 0$, and since $b \in \{0, 1\}$, it follows that $b = (c \bmod G) \bmod 2$.

For any $c_1, \dots, c_l$, with $c_i \leftarrow \text{GB.PollyCracker.Enc}(pk, b_i)$, we have

$$c_i \in b_i + \langle f_1, \dots, f_m \rangle,$$

for an arbitrary symbol polynomial F , by the definition of addition and multiplication of coset, we have

$$F(c_1, \dots, c_l) \in F(b_1, \dots, b_l) + \langle f_1, \dots, f_m \rangle,$$

finally, we have

$$\text{GB.PollyCracker.Dec}(sk, F(c_1, \dots, c_l)) = F(b_1, \dots, b_l) \bmod 2.$$

by the definition (3.2), it follows that **GB.PollyCracker** is perfect fully homomorphic.

The security of GB Polly Cracker is partly based on the Grobener basis assumption [18], which is stated as follows:

Given a Groebner basis generation algorithm **GB.Gen**, the challenger generates a Groebner basis $G = (g_1, \dots, g_t) \xleftarrow{\$} \text{GB.Gen}$. The adversary inquires some samples in the distribution

$$\{f_i \leftarrow f_i - f_i \bmod G : f_i \xleftarrow{\$} \mathbb{K}[x_1, x_2, \dots, x_n]\}$$

and output a set G' . The hardness assumption is say that the advantage of any PPT adversary is negligible. But this hardness assumption is not strictly proven true, and several attacks to GB Polly Cracker seem effective [55].

4.4 GB Polly Cracker with affine/linear transformation

To repair the weakness of GB Polly Vcracker, we propose a modified scheme, which is referred to isomorphism GB Polly Cracker, or IGB Polly Cracker for short.

The public key encryption scheme of IGB Polly Cracker is formally defined as a 3-tuple $\text{IGB.PollyCracker} = \{\text{IGB.KeyGen}, \text{IGBT.Enc}, \text{IGB.Dec}\}$, which are described as follows:

- **IGB.KeyGen**(1^n): Takes as input the security parameter 1^n , runs the Groebner base generation algorithm **GBGen**(1^n) to generate a Groebner basis $G = (g_1, \dots, g_t)$ with $g_i \in K[x_1, x_2, \dots, x_n]$, where $t = c \cdot n$ for a constant $c \geq 1$. Randomly chooses an invertible matrix $\mathbf{A}_1 \in K^{n \times n}$, $\mathbf{A}_2 \in K^{t \times t}$, and a vector

$\mathbf{v} \in K^n$. Then sets

$$\mathbf{y} = \phi(\mathbf{x}) = \mathbf{x}\mathbf{A}_1 + \mathbf{v},$$

and

$$G' = (g'_1(\mathbf{x}), \dots, g'_t(\mathbf{x})) = (g_1(\mathbf{y}), \dots, g_t(\mathbf{y})) \cdot \mathbf{A}_2.$$

Outputs G' as the public key, and keeps G as the secret key.

- IGB.Enc: Takes as input a plaintext $b \in \{0, 1\}$, randomly chooses $h \in \langle g'_1, \dots, g'_t \rangle$, outputs $c = b + h$ as the ciphertext.
- IGB.Dec: Takes as input a ciphertext $c \in \mathbb{K}[x_1, x_2, \dots, x_n]$, sets

$$\mathbf{y} = \phi^{-1}(\mathbf{x}) = (\mathbf{x} - \mathbf{v})\mathbf{A}_1^{-1},$$

and $c'(\mathbf{x}) = c(\mathbf{y})$, outputs $b = (c' \bmod G) \bmod 2$ as the plaintext.

Correctness & homomorphic property:

For any c outputted by IGB.Enc, we have

$$c - b \in \langle g'_1, \dots, g'_t \rangle$$

and

$$c' - b = \phi^{-1}(c - b) \in \langle \phi^{-1}(g'_1), \dots, \phi^{-1}(g'_t) \rangle = \langle g_1, \dots, g_t \rangle.$$

It follows that $b = (c' \bmod G) \bmod 2$.

For any $c_1, \dots, c_l$, with $c_i \leftarrow \text{IGB.Enc}(pk, b_i)$, we have

$$c_i \in b_i + \langle g'_1, \dots, g'_t \rangle,$$

and

$$c'_i = \phi^{-1}(c_i) \in b + \langle \phi^{-1}(g'_1), \dots, \phi^{-1}(g'_t) \rangle = b + \langle g_1, \dots, g_t \rangle.$$

for an arbitrary symbol polynomial F , by the definition of addition and multiplication of coset, we have

$$F(c'_1, \dots, c'_l) \in F(b_1, \dots, b_l) + \langle g_1, \dots, g_t \rangle,$$

finally, we have

$$\begin{aligned} \text{IGB.Dec}(sk, F(c_1, \dots, c_l)) &= (F(c'_1, \dots, c'_l) \bmod G) \bmod 2 \\ &= F(b_1, \dots, b_l) \bmod 2. \end{aligned}$$

by the definition (3.2), it follows that IGB.PollyCracker is perfect fully homomorphic.

Security Analysis:

The security of IGB Polly Cracker does not rely on the Groebner basis hardness assumption, instead, it's security is partly based the hardness of Isomorphism of Polynomial problem (IP)[56],[57], which is as hard as Graph Isomorphism problem (GI), and is believed unlikely to be solve in polynomial time. The IP problem is stated as follows:

Definition 4.20 (Isomorphism of Polynomial (IP) problem) Let m, n be two positive integers, let $F = (f_1, \dots, f_m)$ be a set of polynomial in $\mathbb{F}[x_1, \dots, x_n]$. Let s, t be two bijective affine transformation on K^n , for $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$,

$$\mathbf{y} = s(\mathbf{x}) = \mathbf{x}\mathbf{A}_1 + \mathbf{v},$$

where $\mathbf{A}_1 \in K^{n \times n}$ is invertible and $\mathbf{v} \in K^n$. Then let

$$f'_i(x_1, \dots, x_n) = f_i(y_1, \dots, y_n),$$

$$G = (g_1, \dots, g_n) = t(f'_1, \dots, f'_m) = (f'_1, \dots, f'_m)\mathbf{A}_2$$

where $\mathbf{A}_1 \in K^{n \times n}$ is invertible. Then we say (s, t) is a isomorphism for F to G . The IP problem is given F and G , to find the isomorphism (s, t) . A variant of IP problem is given G to find F .

Furthermore, we introduce the Isomorphism Groebner Basis (IGB) problem as follows:

Definition 4.21 (Isomorphic Groebner Basis (IGB) problem) Given a Groebner basis $G = (g_1, \dots, g_t) \stackrel{\$}{\leftarrow} \text{GB.Gen}$, where $g_i \in K[x_1, \dots, x_n]$. Randomly choose an invertible matrix $\mathbf{A}_1 \in K^{n \times n}$, $\mathbf{A}_2 \in K^{t \times t}$, and a vector $\mathbf{v} \in K^n$. Then set

$$\mathbf{y} = \mathbf{x}\mathbf{A}_1 + \mathbf{v},$$

and

$$G' = (g'_1(\mathbf{x}), \dots, g'_t(\mathbf{x})) = (g_1(\mathbf{y}), \dots, g_t(\mathbf{y})) \cdot \mathbf{A}_2.$$

The IGB problem states that given G' , to find isomorphic Groebner Basis G .

Obviously, if IGB is hard, then IP is hard too, but not vice versa. The security of our IGB Polly Cracker scheme should be finally based on some ideal membership (IM) decision problem. By theorem 4.19, we know if one know a Groebner basis of an ideal, then IM decision problem is easy to solve. But whether IM decision problem is hard if one does not know a Groebner basis of the ideal is still a open problem. A furthermore questions is that whether IM decision problem is hard if one does not know a Groebner basis of the ideal but knows a isomorphic Groebner basis of the ideal. we refer to this problem as ideal membership decision problem with isomorphic Groebner basis (IM-IGB). As follows we give a formal description of IM-IGB hardness assumption, and then reduce the IND-CPA security of our IGB Polly Cracker scheme to IM-IGB assumption.

Definition 4.22 (IM-IGB Assumption) Given a Groebner basis generation algorithm GB.Gen , the challenger generates a Groebner basis $G = (g_1, \dots, g_t) \xleftarrow{\$} \text{GB.Gen}$, where $g_i \in K[x_1, \dots, x_n]$. Challenger randomly choose an invertible matrix $\mathbf{A}_1 \in K^{n \times n}$, $\mathbf{A}_2 \in K^{t \times t}$, and a vector $\mathbf{v} \in K^n$. Then set

$$\mathbf{y} = \mathbf{x}\mathbf{A}_1 + \mathbf{v},$$

and

$$G' = (g'_1(\mathbf{x}), \dots, g'_t(\mathbf{x})) = (g_1(\mathbf{y}), \dots, g_t(\mathbf{y})) \cdot \mathbf{A}_2.$$

then release G' as the isomorphic Groebner basis. The adversary can inquire some sample in the ideal $\langle G' \rangle$. Then for any element a in $\langle G' \rangle \cup 1 + \langle G' \rangle$, the adversary output 1 if he decide $a \in \langle G' \rangle$, otherwise out 0. The IM-IGB hardness assumption states that for any PPT adversary $\mathcal{M}$, the advantage of the adversary is negligible, i.e.,

$$\text{Adv}_{\mathcal{M}} = |\Pr[0 \leftarrow \mathcal{M}(a) : a \xleftarrow{\$} \langle G' \rangle] - \frac{1}{2}| < \frac{1}{c^n},$$

with a constant $c > 1$.

Theorem 4.23 IGB Polly Cracker is IND-CPA secure if the hardness assumption of IM-IGB holds.

Proof: Suppose there is a PPT algorithm $\mathcal{M}$. $\mathcal{M}$ takes as input a set of isomorphic Groebner basis G' and an element a in $\langle G' \rangle \cup 1 + \langle G' \rangle$, and want to decide whether $a \in \langle G' \rangle$ or $a \in 1 + \langle G' \rangle$. Then $\mathcal{M}$ plays as the challenger in CPA game with another PPT algorithm $\mathcal{A}$. The CPA game is played as follows:

- (1) $\mathcal{M}$ publish G' as the public key of an IGB Polly Cracker encryption scheme.
- (2) $\mathcal{A}$ chooses $b_0, b_1 \in \{0, 1\}$, where $b_0 \neq b_1$ and sends it to $\mathcal{M}$.
- (3) $\mathcal{M}$ return a as the challenge ciphertext to $\mathcal{A}$.
- (4) $\mathcal{A}$ makes a decision that a is the encryption of b_i .

Obviously, $\mathcal{M}$ correctly simulate the CPA game with respect to an IGB Polly Cracker encryption scheme. The advantage of $\mathcal{A}$ in this game is:

$$Adv_{\mathcal{A}} = |\Pr[0 \leftarrow \mathcal{A}(a) : a \leftarrow \text{IGB.Enc}(0)]|.$$

Note that “ a is the encryption of b_0 ” is equivalent to “ $a \in 1 + \langle G' \rangle$ ”, therefore

$$Adv_{\mathcal{A}} = Adv_{\mathcal{M}} < \frac{1}{c^n}. \quad (4-1)$$

with a constant $c > 1$. □

4.5 Other Result and Future Works

There is another Polly Cracker style public key encryption scheme that has provable security and support somewhat homomorphic, it is referred to as GB Polly Cracker with noise or PCN for short[18],[19].

In a PCN scheme the secret key is still a Groebner basis G , the public key is a set of polynomial $f_1, \dots, f_m$ such that

$$f_i \in 2\varepsilon_i + \langle G \rangle$$

where $\varepsilon_i \stackrel{\$}{\leftarrow} \chi$ with χ a distribution on $K[x_1, \dots, x_n]/G$.

Given a plaintext $b \in \{0, 1\}$, it encrypts b as

$$c = b + \sum_{i \in S} f_i,$$

where S is a sparse subset of $\{0, 1, \dots, m\}$. Then the decryption of c is to compute

$$(c \bmod G) \bmod 2$$

PCN scheme achieves a provable security via involving noise into the public key, its security is equivalent to that of Regev's LWE scheme. But due to the additional noise, PCN is in fact a kind of DNF scheme which we proposed in chapter 3, and hence it is not a perfect fully homomorphic encryption scheme.

There are also some other Polly Cracker style cryptosystems have been proposed, such as Lattice Polly Cracker[58], and Polly Two[59], but none of them achieve provable security or homomorphic property.

The future works should aim at establishing a perfect fully homomorphic encryption scheme based on multivariate polynomial ring with provable security. Since some computation problem in multivariate polynomial ring are believed enough hard, we should avoid using the trick of adding noise, and achieve a perfect fully homomorphic encryption scheme.

Chapter 5 Apply Somewhat Homomorphic Encryption Scheme in Smart Grid

5.1 Introduction to Smart Grid

Mainly due to the imbalance in power load and the lack of effective real-time diagnosis, the electrical blackout in North America in 2003 affected more than 100 power plants' operations and tens of millions people's lives [60]. In 2006, a blackout firstly happened in Germany and then rapidly spread to many European countries [61], the post-investigation report revealed that the cause of this accident is that most Transmission System Operators cannot access to real-time data, and thus cannot have sufficient inner-communications with each other. Obviously, both accidents illustrate that the traditional power grid becomes more and more unaffordable for nowadays' increasing electricity requirements. Therefore, the smart grid, which is recognized as the next-generation power grid, has been paid significant attention around the world in recent years.

Compared with traditional power grid, smart grid in its architecture is featured with two-way flows, i.e., power flow and information flow, as shown in Figure 5.1. With various sensors deployed along the two-way flows, especially smart meters at the end-users, a huge amount of real-time data regarding the electricity consumption and system status are collected and reported to the *data & control center* (DCC) via the information flow. By analyzing these data, DCC can monitor the grid's operation status, give the advices to optimize the status, and feedback these analysis results to the power flow. By taking these advices, the power plants and transmission operators can correspondingly adjust their power generation and transmission to balance the load in the power grid, and provide more reliable and efficient generation, transmission, distribution, and utilization of electric power.

In order to provide more accurate and fine-grained advices, DCC needs to collect and analyze a large amount of real-time data generated by smart meters. However, transmitting and processing so much data nearly at the same time become the bottleneck of the system. To reduce the communication and computational overhead of DCC, a typical data aggregation architecture in smart grid is widely suggested. In the architecture, a gateway (GW) is established in a residential area to be in charge of all smart meters in this area. In such a way, those smart meters in the area do not directly report their data to DCC, but submit them to the local GW first. After making some preprocessing on these collected data, the GW only needs to send the

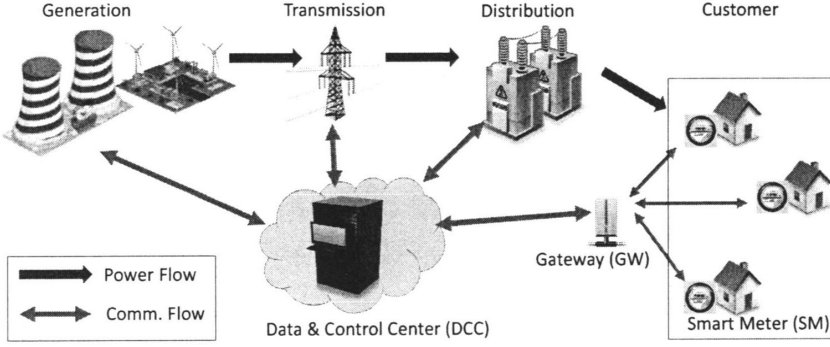


Figure 5.1 Smart grid: the envisioned future grid

processed results to DCC for further analyzing. As a result, the preprocessing at the GW can significantly reduce the communication traffic to DCC but remain the available information desired by DCC. In the architecture, if users' data are transmitted in plaintext, the preprocessing at GW is obvious of no difficulty. However, since the electricity usage data of users are often highly relevant to users' privacy, if there is no security protection, an adversary can easily obtain users' power usage by eavesdropping the communication between smart meters and the GW; and then through the user-behavior model analysis, the adversary can even speculate on the user's private life patterns [62]. Therefore, security and privacy challenge in data aggregation has become one of the biggest concerns in the research of smart grid communications [63, 64]. For example, NIST [65] suggests the privacy should be taken into account in smart grid communication protocol designs, and Netherland recently passed a law to make it compulsory to consider privacy issues in smart meter. Quite recently, several privacy-preserving aggregation schemes [66],[67],[62],[68],[69] have been proposed to provide the aggregation functionality while preserving the users' privacy. Most of them are based on the homomorphic encryption to encrypt users' data, so that an outside attacker or even a compromised GW cannot obtain the user's original data. Meanwhile, because of the homomorphic property, the GW can aggregate all users' data without decryption, and DCC can obtain the aggregated data only through one decryption. However, all the previous aggregation schemes can only support additive aggregation, i.e., the DCC can only obtain the summation of users' data, which greatly narrows down the flexibility of data analysis. As mentioned in [68], it is still an open challenge to compute more complex statistic values of users' data, while without revealing users' privacy.

In this chapter, we make an attempt on the challenging problem mentioned above.

Specifically, based on the lattice cryptographic technique, we propose a new Privacy-preserving Dual-functional Aggregation scheme, named PDA, for the data aggregation in smart grid communications. Our PDA scheme can protect users' data from eavesdropping, while providing dual-functional aggregations. Furthermore, our optimized PDA scheme can further greatly reduce the computational cost and communication overhead of DCC. In summary, the contributions of this work are three-fold as follows: i) PDA uses a homomorphic encryption scheme to encrypt users' data so that the users' privacy can be protected from eavesdropping under the defined attack model; ii) PDA also supports both additive and multiplicative aggregations, which enables DCC to obtain both mean and variance of the users' data with only one report sent by each user; and iii) Additional techniques, including multi-bits ring LWE encryption, encoding integers to polynomials, and superincreasing sequence filling, are integrated in the optimized PDA to further reduce the computational cost, increase bandwidth utilization, and save communication overhead.

5.2 Problem Formalization

We focus on the problem of how the users' private data can be securely reported by users, aggregated by gateways, and read by the DCC in smart grid communications. In what follows, we give a formal description of the system model, security model, and design goals.

5.2.1 System Model

We consider a basic system architecture for smart grid communications, which includes three communication parties: one Data & Control Center DCC, one residential gateway (GW), and a large number of residential users $U = \{U_1, U_2, \dots, U_N\}$, as shown in Fig. 2. Each user $U_i \in U$ is equipped with a smart meter, which has a WiFi link with the gateway (GW) of its residential area, and the gateway (GW) has a more reliable wireline/wireless link with the remote DCC.

Users $U = \{U_1, U_2, \dots, U_N\}$: Each user $U_i, i \in [1, N]$ is equipped with a smart meter (SM). The SM records the real-time and planned electricity usage data of the user, and reports to the DCC via the GW in a certain period, e.g. every 15 minutes.

Gateway (GW): The gateway is a powerful workstation, it has links with the DCC and SMs, Each user's electricity usage data is reported by the corresponding SM to the DCC via the GW, and the DCC can control each SM by distributing system signaling via the GW.

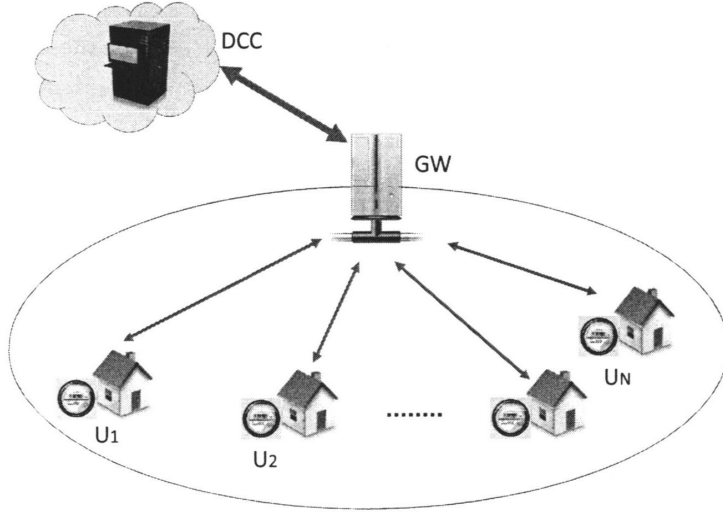


Figure 5.2 System model under consideration

Data & Control Center (DCC): The data & control center is a more powerful entity, it can control the GW, the SMs, and more important, it can receive the data reporting from SMs, analyze these data and optimize the whole system include the power infrastructure.

5.2.2 Security model

In our security model, we consider an attacker can monitor the transmission links, and his goal is to eavesdrop the privacy of the individual user. In addition, we assume that the DCC is trusted and cannot be compromised, but the GW could be compromised in our model. Moreover, there is no communication link between smart meters. Specifically, for the eavesdropping attacker, we assume that he can execute Chosen-Plaintext-Attack (CPA), which means the attacker succeeds if he can distinguish a ciphertext when two corresponding plaintexts are given. Thus, our security goal is to achieve indistinguishable (IND) -CPA secure against the attacker. Note that, an adversary may launch other active attacks in smart grid communications, i.e., bad data injection attack. However, since our focus is on the privacy-preserving aggregation, other attacks are beyond the scope of this work.

5.2.3 Design Goal

Our design goal is to propose a privacy-preserving dual-functional aggregation PDA scheme for smart grid communications which should satisfy the following requirements. First, the proposed PDA scheme should *support dual-functional (mean*

and variance) aggregation. Those traditional aggregation schemes [66],[67],[62] can only support additive aggregation, which is not quite flexible for the smart grid, as in reality DCC usually needs to know both the mean and variance of users' electricity usage data for statistics and analysis. Therefore, in order to satisfy the requirements, traditional aggregation schemes require each user report twice (the ciphertext of electricity usage, and the ciphertext of the square of electricity usage). Different from the traditional aggregation schemes, our proposed PDA scheme is designed to support dual-functional aggregation, which means users only report their encrypted electricity usage once, but the DCC can obtain both the mean and variance of their data. As a result, the communication overhead can be significantly reduced. Secondly, the proposed PDA scheme should also *preserve user privacy* in smart grid communications. That is, even though an attacker can compromise some users, he still cannot gain the data from other uncompromised users.

5.3 Preliminaries

In this section, we will recall some preliminaries, including Ring Learning with Errors (RLWE) problem [22, 36], an RLWE-based somewhat homomorphic encryption scheme [37], and the encoding integers for computation over polynomial ring, which will serve as the basis of our proposed PDA scheme. Before our recalling the preliminaries, we first introduce some used notations as follows:

- $\mathbb{Z}^n$, the set of n -dimension vectors over the integer ring $\mathbb{Z}$.
- $\mathbb{Z}_q$, the ring of $\mathbb{Z}$ modulo q .
- $R \triangleq \mathbb{Z}[x]/\langle f(x) \rangle$, the polynomial ring over $\mathbb{Z}$ modulo $f(x)$.
- $R_q \triangleq \mathbb{Z}_q[x]/\langle f(x) \rangle$, the polynomial ring over $\mathbb{Z}_q$ modulo $f(x)$.
- ϕ , standard linear isomorphism from R to $\mathbb{Z}^n$, i.e. $\forall a = \sum_{i=0}^{n-1} a_i x^i \in R$, $\phi(a) = (a_0, \dots, a_{n-1}) \in \mathbb{Z}^n$. ϕ^{-1} is the invert of ϕ .
- $D_{\mathbb{Z}^n, \sigma}$, n -dimension discrete gaussian distribution over $\mathbb{Z}^n$, with parameter σ , and centered around original point. It is identify with the sum of n orthogonal 1-dimension discrete gaussian distribution $D_{\mathbb{Z}, \sigma}$, which is over $\mathbb{Z}$, with standard deviation σ , and centered around 0, i.e. $\forall e = (e_0, \dots, e_{n-1}) \sim D_{\mathbb{Z}^n, \sigma} \Leftrightarrow e_i \sim D_{\mathbb{Z}, \sigma}$ and $e_i, i = 0, \dots, n-1$ are mutual orthogonal.
- $D_{R, \sigma}$, a distribution over R determined by $D_{\mathbb{Z}^n, \sigma}$, i.e. $\forall a \sim D_{R, \sigma} \Leftrightarrow \phi(a) \sim D_{\mathbb{Z}^n, \sigma}$.

5.3.1 RLWE-based Somewhat Homomorphic (SH) Encryption

In the following, we recall an RLWE-based somewhat homomorphic (SH) encryption scheme, which was first proposed in [37], and later revised more practical in [22].

5.3.1.1 Description of RLWE-based SH Encryption

The RLWE-based SH encryption scheme is mainly comprised of five algorithms: Key Generation (SH.KeyGen), Encryption (SH.Enc), Decryption (SH.Dec), Homomorphic Addition (SH.Add), and Homomorphic Multiplication (SH.Mult), and associated with parameters n, q, t, σ .

- **SH.KeyGen(1^n)**: Sample $s \xleftarrow{\$} D_{R, \sigma}$, and keep it as the private key $sk = \{s\}$. Further sample $a \xleftarrow{\$} R_q$, $e \xleftarrow{\$} D_{R, \sigma}$, and compute the corresponding public key $pk = \{a, b = -(as + te)\}$.
- **SH.Enc(m, pk)**: To encrypt a given message $m \in R_t$, we sample $u, f, g \xleftarrow{\$} D_{R, \sigma}$, and compute the ciphertext as

$$ct = (c_0, c_1) = (bu + tg + m, au + tf) \quad (5-1)$$

- **SH.Dec(ct, sk)**: To decrypt a ciphertext $ct = (c_0, c_1, \dots, c_\delta)$, we firstly compute

$$\tilde{m} = \sum_{i=0}^{\delta} c_i s^i \in R_q$$

and then the message is obtain by taking the computation $\tilde{m} \bmod t$. Here $\delta (\geq 2)$ will increase due to the homomorphic multiplications (discussed below). For an initial ciphertext, i.e. a ciphertext directly obtained by encrypting a plaintext, $\delta = 2$.

- **SH.Add(ct, ct', pk)**: Let $ct = (c_1, c_2, \dots, c_\delta)$ and $ct' = (c'_1, c'_2, \dots, c'_\delta)$ be two ciphertexts of the same length under the same public key pk . (If two ciphertexts do not follow the same length, we can simply pad the short one with proper zeroes). Specifically, the addition is computed by component-wise addition, i.e.,

$$ct_{\text{Add}} = (c_1 + c'_1, c_2 + c'_2, \dots, c_\delta + c'_\delta) \in R_q^\delta$$

- **SH.Mult(ct, ct', pk)**: Let $ct = (c_1, c_2, \dots, c_\delta)$ and $ct' = (c'_1, c'_2, \dots, c'_\gamma)$ be two ciphertexts under the same public key pk . Let v be a symbolic variable, we compute the symbolical expression

$$\left(\sum_{i=0}^{\delta} c_i v^i \right) \cdot \left(\sum_{i=0}^{\gamma} c'_i v^i \right) = \sum_{i=0}^{\delta+\gamma} \tilde{c}_i v^i$$

Then, the multiplication of ct and ct' is $ct_{\text{Mult}} = (\tilde{c}_1, \tilde{c}_2, \dots, \tilde{c}_{\delta+\gamma})$.

Correctness of Homomorphic Properties. Let $\text{ct} = (c_1, c_2, \dots, c_\delta)$ and $\text{ct}' = (c'_1, c'_2, \dots, c'_\delta)$ be two valid ciphertexts, and let $\text{ct}_{\text{Add}} \leftarrow \text{SH.Add}(\text{ct}, \text{ct}', pk)$, i.e. $\text{ct}_{\text{Add}} = (c_1 + c'_1, c_2 + c'_2, \dots, c_\delta + c'_\delta) \in R_q^\delta$. The first step of ct_{Add} decryption yields

$$\tilde{m} = \sum_{i=0}^{\delta} (c_i + c'_i) s^i = \sum_{i=0}^{\delta} c_i s^i + \sum_{i=0}^{\delta} c'_i s^i \quad (5-2)$$

Since ct and ct' are valid ciphertexts, $m + t\tilde{e} = \sum_{i=0}^{\delta} c_i s^i \in R_q$, $m' + t\tilde{e}' = \sum_{i=0}^{\delta} c'_i s^i \in R_q$, we have

$$\tilde{m} = m + m' + t(\tilde{e} + \tilde{e}') \quad (5-3)$$

By properly setting parameter q (i.e., a large q) such that $m + m' + t(\tilde{e} + \tilde{e}') \in R_q$, we guarantee that the second step of decryption is correct, i.e. $\tilde{m} \bmod t$ can correctly output $m + m'$.

Let $\text{ct} = (c_1, c_2, \dots, c_\delta)$ and $\text{ct}' = (c'_1, c'_2, \dots, c'_\gamma)$ be two valid ciphertexts, and let $\text{ct}_{\text{Mult}} \leftarrow \text{SH.Mult}(\text{ct}, \text{ct}', pk)$, i.e. $\text{ct}_{\text{Mult}} = (\tilde{c}_1, \tilde{c}_2, \dots, \tilde{c}_{\delta+\gamma})$ such that

$$\sum_{i=0}^{\delta+\gamma} \tilde{c}_i v^i = \left(\sum_{i=0}^{\delta} c_i v^i \right) \cdot \left(\sum_{i=0}^{\gamma} c'_i v^i \right) \quad (5-4)$$

with a symbolic variable v . The first step of ct_{Add} decryption yields

$$\tilde{m} = \sum_{i=0}^{\delta} \tilde{c}_i s^i = \left(\sum_{i=0}^{\delta} c_i s^i \right) \left(\sum_{i=0}^{\gamma} c'_i s^i \right) \quad (5-5)$$

Since ct and ct' are valid ciphertexts, $m + t\tilde{e} = \sum_{i=0}^{\delta} c_i s^i \in R_q$, $m' + t\tilde{e}' = \sum_{i=0}^{\gamma} c'_i s^i \in R_q$, we have

$$\tilde{m} = mm' + t(\tilde{e}m' + \tilde{e}'m) \quad (5-6)$$

By properly setting parameter q (i.e., a large q) such that $mm' + t(\tilde{e}m' + \tilde{e}'m) \in R_q$, we guarantee that the second step of decryption is correct, i.e. $\tilde{m} \bmod t$ can correctly output $m \cdot m'$.

Lemma 5.1 (See[70]) Let $n \in \mathbb{N}$, for arbitrary real number $\sigma > \omega(\sqrt{\log n})$, we have

$$\Pr[\|x\| > \sigma\sqrt{n} : x \leftarrow D_{z^N, \sigma}] < 2^{-n+1} \quad (5-7)$$

Claim 5.2 (See[22]) Let $n \in \mathbb{N}$, and let $\chi_{R, \sigma}$ be the discrete gaussian distribution over R . For any $s_1, \dots, s_N \leftarrow \chi_{R, \sigma}$ with their l_∞ norm at most η , we have

$$\left\| \sum_{i=1}^N s_i \right\|_\infty \leq 2 \cdot \sqrt{N} \cdot \eta \quad (5-8)$$

holds with overwhelming probability.

According to the above Lemma 5.1 and Claim 5.2, it is easy for us to properly set the required parameters, which will be useful for achieving homomorphic properties.

5.3.1.2 Composite Homomorphic Evaluation and Parameter Setting

The RLWE-based SH encryption scheme that we recalled above can support homomorphic evaluation on a very complex composite circuit, i.e., a composite operation with some additions and multiplications, if the parameter q is set large enough. Specifically, for the RLWE-based SH encryption, once the parameter q is set so large that the complexity of homomorphic evaluation circuit can cover that of the decryption circuit of itself, it can support homomorphic evaluation for arbitrary circuit by using Gentry's bootstrapping architecture [37],[71].

In the following, we will introduce a simple composite homomorphic circuit, which will be used in our PDA scheme design. As an expansion of SH.Add, we first define SH.Sum($ct_1, ct_2, \dots, ct_N, pk$) as

$$\begin{aligned} & \text{SH.Add}(\dots \text{SH.Add}(\text{SH.Add}(ct_1, ct_2, pk), ct_3, pk) \\ & \dots, ct_N, pk) \end{aligned} \quad (5-9)$$

For $i = 1, \dots, N$, let $ct_i = (c_{i,0}, c_{i,1}) \leftarrow \text{SH.Enc}(m_i, pk)$ be the original ciphertext of the plaintext m_i , i.e.,

$$c_{i0} = bu_i + tg_i + m_i, c_{i1} = au_i + tf_i \quad (5-10)$$

Then, we consider the composite circuit SH.QuaSum that defined as the summation of total N homomorphic multiplications of ct_i itself, i.e.,

$$\begin{aligned} & \text{SH.QuaSum}(ct_1, \dots, ct_N, pk) \\ & = \text{SH.Sum}(\text{SH.Mult}(ct_1, ct_1, pk), \\ & \dots, \text{SH.Mult}(ct_N, ct_N, pk), pk) \end{aligned} \quad (5-11)$$

In order to evaluate the homomorphic circuit, i.e., to recover $\sum_{i=1}^N m_i^2$ from the above composite circuit, we first perform the decryption to obtain

$$\sum_{i=1}^N (m_i^2 + 2t\tilde{e}_i m_i) \bmod (f(x), q) \quad (5-12)$$

where $\tilde{e}_i = -(eu_i + f_i s + g_i)$. To guarantee the correctness of decryption, we should set $f(x)$ and q such that

$$\deg(f(x)) > 2\deg(m_i), \quad \left\| \sum_{i=1}^N (m_i^2 + 2t\tilde{e}_i m_i) \right\|_{\infty} < q \quad (5-13)$$

By Lamma5.1 and Claim5.2, a standard bounding shows

$$\left\| \sum_{i=1}^N (m_i^2 + 2t\tilde{e}_i m_i) \right\|_{\infty} \leq 2\sqrt{3N} \cdot t \cdot \sigma^2 n^2 \quad (5-14)$$

As mentioned in [22], we always take the parameter σ to be 8, therefore, by setting $q > 128\sqrt{3N} \cdot t \cdot n^2$, we can guarantee that the decryption on the composite homomorphic evaluation (5-11) returns $\sum_{i=1}^N m_i^2$, i.e.,

$$\text{SH.Dec}(\text{SH.QuaSum}(\text{ct}_1, \dots, \text{ct}_N, pk)) = \sum_{i=1}^N m_i^2 \quad (5-15)$$

5.3.2 Encoding Integers for Computation over Polynomial Ring

Let $m[0], m[1], \dots, m[\delta]$ be the binary expression of an integer $\hat{m}$ in the inverted order, i.e.,

$$\hat{m} = \sum_{i=0}^{\delta} m[i]2^i, m[i] \in \{0, 1\}$$

we define a map $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}[x]$ such that $\forall \hat{m} \in \mathbb{Z}$

$$\varphi(\hat{m}) = m \triangleq \sum_{i=0}^{\delta} m[i]x^i \in \mathbb{Z}[x]$$

and a map $\psi : \mathbb{Z}[x] \rightarrow \mathbb{Z}$ such that $\forall m \in \mathbb{Z}[x]$

$$\psi(m) = \sum_{i=0}^{\delta} m[i]2^i$$

Note that, though ψ and φ are not the inverse to each other, they have the following property.

Property 5.3 Let $\hat{m}, \hat{m}'$ be two integers, and $m = \varphi(\hat{m}), m' = \varphi(\hat{m}')$, then we have

$$\psi(m + m') = \hat{m} + \hat{m}', \psi(mm') = \hat{m}\hat{m}'$$

Therefore, we can apply φ to map a number into $\mathbb{Z}_2[x]$ which is included in $R_q \triangleq \mathbb{Z}_q[x]/\langle f(x) \rangle$, the plaintext space of the RLWE-based SH encryption scheme. At the same time, in order to map the result of the computation on m and m' over $R_q \triangleq \mathbb{Z}_q[x]/\langle f(x) \rangle$ back to a number, we must guarantee the all computations are operated in $\mathbb{Z}[x]$, i.e., we need to set a proper q . Concretely, let $\hat{m}, \hat{m}'$ be two integers, and $m = \varphi(\hat{m}), m' = \varphi(\hat{m}')$. Suppose $\text{ct} \leftarrow \text{SH.Enc}(m)$ and $\text{ct}' \leftarrow \text{SH.Enc}(m')$, and ct_{Add} and ct_{Mult} are respectively the homomorphic sum and homomorphic product of ct and ct' . By setting a proper q , as we discussed above, the decryption algorithm SH.Dec on ct_{Add} and ct_{Mult} will return $((m + m') \bmod f(x)) \bmod t$ and $(mm' \bmod f(x)) \bmod t$, to map these results back to the corresponding integers $\hat{m} + \hat{m}'$ and $\hat{m}\hat{m}'$, we must guarantee

$$\begin{aligned} ((m + m') \bmod f(x)) \bmod t &= m + m' \\ ((mm') \bmod f(x)) \bmod t &= mm' \end{aligned} \quad (5-16)$$

Since $\deg(f(x)) = n$, and $m, m' \in \mathbb{Z}_2[x]$, the above equations hold by constraining $\deg(m) = \deg(m') \leq \lfloor \frac{n}{2} \rfloor - 1$, and setting $t > \deg(m)$. For a more general case, we have the following property:

Property 5.4 Let $\hat{m}_1, \hat{m}_2, \dots, \hat{m}_N$ be N integers, and for $i \in [1, N]$, let $m_i = \varphi(\hat{m}_i)$, and $\text{ct}_i \leftarrow \text{SH.Enc}(m_i)$. Then

$$\psi(\text{SH.Dec}(\text{SH.Sum}(\text{ct}_1, \dots, \text{ct}_N, pk))) = \sum_{i=1}^N \hat{m}_i \quad (5-17)$$

and

$$\psi(\text{SH.Dec}(\text{SH.QuaSum}(\text{ct}_1, \dots, \text{ct}_N, pk))) = \sum_{i=1}^N \hat{m}_i^2 \quad (5-18)$$

hold by constraining $\deg(m) = \deg(m') \leq \lfloor \frac{n}{2} \rfloor - 1$, and setting $t > \deg(m) \cdot N$.

Finally, from the property 5.3 and the property 5.4, we can easily draw the following corollary.

Corollary 5.5 Let $\hat{d}_i, i = 1, 2, \dots, N$, be N integers, and $d_i = \varphi(\hat{d}_i)$, let $\hat{M}$ and $\hat{D}$ be respectively the mean and the squared deviation of $\hat{d}_i, i = 1, 2, \dots, N$, then we have

$$\begin{aligned} \hat{M} &= \frac{1}{N} \psi\left(\sum_{i=1}^N d_i\right) \\ \hat{D} &= \frac{1}{N} \left(\psi\left(\sum_{i=1}^N d_i^2\right) - \hat{M}^2\right) \end{aligned}$$

5.4 Proposed PDA Scheme

In this section, we present our privacy-preserving dual-functional aggregation scheme, called PDA, for smart grid communications, which mainly consists of four phases: system initialization, user report generation, privacy-preserving report aggregation, and secure report reading. For the ease of description, we first present the basic version of PDA, and then enhance it into an optimized one.

5.4.1 Basic PDA Scheme

5.4.1.1 System Initialization

In the phase of system initialization, the DCC first sets up the system parameters κ, k and N , where κ is one security parameter, k is the length of the binary expression of messages reported by users U , and N bounds the users number. The DCC also initializes a homomorphic encryption scheme SH with parameter (n, q, t, σ) , such that $n = \max\{\kappa, 2k - 1\}$, $q > 2\sqrt{3N} \cdot t \cdot \sigma^2 n^2$, and $t > N \cdot k$, and $\sigma = 8$. After that, the DCC runs the algorithm $\text{SH.KeyGen}(1^n)$ to generate a private key $sk = \{s\}$, and the

corresponding public key $pk = \{a, b = -(as + te)\}$, where s, a, e are all drawn from the distribution $D_{R,\sigma}$. Finally, the DCC keeps the private key $sk = \{s\}$ secret, and publishes the public key $pk = \{a, b\}$ to the GW and all users U in a residential area.

In smart grid communications, each reported data is small, e.g., the power assumption in 15 minutes is usually decimal in kilowatt-hour. But we can always transform it to an integer with some effective methods, e.g., applying proper metrics.

5.4.1.2 User Report Generation

In smart grid communication, every 15 minutes [62], each user U_i will report the electricity use data $\hat{m}_i$. Concretely, each user $U_i \in U$ will run the following steps:

- (1) encode the electricity use data $\hat{m}_i$ into $m_i = \varphi(\hat{m}_i) = \sum_{j=0}^{k-1} m_i[j]x^j$;
- (2) use the DCC's public key $pk = \{a, b = -(as + te)\}$ to encrypt m_i into ct_i , i.e., $ct_i = (c_{i0} = bu_i + tg_i + m_i, c_{i1} = au_i + tf_i) \leftarrow \text{SH.Enc}(m_i, pk)$, where u_i, g_i, f_i are all drawn from a given distribution $\chi_{R,\sigma}$;
- (3) report ct_i to the gateway GW.

5.4.1.3 Privacy-Preserving Report Aggregation

After receiving all users' encrypted data ct_i , for $i = 1, 2, \dots, N$, the gateway GW will perform the following privacy-preserving aggregation steps:

- (1) compute the homomorphic summation of $ct_1, ct_2, \dots, ct_N$, i.e., $A \leftarrow \text{SH.Sum}(ct_1, ct_2, \dots, ct_N, pk)$;
- (2) compute $ct'_i \leftarrow \text{SH.Mult}(ct_i, ct_i, pk)$ for each ct_i , and the homomorphic summation of $ct'_1, ct'_2, \dots, ct'_N$, i.e., $A' \leftarrow \text{SH.Sum}(ct'_1, ct'_2, \dots, ct'_N, pk)$;
- (3) report (A, A') to the DCC.

5.4.1.4 Secure Report Reading

Upon receiving (A, A') from the GW, the DCC executes the following steps to recover the data:

- (1) obtain S by decrypting A with the private key $sk = \{s\}$, i.e., $S \leftarrow \text{SH.Dec}(A, sk)$;
- (2) obtain S' by decrypting A' with the private key $sk = \{s\}$, i.e., $S' \leftarrow \text{SH.Dec}(A', sk)$;

- (3) map S and S' to their corresponding integers $\hat{S}$ and $\hat{S}'$ respectively, i.e., $\hat{S} \leftarrow \psi(S)$, and $\hat{S}' \leftarrow \psi(S')$;
- (4) compute the mean $\hat{M}$ and the squared deviation $\hat{D}$ of the user data $(\hat{m}_1, \hat{m}_2, \dots, \hat{m}_N)$, as follows

$$\hat{M} = \frac{1}{N}\hat{S}, \quad \hat{D} = \frac{1}{N}(\hat{S}' - \hat{S}^2)$$

Correctness. The correctness of $\hat{M} = \frac{1}{N} \sum_{i=1}^N \hat{m}_i$, and $\hat{D} = \frac{1}{N} (\sum_{i=1}^N \hat{m}_i - \hat{M})^2$ can be easily verified. From the calculation, we first know $S = \sum_{i=1}^N (t\tilde{e}_i + m_i) \bmod f(x) \bmod t$ and $S' = \sum_{i=1}^N (t\tilde{e}_i + 2t\tilde{e}_i m_i + m_i^2) \bmod f(x) \bmod t$. Based on our proper parameter settings, and the Corollary 5.5, we further know $S = \sum_{i=1}^N m_i$ and $S' = \sum_{i=1}^N m_i^2$. As a result, the correctness follows.

5.4.2 Optimized PDA Scheme

In practical scenarios, each data reported every 15 minutes in smart grid communications is usually very small, and the value can be expressed by only 10 bits. However, for the security reason, the plaintext space should be large enough, e.g., $\{0, 1\}^{192}$ or more [72], which obviously wastes the bandwidth during the encrypted data transmission. Meanwhile, for the fine-grained statistic in smart grid system, the DCC also expects to know the electricity use statistic of each kind of smart appliance. Therefore, there is a great desire to transmit multidimensional data at the same time so as to not only save the bandwidth as a whole but also improve the usability [62]. To meet the requirements, we present our optimized PDA scheme in the following.

5.4.2.1 System initialization

In the system initialization of our optimized PDA scheme, system parameters include κ, k, N , where κ is one security parameter, k is the length of user's data in binary form, and N is the total users number involved in the system. The DCC first sets up a somewhat homomorphic encryption scheme SH with parameter (n, q, t, σ) , such that $n = \kappa$, $q > 2\sqrt{3N} \cdot t \cdot \sigma^2 n^2$, and $t > N \cdot k$, and $\sigma = 8$. Then, DCC runs the algorithm SH.KeyGen(1^n) to generate a private key $sk = \{s\}$, and the corresponding public key $pk = \{a, b = -(as + te)\}$, where s, a, e are drawn from the distribution $D_{R, \sigma}$. Finally, the DCC keeps the private key $sk = \{s\}$ secret, and publishes the public key $pk = \{a, b\}$.

5.4.2.2 User Report Generation

Suppose each time the user $U_i, i = 1, \dots, N$ need to report l data denoted as $d_{i,1}, d_{i,2}, \dots, d_{i,l}$ (after encoded), and each data is a k bits binary number, e.g. $k = 10$. Then the user needs to fill these data bits into a n -bit length blank plaintext m_i , and then encrypt it and report the ciphertext to GW. Finally the DCC will obtain $\sum_{i=1}^N m_i$ by decryption of the aggregated data, and it also needs to recover each $\sum_{i=1}^N d_{i,j}$ from $\sum_{i=1}^N m_i$. Considering the bits extension caused by multiplication over polynomial ring, we can not fill these $d_1, \dots, d_l$ into m_i bit by bit. We should remain enough blank between two data. The following steps give us the detailed process, and the correctness of this process will be proved later.

- (1) Initialized a n bits blank plaintext $m_i = (m_i[0], m_i[1], \dots, m_i[n-1]) = (0, 0, \dots, 0)$, and an index $p = 0$.
- (2) Let $\hat{d}_{i,j}$ denote the j -th data reported by U_i , (the index j starts from 0), encode $\hat{d}_{i,j}$ to $d_{i,j} \triangleq \varphi(\hat{d}_{i,j})$, then we use the following algorithm to fill m_i with each $d_{i,j}$.

• Algorithm1 :

WHILE $(2^{j+1} - 1)(2k - 1) \leq n - 1$, DO:

```

{
     $p \leftarrow (2^j - 1)(2k - 1);$ 
    FOR  $t$  FROM 0 TO  $k - 1$ 
    {
         $m_i[p + t] = d_{i,j}[t];$ 
    }
     $j \leftarrow j + 1;$ 
}
    
```

- (3) encrypt m_i using public key $pk = \{a, b\}$, i.e., $ct_i \leftarrow \text{SH.Enc}(m_i, pk)$.
- (4) report ct_i to GW.

5.4.2.3 Privacy-Preserving Report Aggregation

The aggregation steps of the optimized PDA scheme by GW are the same as those in the basic PDA scheme. Suppose the GW receives all the reported data $ct_i, i = 1, 2, \dots, N$, then GW executes the following steps:

- (1) GW evaluates the homomorphic summation of $ct_1, ct_2, \dots, ct_N$, i.e., $A \leftarrow \text{SH.Sum}(ct_1, ct_2, \dots, ct_N, pk)$.
- (2) For each i , GW evaluates the homomorphic multiplication of ct_i by itself, i.e., $ct'_i \leftarrow \text{SH.Mult}(ct_i, ct_i, pk), i \in [1, N]$.
- (3) GW evaluates the homomorphic summation of $ct'_1, ct'_2, \dots, ct'_N$, i.e., $A' \leftarrow \text{SH.Sum}(ct'_1, ct'_2, \dots, ct'_N, pk)$.
- (4) GW reports A and A' to CC.

5.4.2.4 Secure Report Reading

When the DCC receives A and A' , it executes the following steps to obtain the data:

- (1) First, DCC decrypts A to obtain $\bar{m}$, i.e., $\bar{m} \leftarrow \text{SH.Dec}(A, sk)$. After that, DCC picks out the aggregated data $\bar{d}_j$ from $\bar{m}$ by the following algorithm:

- Algorithm2 :

Initialize $j = 0$ and $p = 0$;

WHILE $(2^{j+1} - 1)(2k - 1) \leq n - 1$, DO:

```
{
     $p \leftarrow (2^j - 1)(2k - 1)$ 
    FOR  $t$  FROM 0 TO  $k - 1$ 
    {
         $\bar{d}_j[t] = \bar{m}[p + t]$ ;
    }
     $j \leftarrow j + 1$ ;
}
```

- (2) Then, DCC decrypts A' to obtain $\bar{m}'$, i.e., $\bar{m}' \leftarrow \text{SH.Dec}(A', sk)$. After that, DCC picks out the the data $\bar{d}'_j$ from $\bar{m}'$ by the following algorithm:

- Algorithm3 :

Initialize $j = 0$ and $p = 0$;

WHILE $(2^{j+1} - 1)(2k - 1) \leq n - 1$, DO:

```

{
   $p \leftarrow (2^{j+1} - 2)(2k - 1)$ 
  FOR  $t$  FROM 0 TO  $2k - 2$ 
  {
     $\bar{d}'_j[t] = \bar{m}'[p + t];$ 
  }
   $j \leftarrow j + 1;$ 
}
    
```

- (3) Next, for each j , DCC maps $\bar{d}_j$ and $\bar{d}'_j$ respectively to their corresponding integers $\hat{d}_j$ and $\hat{d}'_j$, i.e., $\hat{d}_j \leftarrow \psi(\bar{d}_j)$, and $\hat{d}'_j \leftarrow \psi(\bar{d}'_j)$.
- (4) Finally, for each j , DCC computes the mean $\hat{M}_j$ and squared deviation $\hat{D}_j$ of $\hat{d}_{1,j}, \dots, \hat{d}_{N,j}$ as follows,

$$\hat{M}_j = \frac{1}{N} \hat{d}_j, \quad \hat{D}_j = \frac{1}{N} (\hat{d}'_j - \hat{d}_j^2)$$

Correctness. For each j , we will show $\hat{M}_j = \frac{1}{N} \sum_{i=1}^N \hat{d}_{i,j}$ and $\hat{D}_j = \frac{1}{N} \sum_{i=1}^N (\hat{d}_{i,j} - \hat{M}_j)^2$ correct. By the correctness of our basic scheme, we have $\bar{m} = \sum_{i=1}^N m_i$. Since the addition over polynomial will not cause bit extending, and with the algorithm 1 and algorithm 2, it is easy to verify that for each j , $\bar{d}_j = \sum_{i=1}^N d_{i,j}$. Thus, by Corollary 5.5, the correctness of $\hat{M}_j = \frac{1}{N} \sum_{i=1}^N \hat{d}_{i,j}$ is obvious.

Similarly, the correctness of $\hat{D}_j = \frac{1}{N} \sum_{i=1}^N (\hat{d}_{i,j} - \hat{M}_j)^2$ follows the correctness of $\bar{d}'_j = \sum_{i=1}^N d_{i,j}^2$ for each j . Therefore, for the correctness, we just need to verify that $\bar{d}'_j = \sum_{i=1}^N d_{i,j}^2$. For each component $\bar{m}'[r]$ of $\bar{m}'$,

$$\bar{m}'[r] = \sum_{i=1}^N \sum_{\substack{s+t=r \\ s \in [0,r] \\ t \in [0,r]}} m_i[s] m_i[t]$$

Let $p_j = (2^j - 1)(2k - 1)$ and $r \in [0, 2k - 2]$, then by algorithm 3, we have

$$\begin{aligned}
 \bar{d}'_j[r] &= \bar{m}'[2p_j + r] \\
 &= \sum_{i=1}^N \sum_{\substack{s+t=2p_j+r \\ s \in [0, 2p_j+r] \\ t \in [0, 2p_j+r]}} m_i[s] m_i[t] \\
 &= \sum_{i=1}^N \sum_{s \in [0, 2p_j+r]} m_i[s] m_i[2p_j + r - s]
 \end{aligned}$$

Notice that $p_{j+1} = 2p_j + 2k - 1$, and by algorithm 1, for each j , and each $\tau \in [p_j + k, p_{j+1} - 1]$, we have $m[\tau] = 0$.

When $s \in [0, p_{j-1} + k - 1]$, we have

$$\begin{aligned}
 2p_j + r - s &\geq 2p_j - p_{j-1} - k + 1 + r \\
 &= p_j + 2p_{j-1} + 2k - 1 \\
 &\quad - p_{j-1} - k + 1 + r \\
 &= p_j + p_{j-1} + k + r \\
 &\geq p_j + k + r \\
 &\geq p_j + k
 \end{aligned}$$

and

$$\begin{aligned}
 2p_j + r - s &\leq 2p_j + r \\
 &\leq 2p_j + 2k - 2 \\
 &= p_{j+1} - 1
 \end{aligned}$$

therefore $m_i[2p_j + r - s] = 0$.

When $s \in [p_{j-1} + k, p_j - 1]$, we have $m_i[s] = 0$.

In any case, when $s \in [0, p_j - 1]$, we have $m_i[s]m_i[2p_j + r - s] = 0$. Similarly, when $s \in [p_j + k, 2p_j + r]$, we also have $m_i[s]m_i[2p_j + r - s] = 0$. Thus, we can write $\bar{d}_j[r]$ as

$$\begin{aligned}
 \bar{d}_j[r] &= \sum_{i=1}^N \sum_{s \in [p_j, p_j+k-1]} m_i[s]m_i[p_j + r - s] \\
 &= \sum_{i=1}^N \sum_{s' \in [0, k-1]} m_i[p_j + s']m_i[p_j + r - s'] \\
 &= \sum_{i=1}^N \sum_{s' \in [0, k-1]} d_{i,j}[s']d_{i,j}[r - s']
 \end{aligned}$$

which follows $\bar{d}_j = \sum_{i=1}^N d_{i,j}^2$.

5.5 Security Analysis

As mentioned in our security requirement, we mainly focus on protecting the privacy of the user's electricity usage data in this work. Because our proposed PDA scheme is built upon the RLWE based homomorphic encryption scheme, in this section, we first show that the RLWE based homomorphic encryption scheme is IND-CPA secure, and then analyze the security of our proposed PDA scheme in two scenarios based on the results.

The security of the RLWE-based encryption scheme is based on the RLWE assumption [36],[37] which can be stated as follows: Given random numbers $a \in R_q$, $s \in R_q$, a probability distribution χ , and a uniform distribution U , there are two distributions $D_0 = \{(a, as + e_i) : e_i \leftarrow \chi\}$, and $D_1 = \{(a, b) : b \leftarrow U\}$. RLWE decision problem is to decide whether a given sample (a, b) is draw from D_0 or from

D_1 . RLWE assumption is stated that the advantage of arbitrary PPT algorithm to solve RLWE decision problem is negligible, i.e., for arbitrary PPT algorithm $\mathcal{D}$ $|\Pr[\mathcal{D}(a, b) \rightarrow 0] - \frac{1}{2}| \leq \text{negl}(\kappa)$, where κ is the input length of $\mathcal{D}$.

Theorem 5.6 The RLWE-based homomorphic encryption scheme is IND-CPA secure under the RLWE assumption.

Proof: Suppose there exists a PPT attacker $\mathcal{A}$ who can win with an non-negligible advantage ε in the IND-CPA game with respect to a RLWE-based encryption scheme, then we can construct a PPT decider $\mathcal{D}$ to solve the RLWE decision problem as follows.

- (1) $\mathcal{D}$ takes as input (a, b) , it should output 0 if (a, b) is draw from D_0 , otherwise output 1. $\mathcal{D}$ sends (a, b) to $\mathcal{A}$ as the public key of the a RLWE-based encryption scheme.
- (2) $\mathcal{A}$ chooses two plaintext m_0, m_1 , and sends them to $\mathcal{D}$.
- (3) $\mathcal{D}$ randomly chooses a bit i from $\{0, 1\}$, and encrypts m_i as $\text{ct}_i = (bu + tg + m_i, au + tf)$, where u, g, f are randomly draws from R_q . Then $\mathcal{D}$ sends ct_i to $\mathcal{A}$ as a ciphertext
- (4) $\mathcal{A}$ makes a decision that the ciphertext is from the encryption of m_i , and sends $\tilde{i}$ to $\mathcal{D}$.
- (5) $\mathcal{D}$ outputs 0, If $\tilde{i} = i$, otherwise output 1.

If (a, b) is indeed drawn from D_0 , ct_i is a legal ciphertext, i.e., $\mathcal{D}$ is correctly analogue the IND-CPA game. At this time $\mathcal{A}$ can use its ability to guess i correctly with the probability of $\frac{1}{2} + \varepsilon$. If (a, b) is drawn from D_1 , ct_i is not a legal ciphertext, $\mathcal{A}$ has no advantage to guess i correctly with probability more than $\frac{1}{2}$. Therefore, by calling $\mathcal{A}$, $\mathcal{D}$ can solve RLWE decision problem with probability of $\frac{1}{2}\varepsilon$. By the RLWE assumption, there is not such $\mathcal{D}$ that solves RLWE decision problem with non-negligible advantage, thus there is no such $\mathcal{A}$ that wins the corresponding IND-CPA game with non-negligible advantage. The IND-CPA security of the RLWE-based encryption scheme follows. $\square$

According to the above theorem, it is easy to see that an attacker outside the smart grid and even a compromised GW cannot obtain the users' privacy data in our proposed PDA scheme, because the secret key is known only by the DCC. In the following, we will further discuss whether an attacker can abuse the homomorphic property to attack the user privacy.

Suppose an attacker $\mathcal{A}$ eavesdrops the communication link between a user U_i and GW, then he can obtain the ciphertext ct_i sent by U_i . Let the corresponding plain text be m_i , i.e., $ct_i \leftarrow \text{SH.Enc}(m_i, pk)$. Without knowing the secret key, the attacker $\mathcal{A}$ can not decrypt ct_i to obtain m_i . Considering the user's electricity usage data m_i is a small value, $\mathcal{A}$ may try a brute-force attack by exhausting every possible value of m_i , i.e., $\mathcal{A}$ may encrypt every possible m_i to obtain lots of ciphertexts by himself, then by comparing these ciphertexts with the ciphertext eavesdropped, he tries to recover m_i . However, as the homomorphic encryption scheme employed in our proposed PDA scheme is IND-CPA secure, $\mathcal{A}$ still can not recover m_i . Further more, we allow the attacker have the capability to compromise the GW and some users, let the compromised users be $U_1, U_2, \dots, U_t$. Let the users' data be $m_1, m_2, \dots, m_N$, and the corresponding ciphertexts are $ct_1, ct_2, \dots, ct_N$. By eavesdropping the communication between each user and GW, the attacker $\mathcal{A}$ can obtain all these ciphertexts, and by compromising user $U_1, U_2, \dots, U_t$, he can obtain $m_1, m_2, \dots, m_t$. The attacker aims at recovering the users' data $m_{t+1}, m_{t+2}, \dots, m_N$. Since the secret is not involved with aggregation performed by GW, the compromising of GW gives no help to attacker (what the GW can do, is also can be done by himself). As we assumed in our security model, the DCC cannot be compromised, therefore, for any subset $\mathcal{S} \subset \{1, 2, \dots, N\}$, $\sum_{i \in \mathcal{S}} m_i$ cannot be obtained by attacker, this implies that any relation between $m_1, m_2, \dots, m_N$ cannot be obtained by attacker, thus compromising of user $U_1, U_2, \dots, U_t$ also gives no help to attacker. From the analysis above all uncompromised users' data are protected.

5.6 Performance Evaluation

In this section we evaluate the performance of our proposed PDA scheme in terms of computation cost and communication overhead. The comparison between our proposed scheme and basic Paillier-based aggregation scheme shows that our scheme has a lower computation complexity, and the communication overhead of our proposed scheme is also less than that of Paillier-based aggregation scheme when the user number is in a acceptable range.

5.6.1 Computation Cost

Among the four steps of the proposed PDA scheme, the first step, system initialization, is mainly executed by the DCC. Since the DCC is a powerful entity, the initialization is only executed once and the end-users only receive the initialized parameters from DCC via GW, thus the computation cost in this step has little influence over the whole performance. Thus we concern the computation cost in the rest three steps:

report-generation, report-aggregation and report-reading. In report-generation step, each reported data need to be encrypted, the encryption causes the main computation cost. In report-aggregation step, the aggregation only involves some homomorphic operations. And in report-reading step, each received data need to be decrypted before reading, the decryption also causes the main computation cost. From the analysis above, we evaluate three parts of computation cost: encryption cost, homomorphic operation cost, and decryption cost. Besides, considering the mostly homomorphic encryption based aggregation schemes before our PDA scheme are based on Paillier encryption, thus our computation cost evaluation is compared with that of the basic Paillier based aggregation scheme [66],[67],[62],[68].

- **Encryption Cost:** Recall the Paillier encryption scheme as used in [66],[67],[62],[68], let $m \in \{0, 1\}^n$ be a plaintext, then the corresponding ciphertext is $g^m r^{\mathcal{N}} \bmod \mathcal{N}^2$, where $\mathcal{N} = pq$, p, q are two large primes, $\log \mathcal{N} = n$, and g is a generator of $Z_{\mathcal{N}^2}^*$, r is randomly chosen from $\{0, 1\}^n$. In practice, we usually encode g to a binary number with length of $2n$, and use the standard modular exponentiation operation to compute g^m , it is equivalent to $\log m$ times multiplication of two $2n$ bit binary numbers modulo by a $2n$ -bit binary number. Thus the totally computational complexity is $O(2nM_2(2n))$, where $M_2(2n)$ denotes the complexity of multiplication of two binary number with length of $2n$, and $M_2(2n) = O(2n \log 2n \log \log 2n)$ by using Schönhage-Strassen algorithm [73]. Finally, we have the totally computational complexity $O(n^2 \log n \log \log n)$.

While for the same plaintext $m \in \{0, 1\}^n$, the corresponding ciphertext in our proposed scheme is $(bu + tg + m, au + tf)$, where $b = -as + te$ as defined in previous section. Since our scheme supports for aggregation on one time multiplication of two numbers, thus a, s, u, f, g are all $2n$ degree polynomials in $\mathbb{Z}_q[x]$. By using fast fourier transform, the complexity of multiplication of two n degree polynomials is $O(n \log n)$, and since the coefficients of polynomials involved in our proposed scheme is in $\mathbb{Z}_q$, thus the totally complexity is $O(n \log n \cdot \log q \log \log q)$. As system parameter requires $\log q = O(\log N + \log n)$, where N is the number of users that supported by our proposed scheme. In a specific practical scenario, N can not be too large, i.e., we can set $N = O(n)$. Therefore, the totally complexity is $O(n \log^2 n)$, which is much less than the complexity of the Paillier based scheme.

- **Homomorphic Operation Cost:** For additive evaluation, a Paillier based scheme need to compute the product of two $2n$ length binary numbers, whose complexity is $O(n \log n \log \log n)$, and our scheme only need two addition of $2n$ degree polynomial

with coefficients in $\mathbb{Z}_q$, the complexity is also $O(n \log n)$.

• **Decryption Cost:** The most cost computation in decrypting of Paillier based scheme is a modular exponentiation operation, in specific is $c^{\lambda(\mathcal{N})}$, where c is the ciphertext, $\lambda(\mathcal{N}) = \text{LCM}(p-1, q-1)$, thus the complexity is $O(n^2 \log n)$. While the decryption of our scheme only need at most 2 times multiplication 3 times addition of some $2n$ degree polynomials with coefficient in $\mathbb{Z}_q$, the complexity is at most $O(n \log^2 n)$.

Finally, we can safely draw the conclusion that in terms of encryption cost and decryption cost, our proposed scheme is much better than a Paillier based aggregation scheme, while in terms of homomorphic evaluation cost, our scheme is also not worse than a Paillier based aggregation scheme.

5.6.2 Communication Overhead

In smart grid, the number of users is large, and each user reports his data almost in a real-time way, thus the communication overhead between users and GW dominates the cost. As mentioned in our optimized PDA scheme, several user's data are filled in a long message, and after being encrypted it will be sent from the user to GW. We use the metrics of bits per data to evaluate the communication overhead between user and GW.

In our optimized PDA scheme, the length n of the message is determined by the security parameter κ , in fact we set $n = \kappa$. By algorithm 1, we can at most fill $l = \lfloor \log(n + 2k) - \log 2k \rfloor$ data in the n -length message m , where k is the length of each data after encoding into bits. Then user encrypts m to a ciphertext ct , which can be represented as $2n$ symbols in $GF(q)$. In the parameter setting, $q = 256 \cdot N^{\frac{3}{2}} n^2 n'$, where N is total number of users, n' is the largest index of non-zero bit in m . By the analysis above and considering the dual-functional aggregation of our scheme, the communication overhead is $\frac{2n \log q}{2l} = \frac{n \log q}{l}$. As listed in the following table, we first choose the security parameter κ by following the method in [72], and according to different security requirements, we give the corresponding values of other parameters, and the communication overhead.

$n = \kappa$	l	$n' = (2^l - 1)20$	$n \log q / l$
192	3	140	$96 \log N + 1454$
256	3	140	$128 \log N + 2924$
320	4	300	$150 \log N + 3288$

As shown in the last column, the communication overhead increases with the number of users. This is different from Paillier-based aggregation scheme. In the basic Paillier-based aggregation scheme, a user's data m is encrypted to $c = g^m r^N \bmod \mathcal{N}^2$, where $\mathcal{N} = pq$, p, q are two large primes. Thus the communication overhead of the basic Paillier-based aggregation scheme is $2 \log \mathcal{N}$.

By setting $n = 256$ for our PDA scheme, it can provide medium security, which is, as mentioned in [72], equivalent to the security of AES-128, and is enough for ordinary users. While for Paillier-based scheme, to provide the same security, as NIST suggested, we need to choose the RSA modulus $\mathcal{N}$ to be 2048 bits, i.e., $\log \mathcal{N} = 2048$. In this case, the comparison of communication overhead between our PDA scheme and the basic Paillier-based scheme is shown in the following figures

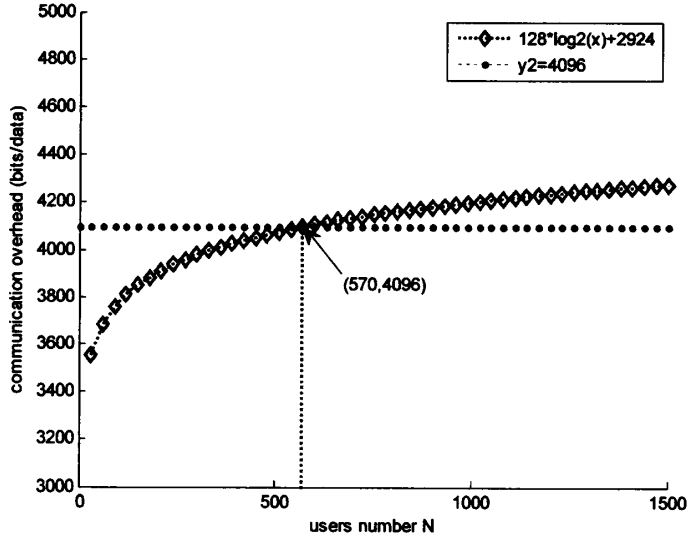


Figure 5.3 Communication overhead comparison (bits/data)

As shown in Figure 5.3, when the users number is less than 570, the communication overhead of our PDA scheme is less than that of a Paillier based aggregation scheme. When the user number is more than 570, the communication overhead of our PDA aggregation scheme is more than that of a Paillier based aggregation scheme. But it increases not fast, as shown in Figure 5.4, when the user number increases from 2000 to 4000, then communication overhead of our PDA scheme increases from 4328 to 4457. 570 is a acceptable number for users in a residential area or a building. If a greater users number is required in some case, our PDA scheme is also recommended, since the advantage in computation complexity can offset the shortcoming in communication

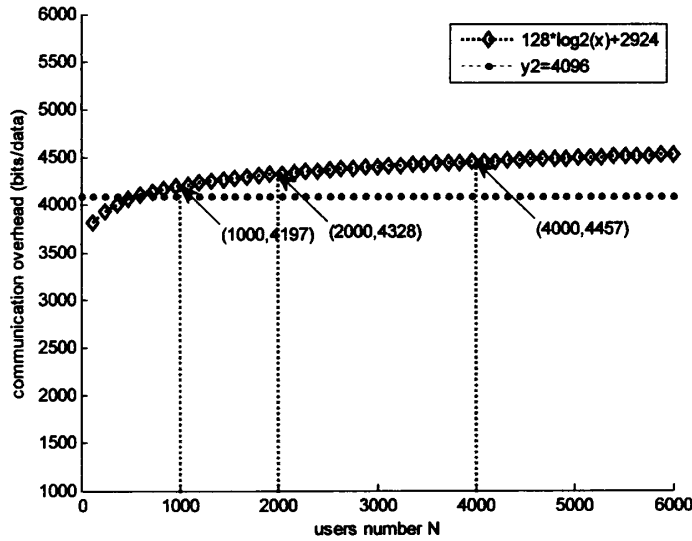


Figure 5.4 Communication overhead comparison (bits/data)

overhead when users number is more than 570.

5.7 Related Works

Privacy-preserving aggregation has been studied for many years, and several privacy-preserving aggregation schemes have been proposed recently. Garcia and Jacobs [66] introduce a privacy-preserving energy-metering aggregation protocol for smart grid communication, where an additive Paillier homomorphic encryption was first recommended in privacy-preserving aggregation scheme. In [67], Li et al. also present a in-network aggregation scheme for smart grid by combining Paillier homomorphic encryption with aggregation tree. In [62], Lu et al. propose an efficient Privacy-preserving aggregation scheme for smart grid, which is also based on homomorphic encryption, and uses a super increasing sequence to support multidimensional aggregation. Shi et al. [68] propose a privacy-preserving aggregation scheme for time-series data, and introduces noise on individual data to preserve the privacy against both outside and insider attackers. Erkin et al. [69] introduce a private computation protocol, which provides aggregation both for spatial data and temporal data. In addition, by broadcasting random numbers between each pair of smart meters via a secure channel, it can also enhance the privacy of the individual user.

There also exist some privacy-preserving aggregation schemes without using homomorphic encryption. Kursawe et al. [74] propose some interactive privacy-preserving aggregation protocols for smart grid, which are basically based on Diffie-Hellman key

exchange protocol and bilinear pairing. Lin et al.[75] designed a multi-dimension aggregation scheme by using super increasing sequence technique, which does not use a public homomorphic encryption scheme, but assumes a symmetric key sharing between communication parties. Different from the above mentioned, in this work, we design our new privacy-preserving dual-functional aggregation scheme from a RLWE-based SH Encryption [37],[22], which can support both additional and multiplicative aggregation for smart grid communications.

Chapter 6 Conclusion and Future Work

6.1 Conclusion

In this dissertation, we review the present somewhat homomorphic encryption scheme. We investigate the capacity of homomorphic evaluation of these somewhat homomorphic encryption scheme by using a multi-layer circuit model. As the basis of fully homomorphic encryption scheme, we propose the dual noise structure, a uniform reputation of somewhat homomorphic encryption scheme. Then based on the uniform model, we point out the different between the concepts of perfect fully homomorphic encryption and generalized fully homomorphic encryption. As follows, we give an explicit description of constructing a generalized fully homomorphic encryption scheme based on a somewhat homomorphic encryption scheme by using bootstrapping, and we also give a condition for the existence of generalized fully homomorphic encryption. Since the complexity of the homomorphic decryption circuit is a crucial metrics in fully homomorphic encryption, we give, from a circuit level, a universal optimization method to reduce the complexity of the homomorphic decryption circuit. Finally, we proposed a somewhat homomorphic encryption scheme based on ideal membership problem, which can be directly used as a fully homomorphic encryption scheme without using bootstrapping.

6.2 Future Work

In the area of fully homomorphic encryption, there still exists some work to do, we point out several direction for the future works.

- (1) **Deep Optimization for FHE:** Optimization to decryption circuit is very important both in theoretic and in practical. Optimization main can be done from two level: algorithm level and circuit level. The optimization in algorithm level is to, for a specific FHE scheme, reduce the complexity of the decryption algorithm, so that the corresponding circuit will be simpler. The optimization in circuit level, as we propose in chapter 3, is a universal method which is directly reducing the complexity of the decryption circuit. Optimization in circuit level is easy to manipulate, and applicable for all FHE scheme, but its effect is limited by the decryption algorithm. So combining algorithm-level and circuit-level optimization will be a valuable research direction.

- (2) **Security of NTRU-FHE:** The NTRU somewhat homomorphic encryption

scheme is the fastest known scheme over ring, but its aggressive parameter setting does ensure the decisional small polynomial ratio assumption hold for a unbounded adversary. However, if this parameter setting is secure for a bounded adversary, then we can relieved to use it. Therefore the security of of NTRU-FHE will be a meaningful work in the future.

- (3) **Perfect FHE:** Perfect FHE scheme is homomorphic encryption scheme that allows anyone to evaluate any homomorphic circuit without using bootstrapping. Therefore perfect FHE schemes are more efficient then generalized FHE schemes. Furthermore, since the homomorphic evaluating need not involve the evaluating key, which is key-dependent, so perfect FHE is believe more secure (noted the security of the perfect FHE itself must be proved). The present FHE schemes with strict provable security are of generalized FHE schemes, so to give a strict security proof for the present perfect FHE schemes or to design new schemes are meaningful works.

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Brief Introduction of Author

1. Basic Information:

Li Chen, male, birth in Xi'an, in Nov. 1982. A Ph.D. candidate in School of Telecommunications Engineering, Xidian University. Specialty is cryptography.

2. Education Experience

2001.09~2005.07 Shaanxi Normal University, Mathematics, Bachelor of Science.

2006.09~2009.04 Xidian University, Cryptography, Master of Science.

2011.09~ Xidian University, Cryptography.

3. Research Results

3.1 Publications:

- [1] Li, Chen; Lu, Rongxing; Li, Hui; Chen, Le; Li, Xiaoqing. Comment on "A Novel Homomorphic MAC Scheme for Authentication in Network Coding"[J]. IEEE Communications Letters, 2014, 18(12): 2129 - 2132.
- [2] Li, Chen; Chen, Le; Lu, Rongxing; Li, Hui; Li. Comment on "An Efficient Homomorphic MAC with Small Key Size for Authentication in Network Coding"[J]. IEEE Transactions on Computers, 2015, 64(3): 882-883.
- [3] Li, Chen; Lu, Rongxing; Li, Hui; Chen, Le; Chen, Jie. PDA: a privacy preserving dual functional aggregation scheme for smart grid communications [J]. Security and Communication Networks, 2015.

3.2 Research Project:

- [1] Twelfth "five-year" National Development Fund in Cryptography. Homomorphic Cryptography Research on Theory and Applications. 2012.1-2013.12. Finished. Participator.
- [2] National Science Foundation of China. Research on the construction methods of provable public key encryption based on hybrid encryption and labeling trick. 2015.1-2017.12. Under research. Participator.
- [3] National Science Foundation of China. Anonymous batch authentication techniques based on group in smart grid. 2013.1-2015.12. Under research. Participator.

作者简介

1. 基本情况

李晨, 男, 陕西西安, 1982 年 11 月出生, 西安电子科技大学通信工程学院密码学专业 2011 级博士研究生。

2. 教育背景

2001.09~2005.07 就读于陕西师范大学, 获得数学与应用数学专业、理学学士学位

2006.09~2009.04 就读于西安电子科技大学, 获得密码学专业硕士学位

2011.09~ 西安电子科技大学密码学专业博士研究生

3. 在学期间的研究成果

3.1 发表的学术论文

[1] Li, Chen; Lu, Rongxing; Li, Hui; Chen, Le; Li, Xiaoqing. Comment on “A Novel Homomorphic MAC Scheme for Authentication in Network Coding”[J]. IEEE Communications Letters, 2014, 18(12): 2129 - 2132.

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3.2 参加科研项目情况:

[1] “十二五”国家密码发展基金, 同态密码理论及其应用研究, 2012.1-2013.12, 已完成, 参与者.

[2] 国家自然科学基金, 基于混合加密和标记技巧的可证明安全公钥加密构造方法研究, 2015.1-2017.12, 在研, 参与者.

[3] 国家自然科学基金, 智能电网中基于群组的匿名批量认证技术, 2013.1-2015.12, 在研, 参与者.